I. Research Paper

Note: [#]Co-first author;

All the following published research papers are available in the public database of PubMed at: **https://pubmed.ncbi.nlm.nih.gov/?term=**

（1） Jun Wang, **Leyna G. Liu,** Wen-Qiao Jiao, Hua Yang, Jing Liu, Dahai Liu：Phenylboronic Acid-Conjugated Chitosan Nanoparticles for High Loading and Efficient Delivery of Curcumin. **Carbohydrate Polymers.** 2021, March 15; 256; 117497. https://doi.org/10.1016/j.carbpol.2020.117497.

（2） Kan He, Jian Zhang, Justin Liu, Yandi Cui, **Leyna G Liu,** Shoudong Ye, Qian Ban, Ruolan Pan, Dahai Liu：Functional Genomics Study of Protein Inhibitor of Activated STAT1 in Mouse Hippocampal Neuronal Cells Revealed by RNA Sequencing. **Aging (Albany NY).** 2021, Mar 24; 13(6): 9011-9027. doi: 10.18632/aging.202749. Epub 2021 Mar 24.

（3） Qian Ban, Li Qiao, Haidong Xia, Baiyi Xie, Justin Liu, Yunhan Ma, Liyi Zhang, Meng Zhang, **Leyna G Liu,** Wenqiao Jiao, Shuting Yang, Zongye Li, Songguo Zheng, Dahai Liu, Junjie Xia, Zhongquan Qi：β-catenin Regulates Myocardial Ischemia/reperfusion Injury Following Heterotopic Heart Transplanation in Mice by Modulating PTEN Pathways. **American Journal of Translational Research.** 2020 Aug 15; 12(8): 4757-4771. eCollection 2020.

（4） Wen-Xing Li, Shao-Xing Dai, San-Qi An, Tingting Sun, Justin Liu, Jun Wang, **Leyna G Liu,** Yang Xun, Hua Yang, Li-Xia Fan, Xiao-Li Zhang, Wan-Qin Liao, Hua You, Luca Tamagnone, Fang Liu, Jing-Fei Huang, Dahai Liu：Transcriptome Integration Analysis And Specific Diagnosis Model Construction for Hodgkin's Lymphoma, Diffuse Large B-cell Lymphoma, And Mantle Cell Lymphoma. **Aging (Albany NY).** 2021, Apr 22; 13(8): 11833-11859. doi: 10.18632/aging.202882. Epub 2021 Apr 22. PMID: 33885377.

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（6） Dahai Liu, Shao-Xing Dai, Kan He, Gong-Hua Li, Justin Liu, **Leyna G Liu,** Jing-Fei Huang, Lin Xu, Wen-Xing Li：Identification of Hub Ubiquitin Ligase Genes Affecting Alzheimer's Disease by Analyzing Transcriptome Data From Multiple Brain Regions. S**cience Progress.** 2021, Jan-Mar; 104(1): 368504211001146. doi: 10.1177/00368504211001146. PMID: 33754896.

（7） An-Ping Yang[#], Leyna G Liu[#], Min-Min Chen[#], Fang Liu, Hua You, Lian Liu, Hua Yang, Yang Xun, Jing Liu, Rui-Xue Wang, David D Brand, Dahai Liu, Song Guo Zheng, Wen-Xing Li. Integrated Analysis of 10 Lymphoma Datasets Identifies E2F8 as a Key Regulator in Burkitt's Lymphoma and Mantle Cell Lymphoma. **American Journal of Translational Research.** 2019, July 30, 11(7): 4382-4396.

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Phenylboronic acid-conjugated chitosan nanoparticles for high loading and efficient delivery of curcumin

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ABSTRACT

In order to achieve high loading and efficient delivery of curcumin, phenylboronic acid-conjugated chitosan nanoparticles were prepared by a simple desolvation method. These nanoparticles exhibited a regular spherical shape with the average size about 200−230 nm and narrow size distribution, which were kinetically stable under physiological condition. Due to boronate ester formation between curcumin and phenylboronic acid groups in the nanoparticles, and the hydrogen bonding interactions between curcumin and nanocarriers, curcumin was successfully loaded into the nanoparticles with high drug loading content. These curcumin-loaded nanoparticles showed pH and reactive oxygen species (ROS)-triggered drug release behavior. *In vitro* cell experiments revealed that the blank nanoparticles were completely nontoxic to cultured cells, and the curcumin-loaded nanoparticles exhibited efficient antitumor efficiency against cancer cells. Moreover, the drug-loaded nanoparticles performed an enhanced growth inhibition in three-dimensional multicellular tumor spheroids. Thus, these nanocarriers would be a promising candidate for curcumin delivery in tumor treatment.

1. Introduction

Nowadays, chemotherapy remains the most common therapy for many types of cancers in clinical usage. Nevertheless, the chemotherapeutic effect is limited by non-specific selectivity and serious side effects of chemotherapy drugs. To overcome these problems, nano drug delivery systems have been developed for delivering anticancer drugs to tumors. They significantly prolong blood circulation and improve drug accumulation at tumor sites (Kumari, Rompicharla, Muddineti, Ghosh, & Biswas, 2018; Miao et al., 2018; Wang, Lv, Deng, Tang, & Chen, 2016; Zhou et al., 2019). As an effective nanoparticulate drug delivery system, they should possess multiple functions, such as biocompatibility, stability, high drug loading and rapid intracellular release profile (Xu et al., 2020; Zhang et al., 2018).

Curcumin (Cur), a hydrophobic polyphenol isolated from *Curcuma longa*, has a wide range of bioactivities including anticancer, anti-inflammatory, antioxidative and immunomodulatory. Studies have shown that curcumin is a potential therapeutic agent for various cancer cells, which inhibits various cell signaling pathways and induces apoptosis in cancer cells (Cheng et al., 2018; Fang et al., 2016; Luo et al., 2017). However, therapeutic use of Cur has not been fully explored, mainly due to its extreme instability under physiological conditions and poor aqueous solubility (Luo et al., 2017; Requejo-Aguilar et al., 2017; van der Vlies, Morisaki, Neng, Hansen, & Hasegawa, 2019). In order to overcome these problems, various polymeric nanocarriers such as micelles, liposomes and other nanoscale particles were prepared for Cur delivery (Chen, Ou, & Tang, 2016; Cheng et al., 2018; Fang et al., 2016; Huang et al., 2018; Zhao, Dong, & Sun, 2019). Generally, Cur is loaded in the polymeric nanoparticles in two ways. One way is that Cur is encapsulated by hydrophobic interaction. These nanocarriers could improve the stability and solubility of Cur and prolong its circulation time, but their delivery efficiency is hindered by low drug loading and premature drug release (Qiao, Liu, Zha, Mao, & Yin, 2019; van der Vlies et al., 2019). The other way is polymer-based prodrugs used for Cur delivery. Through conjugating Cur to the polymers' chains by stimuli-responsive linkages (ester, acetal, disulfide bond, et al), Cur could be delivered with high loading efficiency and release in response to an appropriate stimulus. However, the methods for preparing these prodrugs are usually relatively complex and require multiple synthetic steps. Moreover, they may encounter ineffective drug release in tumor cells due to low sensitivity of some stimuli-responsive linkages (Fang et al., 2016; Qiao et al., 2019; Requejo-Aguilar et al., 2017; Zhao et al.,

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2019). Therefore, it is an imperious demand to develop a facile synthesis method for improving its stability, drug loading capacity and effective drug release.

Cur has the keto-enol structure, and can form boronate ester with phenylboronic acid group (PBA), which could improve the stability of Cur under physiological condition and released Cur upon the oxidative degradation of the boron-carbon bond (Luo et al., 2017; Qiao, Hwang, Kim, & Qi, 2020; van der Vlies et al., 2019). For example, van der Vlies et al. prepared PBA-containing nanoparticles through aqueous dispersion polymerization method. These nanocarriers improved chemical stability of Cur against oxidative degradation and the release of Cur was effectively sustained under physiological condition (van der Vlies et al., 2019). Thus, there has been great interest in utilizing the complexation interaction between PBA and Cur to design drug carriers that can achieve high-efficiency loading and efficient release of Cur in tumor cells.

Apart from these as mentioned above, the biocompatibility is another key consideration for designing drug delivery system. For this purpose, various natural biodegradable polysaccharide, such as cellulose (Yang et al., 2020), chitosan (Xie & Liu, 2020; Yu, Ma, Ye, Li, & Zhang, 2020), hyaluronic acid (Huang & Huang, 2018) were used for the development of novel drug delivery carriers. Among these polysaccharides, chitosan oligosaccharide has received increasing attention due to excellent biocompatibility and biodegradability, good water solubility, and high chemical reactivity (Chai et al., 2020). Therefore, chitosan oligosaccharide has been modified to prepare various nanocarriers for drug delivery (Hu, Liu, Du, & Yuan, 2009; Wang et al., 2019).

Cur having two phenol groups may form the hydrogen bonding interactions with chitosan, which contains a lot of hydroxyl and amino groups. Herein, we developed a convenient strategy to prepare the stable nanoparticles based on phenylboronic acid-conjugated chitosan oligosaccharide to realize high loading and efficient release of Cur through two types of interactions including complexation of Cur-PBA and the hydrogen bonding interactions between Cur and nanocarriers (Scheme 1). The chemical components, size, zeta potential and morphology of the chitosan-based nanoparticles were characterized. The kinetic stability of these nanoparticles was also evaluated in different conditions. Furthermore, the drug loading and *in vitro* drug release behavior of Cur from the nanoparticles were studied in detail. Finally, the biocompatibility of blank nanoparticles and *in vitro* antitumor activity of Cur-loaded nanoparticles were determined against HepG2 cells using MTT assays. The tumor growth inhibition of Cur-loaded nanoparticles was further evaluated by using 3D multicellular tumor spheroids (MCTS) as *in vitro* tumor model.

2. Materials and methods

2.1. Materials

Chitosan oligosaccharide (Cs, Mw: 3000−6000 Da, Deacetylation degree = 80 %) was purchased from Hefei Bomei Biotech Co., Ltd. 3-Carboxyphenylboronic acid (99 %), ethylenediaminetetraacetic acid disodium salt (EDTA·2Na, 98 %), curcumin (Cur, >97.0 %), N-(3-dimethyminopropy)-N'-ethylcarbodiimide hydrochloride (EDC, 98 %), N-hydroxysuccinimide (NHS, 98 %) were purchased from Aladdin Technology Co. 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT, 98 %) and 4,6-diamidino-2-phenylindole (DAPI, ≥98 %) were obtained from Sigma-Aldrich. The Dulbecco's modified eagle media (DMEM) and cell culture medium (RPMI 1640) were purchased from Gibco, USA. Human liver carcinoma (HepG2) cells were purchased from KeyGEN BioTECH (Nanjing, China).

2.2. Chitosan modified with 3-carboxyphenylboronic acid (PBA-Cs)

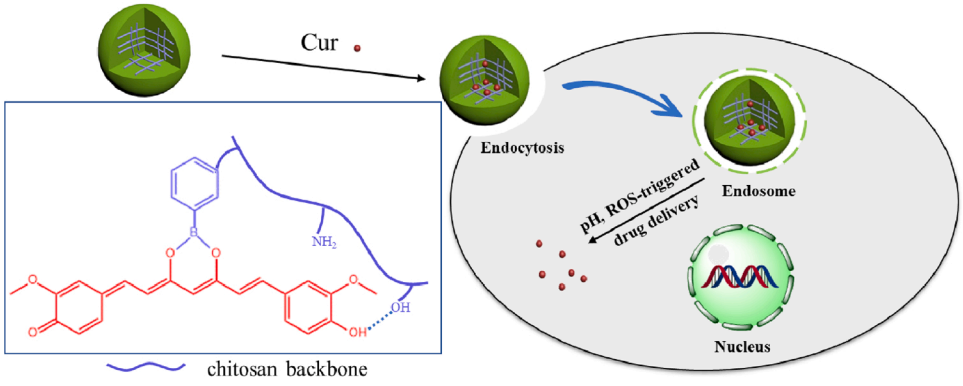
PBA-Cs was prepared by amide reaction between carboxyl group of 3-carboxyphenylboronic acid and amino groups of Cs. Briefly, a certain amount of 3-carboxyphenylboronic acid (1 equiv), EDC (1.2 equiv) and NHS (1.2 equiv) were added in 20 mL DMSO and reacted for 1 h at room temperature. Then this mixture solution was added into 20 mL aqueous solution of chitosan oligosaccharide (1 g), and the mixture was reacted at room temperature for 24 h. The crude product was dialyzed and lyophilized to obtain yellow powder PBA-Cs. Different PBA-Cs samples were prepared by charging various amounts of 3-carboxyphenylboronic acid. The resulting chitosan derivatives were noted as PBA-Cs1 (mole ratio of 3-carboxyphenylboronic acid to amine = 1: 6), and PBA-Cs2 (1: 4). The PBA-Cs samples were subjected to ^{1}H-NMR analysis and Fourier transform infrared spectroscopy (FTIR).

2.3. Preparation of PBA-Cs NPs

Cs NPs were obtained using previously reported technique with some modifications (Wang et al., 2014). Briefly, 3 mL of acetone was slowly dropped into 2 mL of Cs aqueous solution (5 mg/mL) containing 1.25 mg EDTA·2Na to form milky, uncross-linked nanoparticles. Next, 20 μL of glutaraldehyde solution was added into this mixture, and cross-linked for 4 h. Finally, Cs NPs were purified by dialysis against distilled water for 48 h using a dialysis bag (MWCO 14 kDa).

PBA-Cs1 NPs were prepared as follows: 3.2 mL acetone was slowly dropped into 1 mL PBA-Cs1 aqueous solution (20 mg/mL) to form uncross-linked nanoparticles. Next, 20 μL of glutaraldehyde solution was added into this mixture, and cross-linked for 4 h. Finally, PBA-Cs1 NPs were purified by dialysis against distilled water for 48 h using a dialysis bag (MWCO 14 kDa).

PBA-Cs2 NPs were prepared as follows: 2 mL acetone was slowly



Scheme 1. Schematic illustration of the formation of Cur-loaded nanoparticles and their pH, ROS-triggered drug release.

dropped into 1 mL PBA-Cs2 aqueous solution (9 mg/mL) to form uncross-linked nanoparticles. Next, 20 μL of glutaraldehyde solution was added into this mixture, and cross-linked for 4 h. Finally, PBA-Cs2 NPs were purified by dialysis against distilled water for 48 h using a dialysis bag (MWCO 14 kDa).

2.4. Characterization

The ^{1}H-NMR experiments were recorded by an AVANCEII 400 MHz NMR spectrometer using deuterium oxide as the solvent. The FTIR spectra of Cs and PBA-Cs samples were determined using a FTIR Spectrometer (NEXUS-870, Nicolet Instrument Co·USA) using the potassium bromide pellet method. The size, polydispersity index (PDI) and surface zeta potential of the different nanoparticles were measured by dynamic light scattering (DLS) using Nano-ZS90 apparatus (Malvern, United Kingdom) equipped with a 633 nm He-Ne laser. All the measurements were carried out at 25 °C and repeated for three times. To investigate the physiological stability of the nanoparticles, the nanoparticles were re-suspended in water, saline, PBS, and RPMI 1640, and incubated for 24 h. In addition, PBA-Cs were re-suspended in the solution of different pH (4.5, 5.5, 6.5, 7.5 and 8.5). After incubation for 24 h, the size, distribution and scattering intensity of these nanoparticles were measured by DLS. The morphology of the nanoparticles was observed by scanning electron microscope (SEM, HITACHI S-4800, Japan). Briefly, one drop of the nanoparticles' suspension was placed on a silicon wafer and air-dried at room temperature and coated with a thin layer of gold before observation.

2.5. Preparation of Cur-loaded nanoparticles

To explore the drug-loading of three different nanoparticles (Cs NPs, PBA-Cs1 NPs, and PBA-Cs2 NPs), 200 μL of a solution of Cur (2 mg) in DMSO was added into 1 mL of three types of nanoparticles suspension (10 mg), respectively. The mixture solution was stirred at room temperature for 12 h in the dark. Then, the mixture was centrifuged at 1×10^4 rpm for 10 min to remove un-loaded Cur, and the sediment was re-dispersed in 2 mL deionized water to obtain Cur-loaded nanoparticles, which were denoted as Cs/Cur NPs and PBA-Cs/Cur NPs, respectively. The concentration of un-loaded Cur in the supernatant was determined by UV spectrophotometer (UV-1900, Shimadzu) at 425 nm. The drug loading content (DLC) and drug loading efficiency (DLE) were calculated using the following equation:

$$\text{DLC}(\%) = (\text{weight of Cur in NPs})/(\text{weight of Cur-loaded NPs}) \times 100\ \%$$

$$\text{DLE}(\%) = (\text{weight of Cur in NPs})/(\text{weight of the feeding Cur}) \times 100\ \%$$

To study the effect of drug concentration on drug-loading of PBA-Cs NPs, different amounts of Cur (1 mg, 2 mg, 3.3 mg) was dissolved in 200 μL DMSO, respectively. The other steps were the same as above.

To further evaluate Cur-loading in PBA-Cs NPs, Cur was dissolved in 10 % DMSO solution (V_{DMSO}: V_{H2O} = 1:9) at a concentration of 30 μg/mL considering the poor water solubility of Cur. The absorbance of Cur, PBA-Cs NPs and PBA-Cs/Cur NPs (Cur dosage: 30 μg/mL) in water was recorded by UV–vis spectrophotometer.

2.6. In vitro release of Cur

Cur release profiles from PBA-Cs NPs were measured under three different conditions (pH 7.4, pH 5.5, pH 5.5 and 5 mM H_2O_2). Briefly, 1 mL of Cur-loaded nanoparticles with known Cur concentration was transferred into a dialysis bag (MWCO 3.5 kDa) and immersed into 5 mL of PBS. Each sample was then placed into a 50 mL centrifuge tube and slowly agitated (100 rpm) at 37 °C in the dark. At regular time-points, 5 mL of buffer solution was removed and replaced with 5 mL fresh PBS buffer solution. The release amount of Cur was determined by UV-spectrophotometry at 425 nm. The experiments were conducted in triplicate, and the results presented are the average data.

2.7. Cell viability assay

HepG2 cells (5000 cells per well) were seeded into a 96-well plate and cultured at 37 °C with 5 % CO_2 for 24 h. The medium was replaced by 200 μL of fresh medium containing free Cur and Cur-loaded nanoparticles at final Cur concentration range from 1.5625 to 50 μg/mL or PBA-Cs NPs at final concentration range from 31.25 to 1000 μg/mL. After incubation for 48 h, the medium was removed, and 200 μL of fresh medium containing MTT (0.5 mg/mL) were added to each well and incubated in darkness for 4 h. Finally, the supernatants were withdrawn, and 150 μL DMSO was added in each well to dissolve the blue formazan crystals. The plates were vibrated for 5 min in the dark, and then the absorbance of formazan in each well was measured at 570 nm using microplate reader (SpectraMax M2e, Molecule Devices). Cell viability was calculated by the following formula: Cell viability = (OD570(samples)/OD570(control)) × 100 %. Data were represented as mean ± standard deviation (n = 3).

2.8. In vitro cellular uptake

Qualitative analysis of the cellular uptake of different Cur formulations was observed by confocal laser scanning microscopy (CLSM, FluoView TM FV1000, Olympus). In brief, HepG2 cells (1×10^5 cells per well) were seeded into a 6-well plate which containing a microscope coverslip on the bottom of per well. After adhering for 24 h, the culture medium of each well was replaced by 2 mL fresh medium containing PBA-Cs/Cur NPs (drug dosage: 25 μg/mL). After co-cultured for 4 h, the medium was removed and the cells were washed with PBS twice and fixed with 4% paraformaldehyde for 10 min. The cells were then washed with PBS twice. The fluorescence images of cells were observed by CLSM.

The cellular uptake of PBA-Cs/Cur NPs (drug dosage: 25 μg/mL) were further quantified using flow cytometry. Briefly, HepG2 cells (1×10^5 cells per well) were seeded into a 6-well plate and allowed to adhere for 24 h. The medium was removed, the cells were washed with PBS twice and the culture medium of each well was replaced with the fresh medium containing PBA-Cs/Cur NPs (drug dosage: 25 μg/mL). After co-cultured for 4 h, the Cur-containing medium was removed. The cells were washed with PBS twice and digested by trypsin solution for 5 min. The cells suspension was centrifuged at 1000 rpm for 10 min. Subsequently, the upper medium was removed and the cells were re-suspended by 1 mL PBS buffer solution. The green fluorescence intensity (FL1, 515−545 nm) of Cur was determined by flow cytometry (BD FACS Calibur).

2.9. Growth inhibition in HepG2 MCTS

Three-dimensional cultured multicellular tumor-like spheroids (MCTS) based on HepG2 cells were cultured as described in the previous work (Wang, Xu, Xu, Fang, & Tang, 2020). Briefly, HepG2 cells (5×10^5 cells) were added into a poly-HEMA-coated cell culture flask containing 5 mL fresh medium. The medium was replaced every other day, and HepG2 MCTS with diameter range from 100 to 150 μm formed spontaneously after 7 days. MCTS were co-cultured with PBA-Cs NPs, free Cur and PBA-Cs/Cur NPs (final drug dosage: 25 μg/mL), respectively. The images of MCTS were recorded by optical microscope every two days. The regular changes of the MCTS's diameter could reveal the cytotoxicities of all formulations to MCTS.

3. Results and discussion

3.1. Preparation and characterization of PBA-Cs and PBA-Cs NPs

The synthetic route of PBA-Cs was presented in Fig. 1a. PBA-Cs was

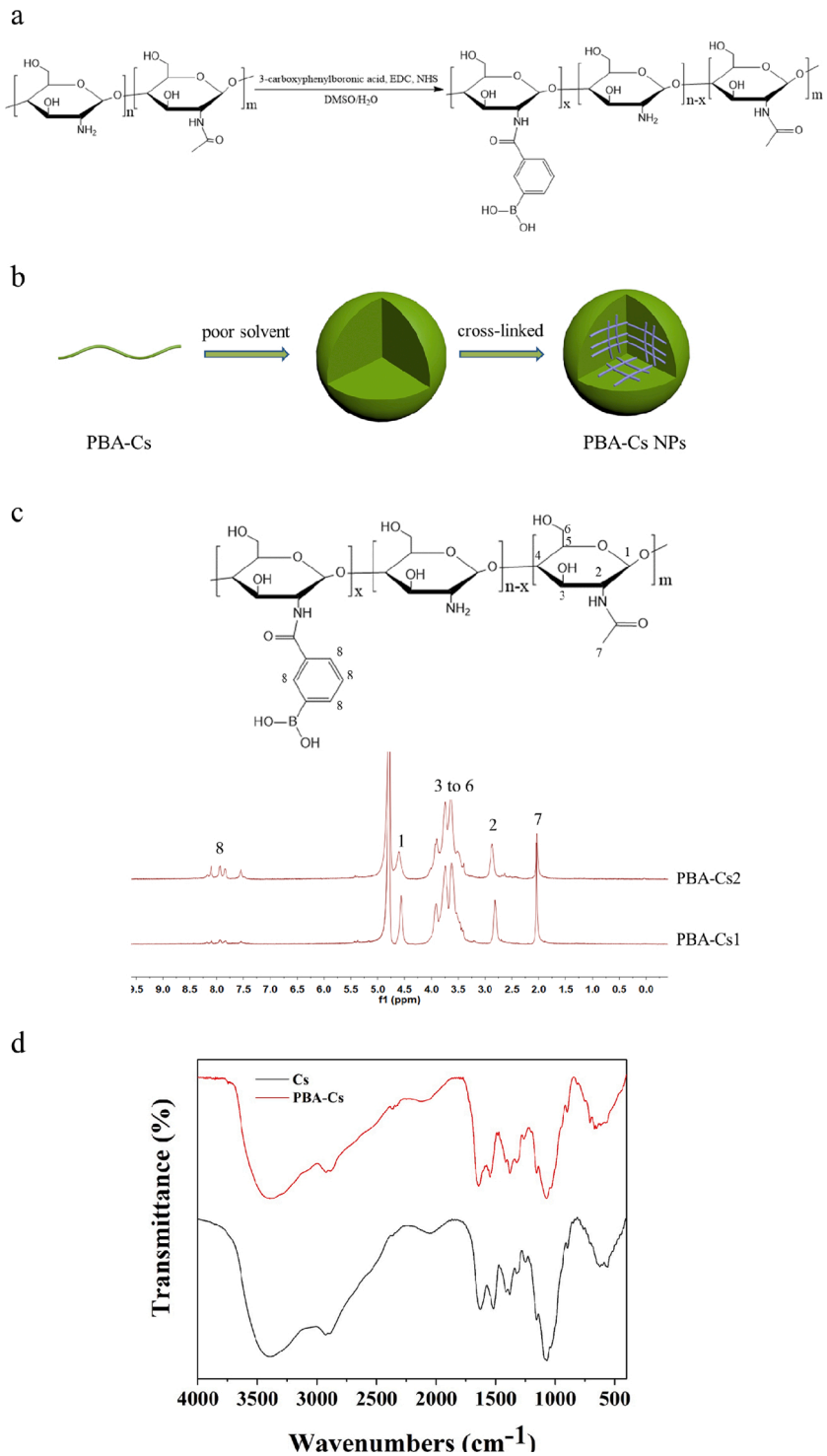


Fig. 1. The synthesis route of PBA-Cs (a) and its nanoparticles (b); ^{1}H NMR spectra of Cs and PBA-Cs in deuterium oxide (c); FTIR spectra of Cs and PBA-Cs (d).

synthesized through amide reaction at different feed ratios. The chemical structure of PBA-Cs1 and PBA-Cs2 was confirmed by ^{1}H NMR spectra, as shown in Fig. 1c. The ^{1}H NMR spectra of PBA-Cs showed chemical shifts at 7.4–8.3 ppm, referring to the aromatic protons. The integral ratio between the peaks at 7.4–8.3 ppm and at 3.5–4.1 ppm (3–6) determined the degree of phenylboronic acid substitution to be 7.3 % and 17.6 %, respectively. The modification of PBA was also detected by FTIR, as shown in Fig. 1d. The C═O (amide I) stretching vibration absorption peak arose at 1641 cm^{-1} for PBA-Cs. The absorption bands at 707 and 1324 cm^{-1} were due to in-plane bending vibration of benzene and stretching vibration of boric acid, respectively, indicating successful synthesis of PBA-Cs. Fig. 1b showed the synthesis route of PBA-Cs NPs, which were prepared by a simple desolvation method and then cross-linked by glutaraldehyde solution. During the preparation of nanoparticles, the effect of cross-linking agent glutaraldehyde on physical characteristics of the nanoparticles was also studied (Gupta & Jabrail, 2006, 2007). The zeta potential of the nanoparticles decreased with increase of glutaraldehyde volume, which indicated that many amino groups of the nanoparticles were consumed. Further increase of glutaraldehyde volume resulted in large particles or precipitate because

the particles were cross-linked with each other. The particles size and size distribution of three different nanoparticles (Cs NPs, PBA-Cs1 NPs, and PBA-Cs2 NPs) were measured by DLS. As shown in Fig. 2a–c, they showed a small average size 200−230 nm and a relatively narrow distribution. The ratio of PBA modification (0, 7.3 %, 17.6 %) has a slight influence on the average size of three different nanoparticles, and the corresponding average size was 204.9 nm, 208.1 nm, and 226.9 nm, respectively, while PDI was 0.252, 0.158, and 0.193. Besides, the size and morphology of the nanoparticles were also analyzed with SEM images. As shown in Fig. 2d–f, three different nanoparticles were well dispersed with spherical shape. Interestingly, the average sizes of Cs NPs and PBA-Cs1 NPs were ~ 100 nm, and smaller than the average size of PBA-Cs2 NPs (~ 150 nm), which was attributed to rigidity of PBA-Cs2. Moreover, their average sizes were smaller than the sizes measured by DLS due to the dehydration during observation.

3.2. Stability of PBA-Cs NPs

The stability of PBA-Cs1 NPs and PBA-Cs2 NPs in the buffer solution at different pH values was also investigated by DLS. As shown in Fig. 3a and b, their average sizes and count rates only gave a slight change, indicating PBA-Cs1 NPs and PBA-Cs2 NPs were stable at different pH values, even in mild acid buffer solution (such as pH 5.5 and 6.5). In addition, the change of zeta potential of PBA-Cs1 NPs and PBA-Cs2 NPs was evaluated. As shown in Fig. 3c. the surface potential of these nanoparticles was positive and decreased with the increase of pH value. In acid buffer solution, the zeta potential of the nanoparticles increased, which indicated that the amino groups of phenylboronic acid-conjugated chitosan chains were protonated. While at high pH value, the zeta potential of the nanoparticles decreased, which was attributed to two effects: deprotonation of amino groups and anionization of phenylboronic acid. Interestingly, PBA-Cs2 NPs displayed relatively lower zeta potential than PBA-Cs1 NPs at corresponding pH values because the conjugation of PBA to chitosan consumed many amino groups. Besides, both the nanoparticles showed a slight change in size under various physiological conditions (water, PBS, saline and RPMI 1640), as shown in Fig. 3d and e. They showed a good stability in these environments. While the size of PBA-Cs1 NPs increased slightly in RPMI 1640 due to the relatively higher zeta potential, which resulted in the electrostatic interactions between the nanoparticles and the composition of culture media. These results indicated that PBA-Cs NPs were kinetically stable and suitable for a drug carrier.

3.3. Cur loading and in vitro release

The previous studies proved that boronate ester formation between Cur and PBA could prompt the encapsulation of Cur into nanoparticles (Luo et al., 2017; van der Vlies et al., 2019). Herein, we demonstrated a convenient strategy to prepare the stable nanoparticles with high drug loading of Cur through the incorporation of the complexation interactions between PBA and Cur. DLC and DLE were investigated for the nanoparticles, as shown in Tables 1 and 2. At mass ratio (NPs/Cur) = 5:1, Cur was loaded in Cs NPs and DLC was 40.1 %, indicating the hydrogen bonding formation between Cur and chitosan, which contained many hydroxyl and amino groups. In addition, the DLC of phenylboronic acid-conjugated chitosan nanoparticles increased from 11.0 % to 15.1 % with the increase of modification ratio of PBA from 7.3 % to 17.6 %, while DLE increased from 61.6 % to 83.5 %. These results demonstrated that the hydrogen bonding formation between Cur and nanocarriers, and boronate ester formation between Cur and PBA in the nanoparticles, could contribute to delivering Cur with high efficiency loading (Fig. 4a).

The particles size of Cur-loaded nanoparticles only gave a slight change after Cur-loading. We selected PBA-Cs2 NPs (aliased as PBA-Cs) for further experiments due to the higher DLC. To study the effect of drug concentration on drug loading of PBA-Cs NPs, we found that the DLC increased from 8.38 % to 18.3 % with the increase of Cur feeding ratio from 10:1 to 3:1, while DLE decreased from 91.5 % to 67.3 % because there were no more interactions between Cur and the nanoparticles. In addition to prodrug, the drug encapsulating capacity of polymeric nanocarriers was limited. For example, Cur loading content was 10.2 % for poly(lactic-co-glycolic acid) copolymer (PLGA) nanoparticles, 14.0 % for galactosylated BSA nanoparticles, 9.9 % for 4-(hydroxymethyl) phenylboronic acid (HPBA)-modified poly(ethylene glycol) (PEG)-grafted poly(acrylic acid) polymer (PPH) micelles, 12.95 % for poly(ethylene glycol)-poly(ε-caprolactone) (MPEG-PCL) micelles

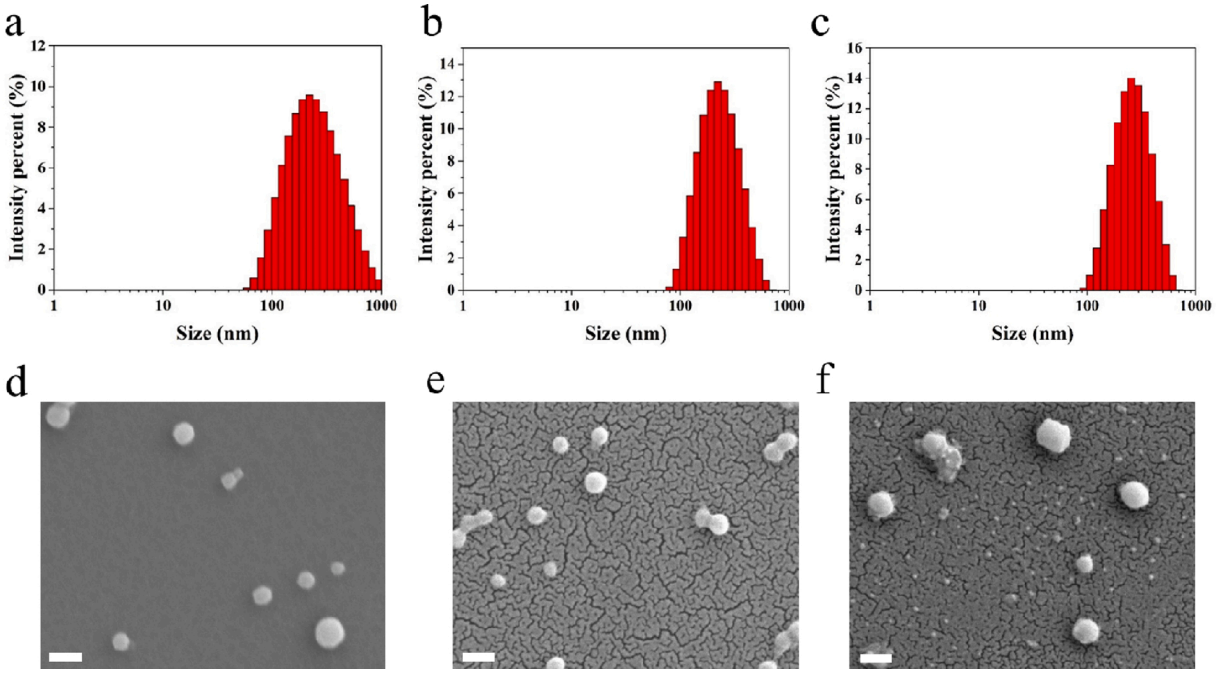


Fig. 2. Size and distribution of Cs NPs (a), PBA-Cs1 NPs (b), and PBA-Cs2 NPs (c); SEM images of Cs NPs (d), PBA-Cs1 NPs (e), and PBA-Cs2 NPs (f). The scale bar is 200 nm.

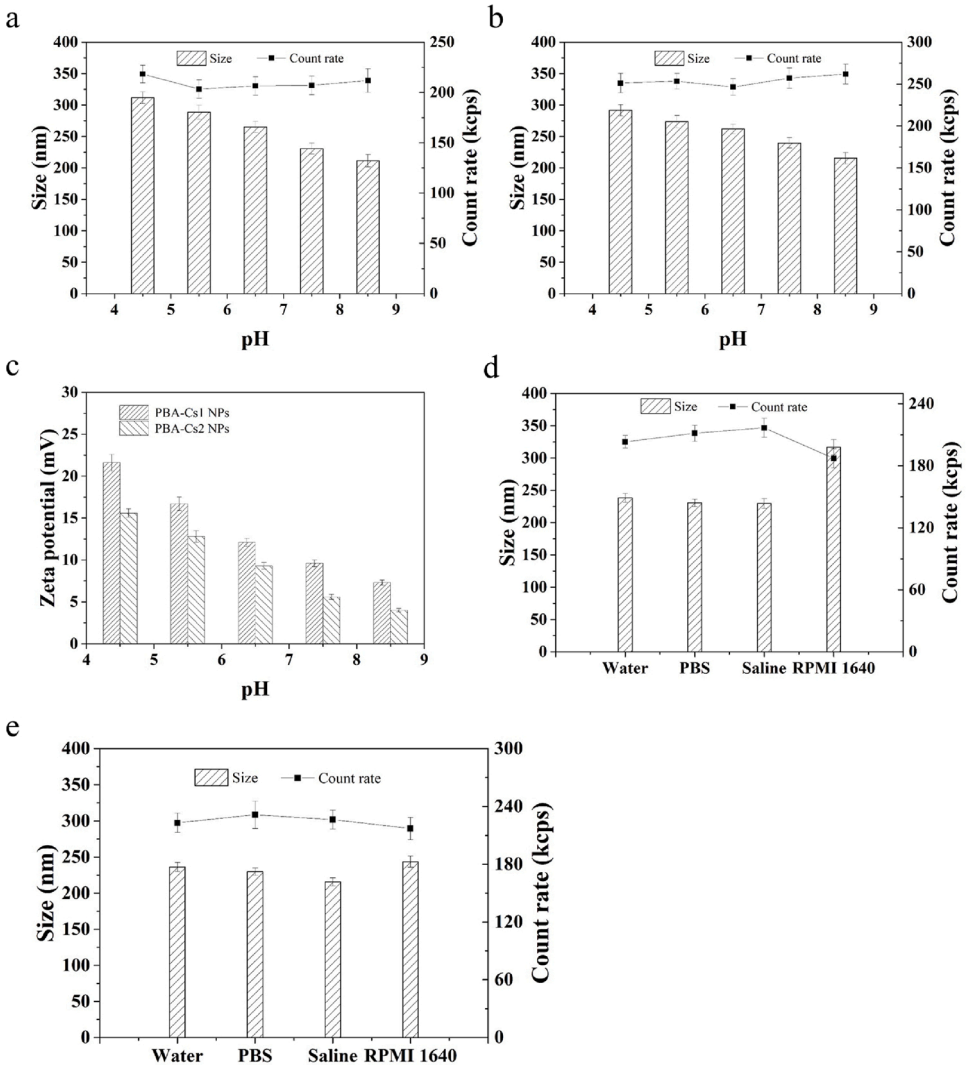


Fig. 3. The stability of PBA-Cs1 NPs (a) and PBA-Cs2 NPs (b) at different pH values; Zeta potentials of PBA-Cs1 NPs and PBA-Cs2 NPs (c) at different pH values; The stability of PBA-Cs1 NPs (d) and PBA-Cs2 NPs (e) in water, PBS, saline and RPMI 1640.

Table 1
Characterization of Cur-loaded Cs NPs, PBA-Cs1 NPs, and PBA-Cs2 NPs.

NPs	Drug loading content (%)	Drug loading efficiency (%)	Size (nm)	PDI
Cs NPs	7.4	40.1	227.5	0.163
PBA-Cs1 NPs	11.0	61.6	236.7	0.192
PBA-Cs2 NPs	15.1	83.5	243.5	0.173

Mass ratio (NPs/Cur) = 5:1, PDI: polydispersity index.

Table 2
Characterization of Cur-loaded PBA-Cs NPs.

Mass ratio (NPs/drug)	Drug loading content (%)	Drug loading efficiency (%)	Size (nm)	PDI
3:1	18.3	67.3	259.3	0.187
5:1	15.1	83.5	243.5	0.173
10:1	8.38	91.5	234.7	0.157

(Huang et al., 2018; Luo et al., 2017; Ning et al., 2018). In the study, the loading content of Cur was up to 18.3 %. This was attributed to cooperative effect of hydrogen bonding and boronate ester bonding that facilitated drug loading.

In the preparation of the Cur-loaded nanoparticles, 1 mg Cur was dissolved in 2 mL of acetone, and slowly dropped into 1 mL of PBA-Cs2 aqueous solution (9 mg/mL) to form uncross-linked nanoparticles. Next, 20 μL of glutaraldehyde solution was added into this mixture, and cross-linked for 4 h. The average size of Cur-loaded nanoparticles was about 200 nm. However, further increase of the amount of Cur (2 mg, 3.3 mg) resulted in big particles (above 300 nm) or precipitate maybe due to hydrophobicity of PBA-Cs/Cur complexes.

As showed in Fig. 4b, PBA-Cs/Cur NPs exhibited a maximum absorption intensity at 460 nm, while Cur displayed an intense optical absorption at 425 nm. This red shift in the absorption peak indicated boronate ester bond formation between Cur and PBA groups in the nanoparticles, which was in conformity with the previous results (Luo et al., 2017; van der Vlies et al., 2019). The keto-enol structure of Cur complexed with PBA by forming a six-membered ring with a tetrahedral symmetry at the boron center, which resulted in red shift (van der Vlies et al., 2019).

Cur release profiles from PBA-Cs/Cur NPs were measured under different conditions (pH 7.4, 5.5, and pH 5.5 containing 5 mM H_2O_2), respectively. As shown in Fig. 4c, Cur release from PBA-Cs/Cur NPs

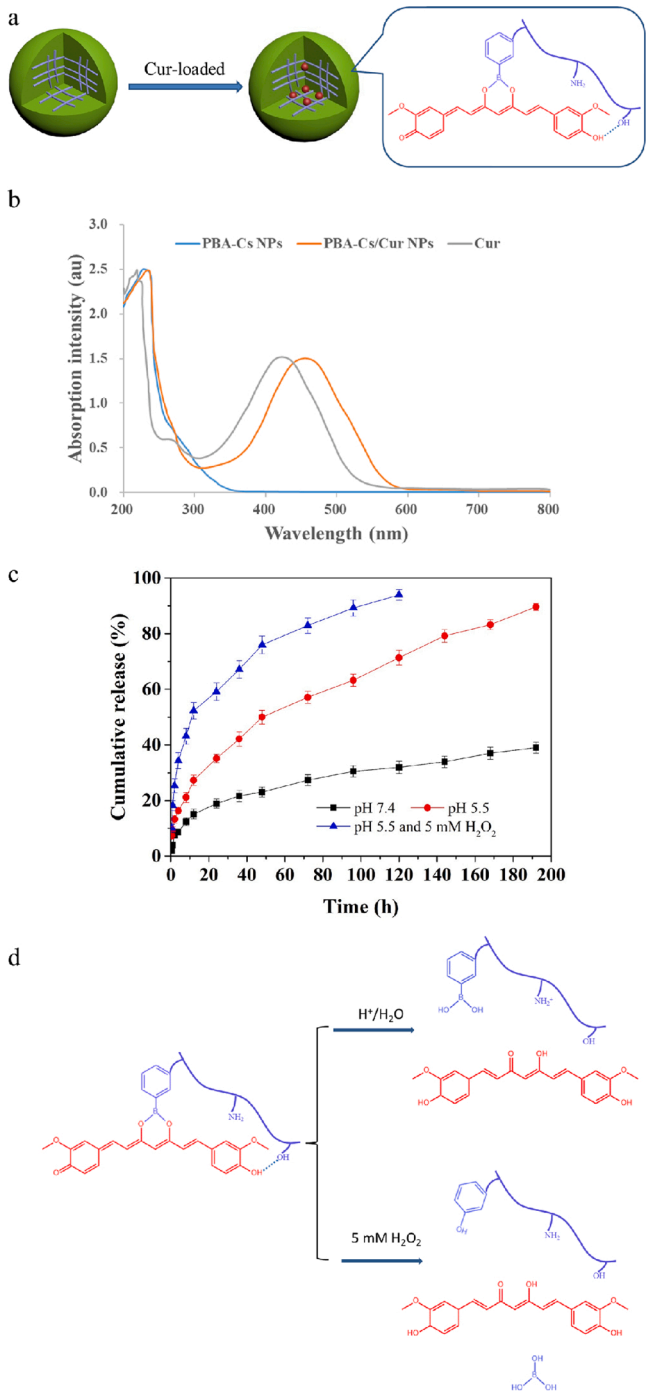


Fig. 4. Schematic illustration of the preparation of Cur-loaded nanoparticles using two types of interactions (a); UV–vis spectra of PBA-Cs NPs, Cur and PBA-Cs/Cur NPs in water (Cur = 20 μg/mL, Cur was solubilized with 10 % DMSO) (b); *In vitro* release profiles of Cur from PBA-Cs/Cur NPs under different conditions (c); Schematic illustration for the acid and ROS-triggered drug release of Cur from PBA-Cs/Cur NPs (d).

displayed a pH and ROS-dependent profile. At pH 7.4, only 39.1 % of Cur was released from PBA-Cs/Cur NPs within 196 h, indicating the boronate ester bonds were quite stable. Compared to other nanoparticles based on phenylboronic acid (van der Vlies et al., 2019), the drug release rate was slower at pH 7.4, which probably reflected the strong interactions between Cur and nanocarriers. While 89.6 % of Cur was released from PBA-Cs/Cur NPs at pH 5.5 within 196 h, and 94.1 % at pH 5.5 containing 5 mM H_2O_2 within 120 h. As PBA-Cs NPs were stable at different pH values, even in mild acid buffer solution, this accelerated drug release behavior under mildly acidic and ROS condition was attributed to two effects (Fig. 4d): the first was the decline of the hydrogen bonding interactions between Cur and the nanocarriers at low pH value; the second was the destruction of boronate ester bond by hydrogen peroxide and acid (Song, Du, & Li, 2014; van der Vlies et al., 2019; Zhang et al., 2011). In addition, the method for preparing these nanocarriers was simple and did not need multiple synthetic steps. Therefore, these nanocarriers could be suitable for tumor drug delivery.

3.4. *In vitro cytotoxicity*

The cytotoxicity of PBA-Cs NPs and PBA-Cs/Cur NPs were evaluated by MTT assay. The results were presented in Fig. 5a and b. the cell viability of PBA-Cs NPs did not display significantly cytotoxicity even after 48 h co-cultured (Fig. 5a). The cell viabilities were almost higher than 90 %, indicating that phenylboronic acid-conjugated chitosan nanoparticles were biocompatible. On the other hand, a dose-dependent cytotoxicity for free Cur and PBA-Cs/Cur NPs were evaluated (Fig. 5b). Cell viability decreased gradually with the increase of Cur concentration. Cell viability decreased to 33.3 % after PBA-Cs/Cur NPs treatment for 48 h. PBA-Cs/Cur NPs possessed lower cytotoxicity than free Cur, which was due to the controlled and slow drug release of nano-carriers in cytoplasm.

3.5. *In vitro cellular uptake*

To confirm the feasibility of cellular internalization and intracellular dug release of PBA-Cs/Cur NPs, cellular uptake experiments were evaluated on HepG2 cells using CLSM. As shown in Fig. 6a, emission of strong green fluorescence from PBA-Cs/Cur NPs was located in the cytoplasm and nucleus region, indicating PBA-Cs/Cur NPs could be rapidly internalized into the cells by endocytosis. The cellular uptake of free Cur and PBA-Cs/Cur NPs was quantified by flow cytometry. As shown in Fig. 6b, PBA-Cs/Cur NPs showed higher fluorescence intensity than free Cur under the same conditions, which was consistent with the results of CLSM. The mechanism may be that PBA groups, as a tumor-homing ligand, could react with the sialic acid groups on the surface of cancer cells to form annular boronate ester, which improved PBA-Cs/Cur NPs' cellular uptake (Wang, Wu, & Jiang, 2015).

3.6. *Growth inhibition study in HepG2 MCTS*

To investigate cytotoxicity of PBA-Cs NPs, free Cur and PBA-Cs/Cur NPs in HepG2 MCTS. The HepG2 MCTS were co-incubated with various formulations at 37 °C for 7 days and the representative images were observed by optical microscope every other day. As shown in Fig. 7a. The MCTS treated with PBA-Cs NPs showed a continuous increase in terms of diameter due to the rapid proliferation of cells. On the contrary, free Cur and PBA-Cs/Cur NPs inhibited the growth of MCTS in a certain extent during a week. PBA-Cs/Cur NPs could accumulate and deliver Cur into the periphery cells of MCTS, leading to the cell death and destruction of MCTS. Fig. 7b showed the average diameter of MCTS during the treatment. For PBA-Cs NPs, the average diameter of MCTS was 132.2 μm at day 1, while the average diameter increased to 217.5 μm at day 7. Conversely, MCTS treated with free Cur and PBA-Cs/Cur NPs displayed a decrease in spheroids diameter from 130.1 to 112.3 μm and from 129.1 to 90.9 μm during a weak, respectively. At day 7, the average diameter of PBA-Cs/Cur NPs treated MCTS was shorter than free Cur, which could be attributed to the higher penetration ability of PBA-Cs/Cur NPs and rapid drug release.

4. Conclusion

In this study, phenylboronic acid-conjugated chitosan nanoparticles were prepared by a simple desolvation method. The nanoparticles

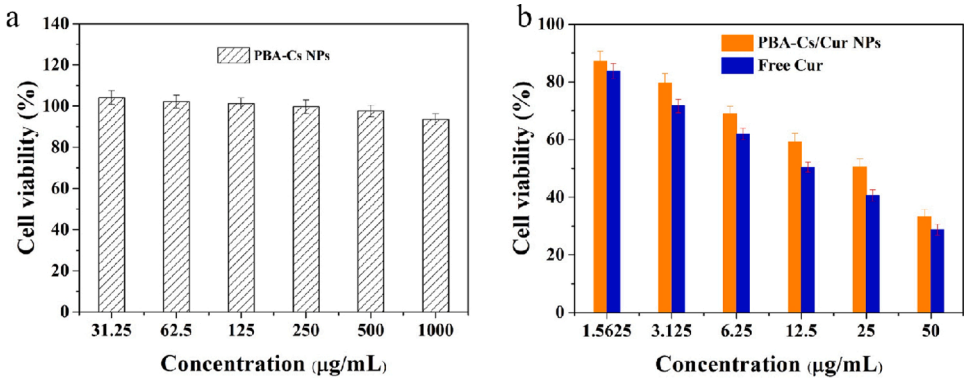


Fig. 5. *In vitro* cytotoxicity of PBA-Cs NPs (a), PBA-Cs/Cur NPs and free Cur (b) after incubated with HepG2 cells for 48 h.

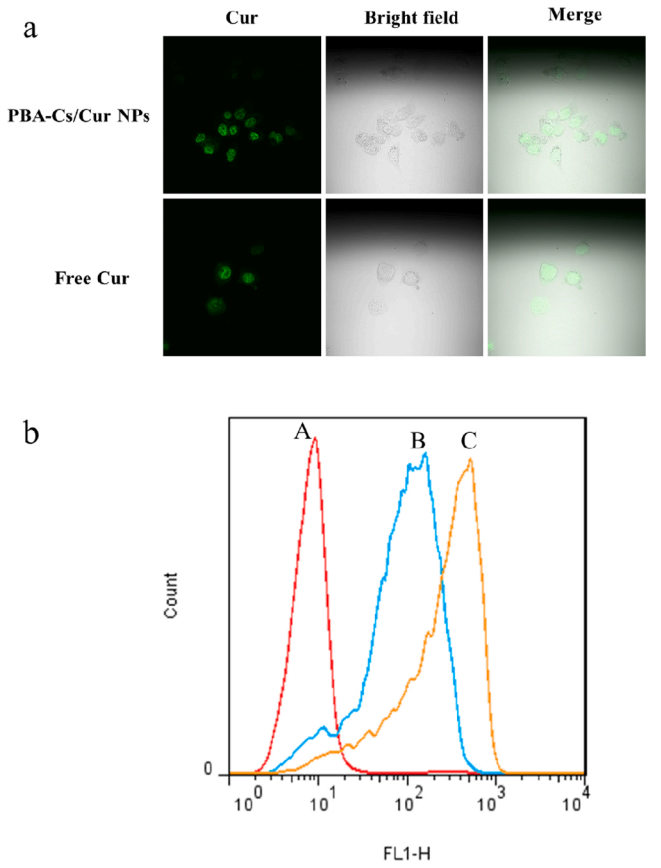


Fig. 6. CLSM images of HepG2 cells after incubated with PBA-Cs/Cur and free Cur NPs for 4 h (a); Cellular uptake of Control (A), Free Cur (B), and PBA-Cs/Cur NPs (C) for 4 h measured by flow cytometry (b).

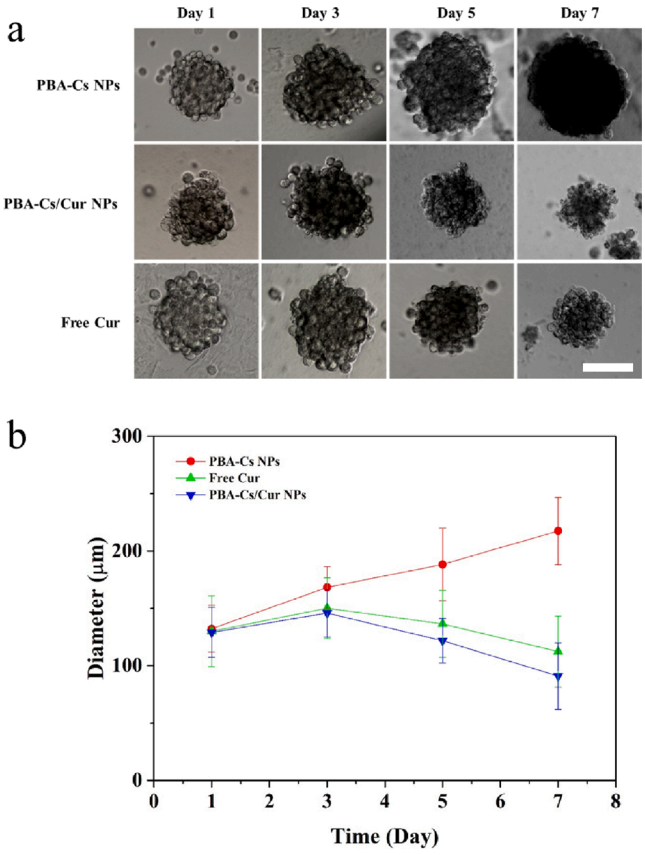


Fig. 7. Representative images of HepG2 MCTS treated with PBA-Cs NPs, free Cur and PBA-Cs/Cur NPs (a); HepG2 MCTS diameter change after various treatment (b). The scale bar is 100 μm.

displayed good stability in the physiological environment. Due to boronate ester formation between Cur and PBA groups in the nanoparticles, and the hydrogen bonding interactions between drug and nanocarriers, Cur was effectively loaded into PBA-Cs NPs with high DLC and DLE. *In vitro* drug release experiments showed that Cur release rate increased significantly in a mildly acidic environment containing 5 mM H_2O_2. PBA-Cs NPs exhibited good biocompatibility, and PBA-Cs/Cur NPs exhibited excellent anticancer activity toward HepG2 cells. 3D MCTS growth inhibition experiment demonstrated PBA-Cs/Cur NPs performed better growth inhibition than free Cur. Therefore, the phenylboronic acid-conjugated chitosan nanocarriers could be used as a promising platform for tumor therapy.

CRediT authorship contribution statement

Jun Wang: Conceptualization, Methodology, Investigation, Writing - original draft. **Leyna G. Liu:** Data curation, Writing - review & editing. **Wen-Qiao Jiao:** Investigation, Data curation. **Hua Yang:** Investigation, Visualization. **Jing Liu:** Investigation, Validation. **Dahai Liu:** Writing - review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.carbpol.2020.117497.

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Research Paper

Functional genomics study of protein inhibitor of activated STAT1 in mouse hippocampal neuronal cells revealed by RNA sequencing

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ABSTRACT

Protein inhibitor of activated STAT1 (PIAS1), a small ubiquitin-like modifier (SUMO) E3 ligase, was considered to be an inhibitor of STAT1 by inhibiting the DNA-binding activity of STAT1 and blocking STAT1-mediated gene transcription in response to cytokine stimulation. PIAS1 has been determined to be involved in modulating several biological processes such as cell proliferation, DNA damage responses, and inflammatory responses, both *in vivo* and *in vitro*. However, the role played by PIAS1 in regulating neurodegenerative diseases, including Alzheimer's disease (AD), has not been determined. In our study, significantly different expression levels of PIAS1 between normal controls and AD patients were detected in four regions of the human brain. Based on a functional analysis of *Pias1* in undifferentiated mouse hippocampal neuronal HT-22 cells, we observed that the expression levels of several AD marker genes could be inhibited by *Pias1* overexpression. Moreover, the proliferation ability of HT-22 cells could be promoted by the overexpression of *Pias1*. Furthermore, we performed RNA sequencing (RNA-seq) to evaluate and quantify the gene expression profiles in response to *Pias1* overexpression in HT-22 cells. As a result, 285 significantly dysregulated genes, including 79 upregulated genes and 206 downregulated genes, were identified by the comparison of *Pias1*/+ cells with WT cells. Among these genes, five overlapping genes, including early growth response 1 (*Egr1*), early growth response 2 (*Egr2*), early growth response 3 (*Egr3*), FBJ osteosarcoma oncogene (*Fos*) and fos-like antigen 1 (*Fosl1*), were identified by comparison of the transcription factor binding site (TFBS) prediction results for STAT1, whose expression was evaluated by qPCR. Three cell cycle inhibitors, p53, p18 and p21, were significantly downregulated with the overexpression of *Pias1*. Analysis of functional enrichment and expression levels showed that basic region leucine zipper domain-containing transcription factors including zinc finger C2H2 (zf-C2H2), homeobox and basic/helix-loop-helix (bHLH) in several signaling pathways were significantly involved in PIAS1 regulation in HT-22 cells. A reconstructed regulatory network under PIAS1 overexpression demonstrated that there were 43 related proteins, notably Nr3c2, that directly interacted with PIAS1.

INTRODUCTION

PIAS1, named as protein inhibitor of the activated STAT1, which is one of E3-type small ubiquitin-like modifier (SUMO) ligases and plays important roles in diverse cellular pathways, such as the STAT, p53 and steroid hormone signaling pathways. Based on the results of *in vivo* and *in vitro* experiments, PIAS1 was reported to regulate innate immune responses mediated by the interferon (IFN)-gamma- or -beta-inducible genes by inhibiting the STAT pathway [1]. The STAT1 protein is an essential transcription factor functioning in inflammatory activation in Alzheimer's disease (AD) brains [2–5]. PIAS1 may function as an E3 ligase for ligand-activated nuclear receptor peroxisome proliferator-activated receptor (PPAR) gamma, which is modified by SUMO-1 and is a potential target for apoptosis-induction therapy in cancer cells [6]. Based on a previous study of *Pias1*-/- cells that employed microarray analysis, PIAS1 was also identified as a novel negative regulator of NF-κB that is associated with immunity and homeostasis in a number of human illnesses, such as chronic inflammatory diseases and cancer [7–9]. PIAS1 acts as one of the positive transcriptional coregulators of GATA-3, and its molecular interaction is reported to be important for Th2 immune responses by regulating IL-13 [10]. Two members of the PIAS family proteins, PIAS1 and PIAS3, were demonstrated to interact with TATA-binding protein (TBP) [11]. It has been showed that PIAS1 associates with protein-tyrosine phosphatase 1B (PTP1B) in mammalian fibroblasts and catalyzes sumoylation of PTP1B, thereby regulating metabolism and cell proliferation [12]. PIAS1 can promote the proliferation of fibroblasts, smooth muscle cells, and intestinal epithelial cells by interacting with the Kruppel-like factor 5 (KLF5) and enhancing its ability [13]. PIAS1 was reported to confer DNA-binding specificity to the homeoprotein Msx1 in regulating myoblast differentiation through repression of the myogenic regulatory gene MyoD [14]. An epigenetic mechanism has been elucidated that PIAS1 might recruit DNA methyltransferases and heterochromatin protein 1 by binding to the Foxp3 promoter during the natural regulatory T cell differentiation [15]. It is reported that the suppression of PIAS1 abolished the ability of arsenic trioxide, an effective treatment of acute promyelocytic leukemia (APL), to trigger apoptosis in APL cells [16]. During the adipogenesis, PIAS1 plays a dynamic role by promoting the SUMOylation of CCAAT/enhancer-binding protein beta (C/EBPbeta) [17]. The epigenetic pathway associated with PIAS1 may play important roles in regulating the hematopoietic stem cells self-renewal and differentiation [18]. A critical function of PIAS1 in controlling insulin sensitivity by inhibiting the inflammatory cascade has also been observed in adipose tissue [19]. The phosphorylation of PIAS1 was reported to mediate SUMOylation of Elk-1, which may function in promoting neuronal survival in APP/PS1 mice [20].

PIAS1 may serve to regulate the accumulation of Huntingtin (HTT) proteins and its modulation in neurons, which can alter Huntington's disease (HD)-associated phenotypes *in vivo* [21, 22]. However, the pathological and molecular functions of PIAS1 in other neurodegenerative disorders has not been studied, especially for AD. Therefore, it is particularly important to explore the function of PIAS1 in AD and analyze its molecular regulation mechanism.

RESULTS AND DISCUSSION

Expression and regulation of PIAS1 in AD and HT-22 cells

Based on analyzing the differences of the PIAS1 gene expression levels in four human brain regions between the normal controls and AD patients, the expression of PIAS1 was found to be lower in both the inferior frontal gyrus (p = 0.0342) and parahippocampal gyrus (p = 0.0882) in AD samples. In the frontal cortex, the expression of PIAS1 was detected to be significantly higher (p = 0.0315) in AD samples. No significant difference was identified between the control and AD samples in the superior temporal gyrus. These results are presented in Figure 1.

By transfecting the recombinant plasmid, the stable HT-22 cells under the overexpression of *Pias1* (*Pias1*/+) was established. The qRT-PCR results showed that through the over expression experiment, we have significantly increased the expression of *Pias1* (Figure 2A). According to the results of Western blotting (WB), the protein expression of PIAS1 was also found to be higher in *Pias1*/+ cells that is labeled as ~ 72 kDa protein (Figure 2B). The CCK-8 assay results showed that the proliferation of HT-22 cells was enhanced in *Pias1*/+ cells (Figure 2C). The neurite outgrowth of HT-22 cells could also be enhanced by the overexpression of *Pias1*. In a functional study of PIAS1 in the regulation of cell proliferation in human prostate cancer (HPC), PIAS1 was reported to be increased in HPC and to enhance cell proliferation through inhibition of cyclin-dependent kinase inhibitor 1A (P21) [23]. The activation of the PIAS1-modulated Smad2/4 complex was demonstrated to mediate zinc-induced apoptosis of prostate cancer cells [24]. PIAS1 has promotive effects on cell proliferation by upregulating the activity of KLF5 in various cell types [13].

To explore the functional roles of PIAS1 in AD, some marker genes such as neuron-specific nuclear proteins including neuronal differentiation 1 (*Neurod1*) and *NeuN* encoded by FOX3, mitogen-activated protein kinase 2 (*Mapk2*), gamma-secretase activating protein (*Gsap*), microtubule associated protein tau (*Mapt*), amyloid beta precursor protein (*App*), beta-site APP-cleaving enzyme 1 (*Bace1*) and beta-site APP-cleaving enzyme 2 (*Bace2*) were investigated to determine their expression changes in response to overexpression of *Pias1*. All of these genes expressions were shown to be significantly decreased in the *Pias1*/+ group (Figure 2D). Moreover, according to the immunofluorescence analysis, it was showed that the axonal outgrowth of HT-22 cells could be increased by the overexpression of *Pias1* (Figure 2E). These results suggested that *Pias1* might have potential functions in the regulation of AD and neuronal development.

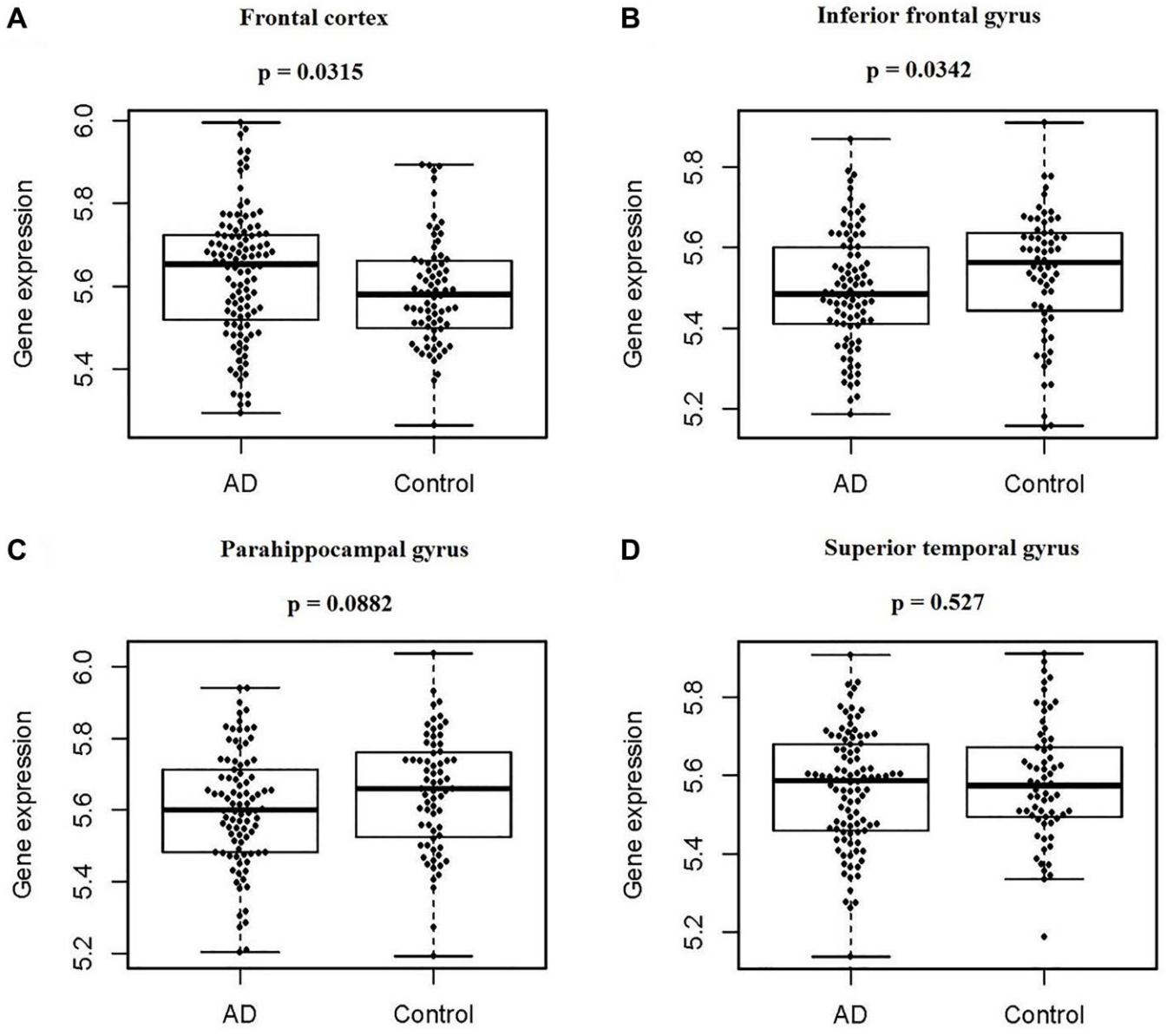

Figure 1. PIAS1 gene expression in four regions of human brains between normal controls and AD patients. Based on the Synapse database, the difference in PIAS1 gene expression in four regions of human brains between normal controls and AD patients was determined by using the Wilcoxon signed-rank test. (**A**) For the frontal cortex, there were 111 cases and 76 controls, and PIAS expression was significantly higher in AD patients, with a *p* value of 0.0315. (**B**) For the inferior frontal gyrus, there were 90 cases and 64 controls, and PIAS expression was significantly lower in AD patients, with a *p* value of 0.0342. (**C**) For the parahippocampal gyrus, there were 90 cases and 68 controls, and PIAS expression was not significantly changed, with a *p* value of 0.0882. (**D**) For the superior temporal gyrus, there were 102 cases and 65 controls, and PIAS expression was not significantly changed, with a *p* value of 0.527.

Differentially expressed genes analysis in response to *Pias1* overexpression in HT-22 cells

To screen the downstream target genes in HT-22 cells by overexpressing *Pias1*, we performed RNA-seq to obtain the transcriptome data of two cell types, WT cells and *Pias1*/+ cells. The sequencing data quality is shown in Table 1. Our data were of high quality, exhibiting a Q20 value greater than 95% and a Q30 value greater than 90%. Based on the data preprocessing and transcriptome assembly, a total of 21,862 annotated transcripts were identified in this study. According to the differentially expressed genes (DEGs) analysis, there were 285 significantly dysregulated genes, including 79 upregulated genes (FC > 2, $p < 0.05$) and 206 downregulated genes (FC < 1/2, $p < 0.05$), as determined by the comparison of the *Pias1*/+ group to the WT group. The details of the up- and downregulated DEGs are shown in Supplementary Table 1. The PCA of sequencing samples, as well as the distribution and clustering of DEGs, are shown in Figure 3.

To test the accuracy of screening DEGs by high-throughput sequencing, we selected 11 upregulated genes and 18 downregulated genes for the following qRT-PCR validation. The results showed that 7 out of 11 upregulated genes were significantly identified in the *Pias1*/+ group, they were *A4gnt*, *Cldn6*, *Gal3st2*, *Cnnm1*, *Rnf151*, *Ccr7* and *Chad* (Figure 4A). Human alpha1,4-N-acetylglucosaminyltransferase (*A4gnt*) was reported to function in O-glycan biosynthesis [25]. The expression of Claudin-6 (*Cldn6*) in early development was reported to be relevant to definitive endoderm derivatives [26]. In addition, 9 out of 18 downregulated genes were significantly identified in the *Pias1*/+ group, they were *Rflna*, *Rbm20*, *Mapk15*, *Egr1*, *Fam209*, *Mettl24*, *Shcbp1l*, *Fos* and *S100b* (Figure 4B). We further examined the PIAS1 regulation of cell cycle inhibitors, including p15, p18, p21, p27, p53 and p57. Among these genes, three inhibitors, p53, p18 and p21, were significantly downregulated with the overexpression of Pias1 (Figure 4C).

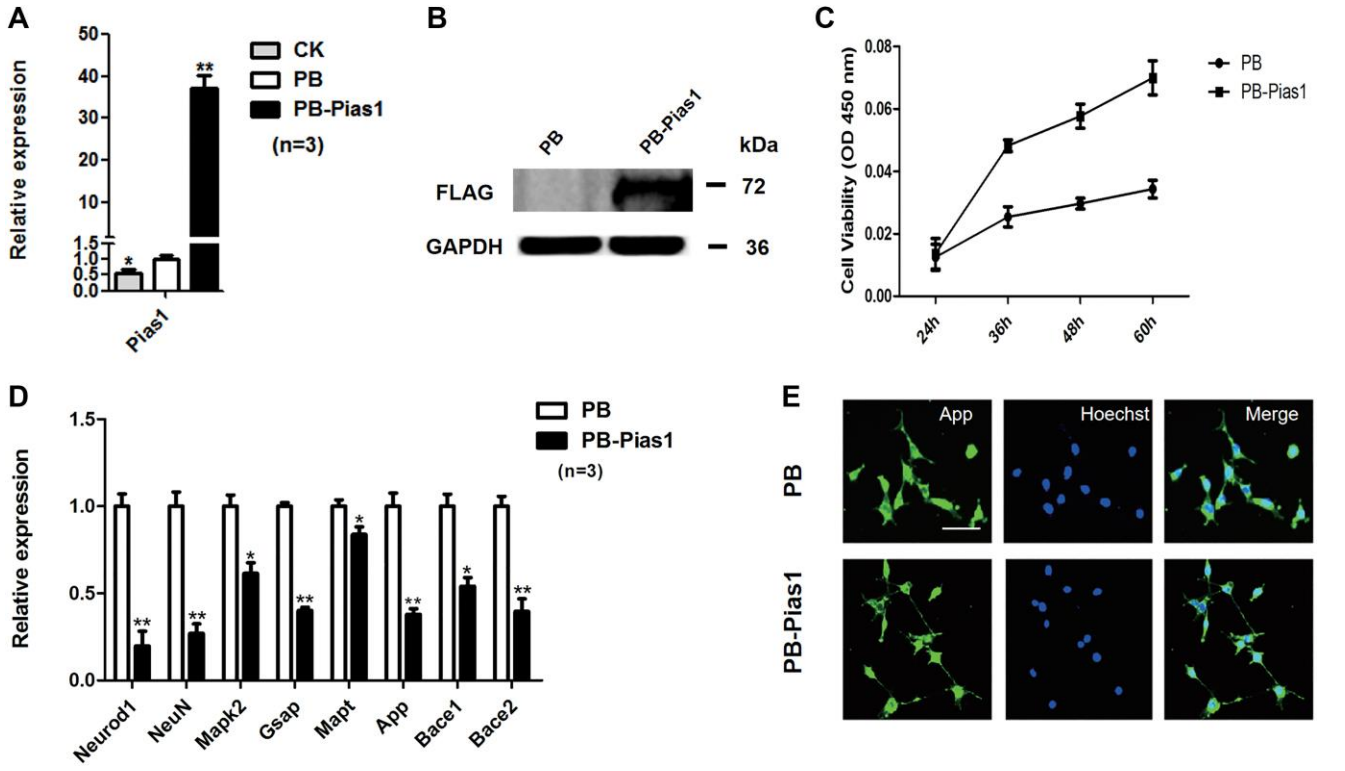

Figure 2. Overexpression of *Pias1* regulates AD marker gene expression and cell proliferation in HT-22 cells. (**A**) qRT-PCR analysis of *Pias1* mRNA expression in the hippocampal neuronal HT22 cells transfected with the indicated transgenes. Data represent the mean ± SD of three biological replicates. $^{*}p < 0.05$, $^{**}p < 0.01$ versus PB vector. The control check (CK) group represents the baseline expression of Pias1 in HT-22 cells. (**B**) Western blot analysis of PIAS1 in the hippocampal neuronal HT22 cells with stable *Pias1* transgenic expression. (**C**) Cell proliferation was determined by CCK-8 assays in HT-22 cells transfected with PiggyBac vector (PB) or PB-Pias1. Data represent the mean ± SD. $^{*}p < 0.05$, $^{**}p < 0.01$. (**D**) qRT-PCR analysis of *Neurod1, NeuN, Mapk2, Gsap, Mapt, App, Bace1* and *Bace2* expression levels in PB and PB-*Pias1* hippocampal neuronal HT22 cells. Data are presented as the mean ± SD of three independent experiments. $^{*}p < 0.05$, $^{**}p < 0.01$. (**E**) Immunofluorescence staining of APP in the hippocampal neuronal HT22 cells overexpressing PB-*Pias1*. Bar: 100 μm.

Table 1. The quality control of RNA-seq data.

Sample	Raw reads	Clean reads	Clean reads %	Q20 %	Q30 %
PB-1	48213112	45042086	93.42	97.38	92.97
Pias1-1	45822220	42887740	93.59	97.36	92.89
PB-2	50567794	47369520	93.67	97.27	92.71
Pias1-2	48265814	45448864	94.16	97.40	92.97

Signaling pathway and gene ontology enrichment under the overexpression of *Pias1* in HT-22 cells

Based on GSEA, 30 significantly enriched KEGG pathways were identified to be involved in the regulation of *Pias1* in HT-22 cells, which are shown in Figure 5. Among these pathways, the top two significantly enriched pathways were the TNF signaling pathway (FDR=4.25E–08) in terms of environmental information processing and the IL-17 signaling pathway

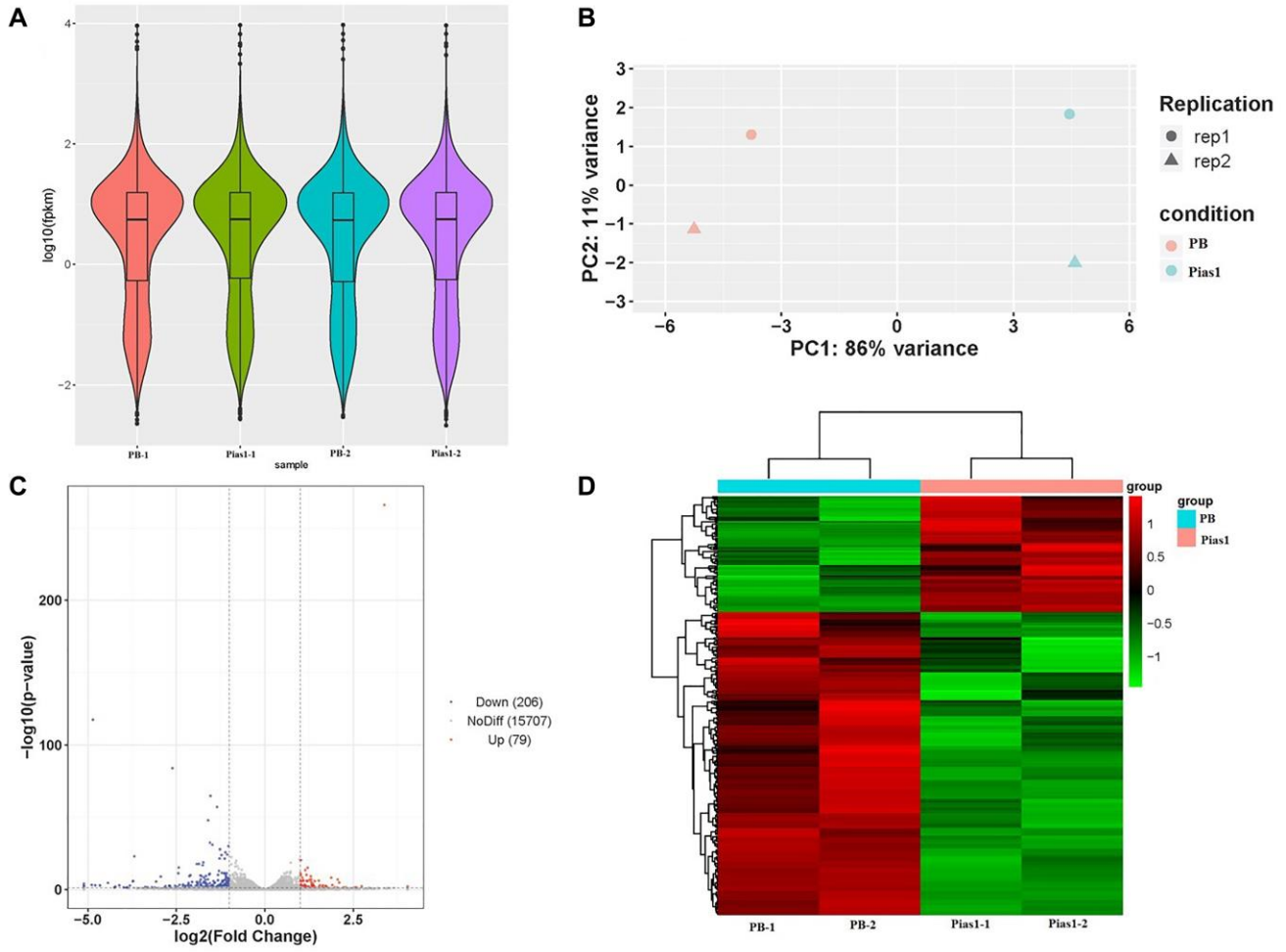

Figure 3. RNA-seq data quality and DEG analysis. (**A**) The violin plot based on values of log10 (fpkm) showed the whole gene expression distribution of four groups, including PB-1, PB-2, Pias1-1 and Pias1-2. (**B**) The PCA plot showed how well the four sequencing samples (PB-1, PB-2, Pias1-1 and Pias1-2) are separated from each other. (**C**) The scatter plot showed the results of DEGs based on the values of log2 (fold change) and -log10 (*p* value), including 206 downregulated genes, 79 upregulated genes and 15707 genes with no significant differences. (**D**) The heatmap showed the expression patterns of 285 significantly dysregulated genes with FC greater than 2 and *p* value less than 0.05, including 79 upregulated genes and 206 downregulated genes, by comparison of the *Pias1*/+ group to the WT group.

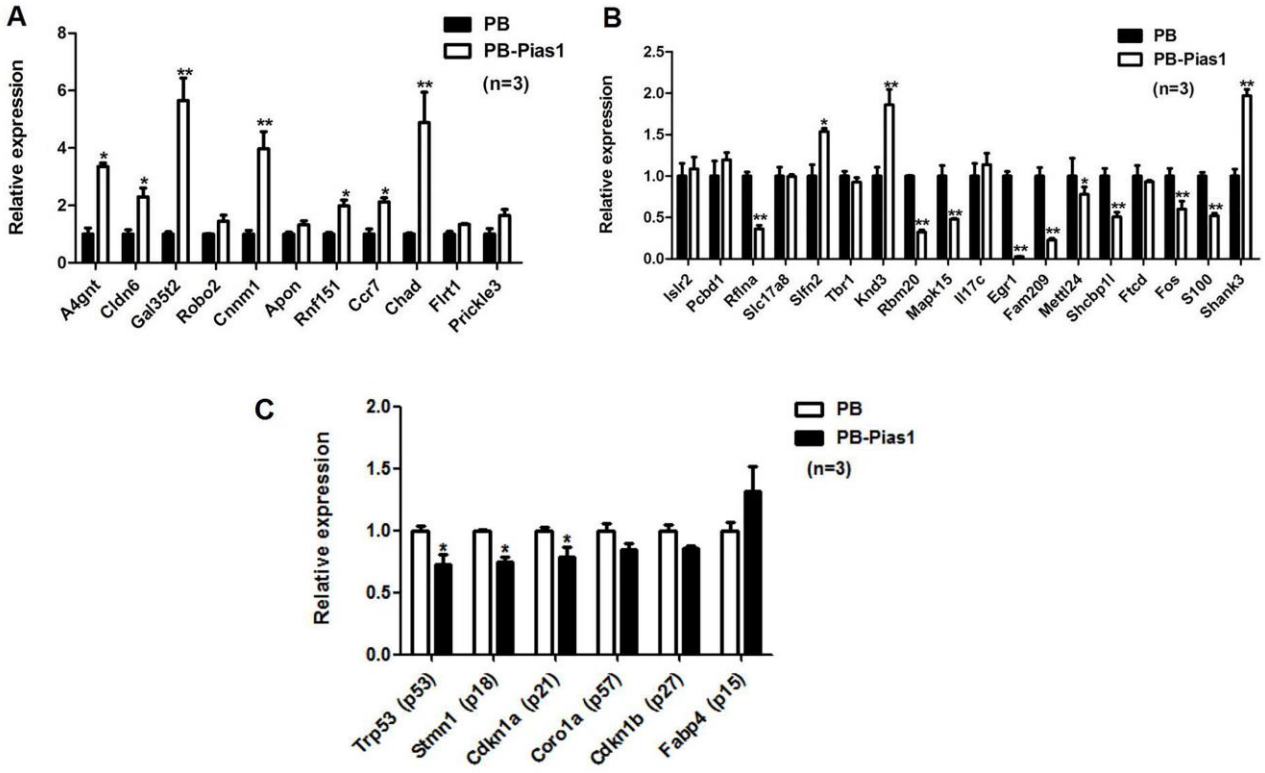

Figure 4. Validation of target gene expression by qRT-PCR. (**A**) For upregulated DEGs, 11 genes were selected for the following qRT-PCR validation. (**B**) For downregulated DEGs, 18 genes were selected for the following qRT-PCR validation. (**C**) The PIAS1 regulation of cell cycle inhibitors, including p15, p18, p21, p27, p53 and p57, was tested by qRT-PCR. Among them, three inhibitors, p53, p18 and p21, were significantly downregulated under the overexpression of *Pias1*. $^{*}p < 0.05$, $^{**}p < 0.01$.

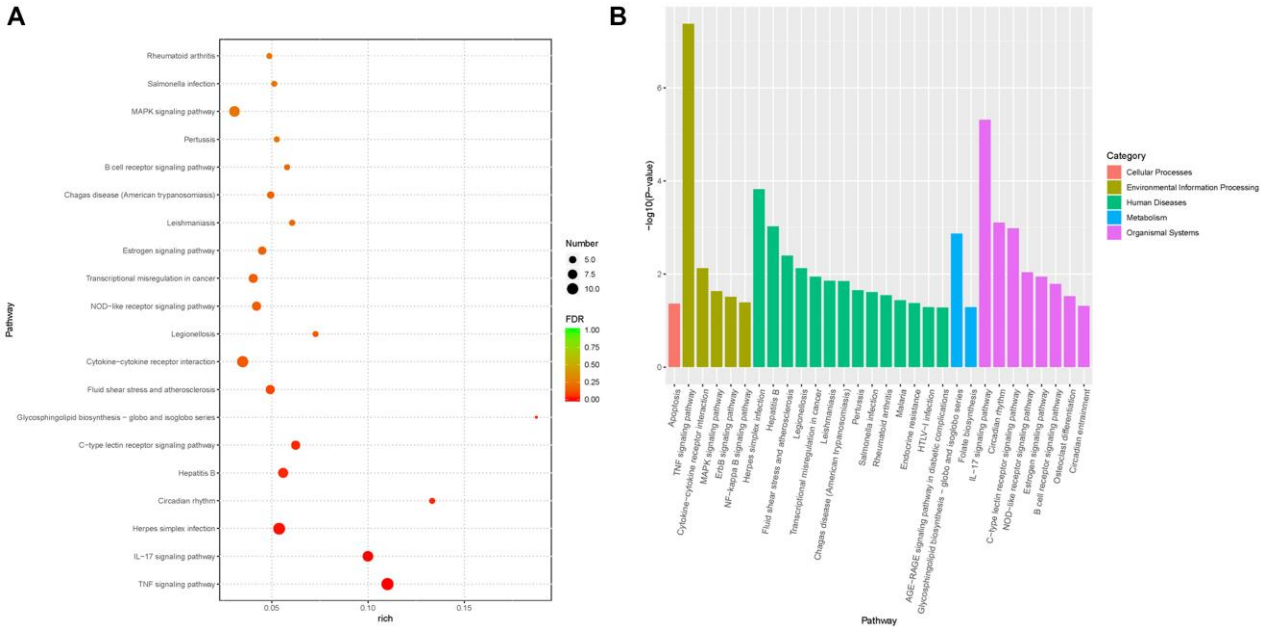

Figure 5. Functional enrichment analysis of DEGs from RNA-seq data. (**A**) The plot showed the top 20 significantly enriched KEGG pathways of differentially expressed genes (DEGs) associated with *Pias1* regulation with an FDR value cutoff of 0.05. Round size represents the gene count of each pathway, and the color represents the significance level. (**B**) The bar chart showed the categories of significantly enriched pathways. There were five pathway categories including cellular processes (red), environmental information processing (green yellow), human diseases (green), metabolism (blue) and organismal systems (purple).

Table 2. Significantly enriched signaling pathways and dysregulated genes.

Pathway	FDR	Up_genes	Down_genes
TNF signaling pathway	4.25E–08	–	Fas, Creb5, Gm5431, Jun, Lif, Junb, Fos, Ptgs2, Cxcl1, Vcam1, Ccl2, Mmp9
IL-17 signaling pathway	4.88E–06	–	Mapk15, Fosl1, Il17c, Jun, Fos, Ptgs2, Cxcl1, Ccl2, Mmp9
MAPK signaling pathway	2.34E–02	Hspa1l	Fas, Jun, Dusp6, Dusp1, Nr4a1, Fos, Areg, Epha2
ErbB signaling pathway	3.09E–02	–	Jun, Nrg1, Areg, Hbegf
NF-kappa B signaling pathway	4.11E–02	Blnk	Ptgs2, Vcam1, Plau

(FDR=4.88E–06) in terms of organismal systems. In addition, other signaling pathways, including the MAPK signaling pathway (FDR=2.34E–02), the ErbB signaling pathway (FDR=3.09E–02) and the NF-kappa B signaling pathway (FDR=4.11E–02), were also identified as being significantly associated with *Pias1* regulation. The genes involved in each pathway are shown in Table 2.

Moreover, we have also identified several significantly enriched GO terms, which are listed in Supplementary Table 2. The top three biological process (BP) terms were response to lipid (GO:0033993), regulation of cell population proliferation (GO:0042127) and response to steroid hormone (GO:0048545). The molecular function (MF) terms were primarily related to DNA-binding transcription factor activity, steroid hormone receptor activity and transcription regulator activity. The cellular component (CC) term of the transcription factor AP-1 complex (GO:0035976) was significantly enriched with the target genes *Fos*, *Jun* and *Junb*.

Transcriptional regulation of genes related to the overexpression of *Pias1* in HT-22 cells

To further investigate transcription factor (TF) gene regulation associated with the overexpression of *Pias1* in HT-22 cells, we analyzed the enrichment of transcription factors regulating target DEGs (Figure 6). As a result, the most strongly enriched TFs were zinc finger C2H2 (zf-C2H2), homeobox and basic/helix-loop-helix (bHLH) proteins. More than 100 genes were encoded or regulated by these three enriched TFs. Using the SMART tool to analyze the proteins with the same domain composition that have at least one copy of each of the domains of the query, we identified the significant domains associated with target proteins. The results are shown in Table 3. The first domain was the basic region leucine zipper (FDR=1.00E–03) with the target factors Creb5, Dbp, Fos, Fosl1, Jun, Junb and Tef. The second domain was the leucine-rich repeat C-terminal domain (FDR=1.60E–03) with the target factors Chad, Flrt1, Gp1ba, Igsf10, Islr, Islr2, Lrrc19 and Tlr2. The third domain was tumor necrosis factor receptor or nerve growth factor receptor repeats (FDR=1.60E–03) with the target factors Eda2r, Fas, Tnfrsf14, Tnfrsf18 and Tnfrsf26.

According to the transcription factor binding site (TFBS) prediction, we identified 659 TFBS genes for STAT1. Based on the comparison of 206 downregulated DEGs and 79 upregulated DEGs by overexpressing PIAS1, there were 18 overlapping genes, including one upregulated gene, PIAS1, and 17 downregulated genes. The details of this analysis are shown in Figure 7A. The expression levels of five TFBS genes were identified to be significantly decreased in response to PIAS1 overexpression. They were early growth response 1 (*Egr1*), early growth response 2 (*Egr2*), early growth response 3 (*Egr3*), FBJ osteosarcoma oncogene (*Fos*) and fos-like antigen 1 (*Fosl1*) (Figure 7B). The immediate early gene *Egr1* (also known as *Zif268*) is required for the maintenance of late long-term potentiation (LTP) in the dentate gyrus of the hippocampus and for the expression of long-term memory [27]. *Egr1* was found to have functions in the activation of β-secretase 1 (BACE–1) and acceleration of amyloid-β peptide (Aβ) synthesis in the AD patients brains, which was suggested as one of potential therapeutic candidates for the treatment of AD [28, 29]. During long-term memory formation in the context of the preexposure facilitation effect (CPFE), both *Egr1* and *Fos* (also known as *c-Fos*) have been identified as differentially expressed immediate early genes [30]. *Egr2* has been reported to be a downstream target of NF-κB in neurons, which plays an important role in peripheral nervous system myelination [31, 32]. Not completely consistent with *Egr1*, the role of *Egr3* in learning and memory processing has also been revealed [33]. *Fosl1* is an immediate-early gene that has been identified as a target of the NMDAR-Wnt/β-catenin signaling pathway in the hippocampus [34, 35].

Table 3. SMART protein domains.

Domains	FDR	Target proteins
basic region leucin zipper	1.00E–03	Creb5, Dbp, Fos, Fosl1, Jun, Junb, Tef
Leucine rich repeat C-terminal domain	1.60E–03	Chad, Flrt1, Gp1ba, Igsf10, Islr, Islr2, Lrrc19, Tlr2
Tumor necrosis factor receptor / nerve growth factor receptor repeats	1.60E–03	Eda2r, Fas, Tnfrsf14, Tnfrsf18, Tnfrsf26
PAS domain	1.70E–03	Arnt2, Pde8a, Per1, Per2, Per3
Epidermal growth factor-like domain	1.70E–03	Adam33, Areg, Celsr1, Cspg5, Gpr179, Hbegf, Nrg1, Plau, Ptgs2, Susd1, Tnfrsf18
c4 zinc finger in nuclear hormone receptors	1.70E–03	Nr1d1, Nr1d2, Nr1h4, Nr3c2, Nr4a1, Nr6a1
Ligand binding domain of hormone receptors	1.70E–03	Nr1d1, Nr1d2, Nr1h4, Nr3c2, Nr4a1, Nr6a1
Leucine rich repeat N-terminal domain	5.20E–03	Amigo3, Chad, Flrt1, Gp1ba, Igsf10, Islr, Islr2
Leucine-rich repeats, typical (most populated) subfamily	5.40E–03	Amigo3, Chad, Flrt1, Gp1ba, Igsf10, Islr, Islr2, Tlr2
Motif C-terminal to PAS motifs (likely to contribute to PAS structural domain)	6.70E–03	Arnt2, Per1, Per2, Per3
Leucine-rich repeats, outliers	9.40E–03	Amigo3, Chad, Flrt1, Gp1ba, Igsf10, Islr, Islr2, Lrrc19, Lrrc73, Tlr2

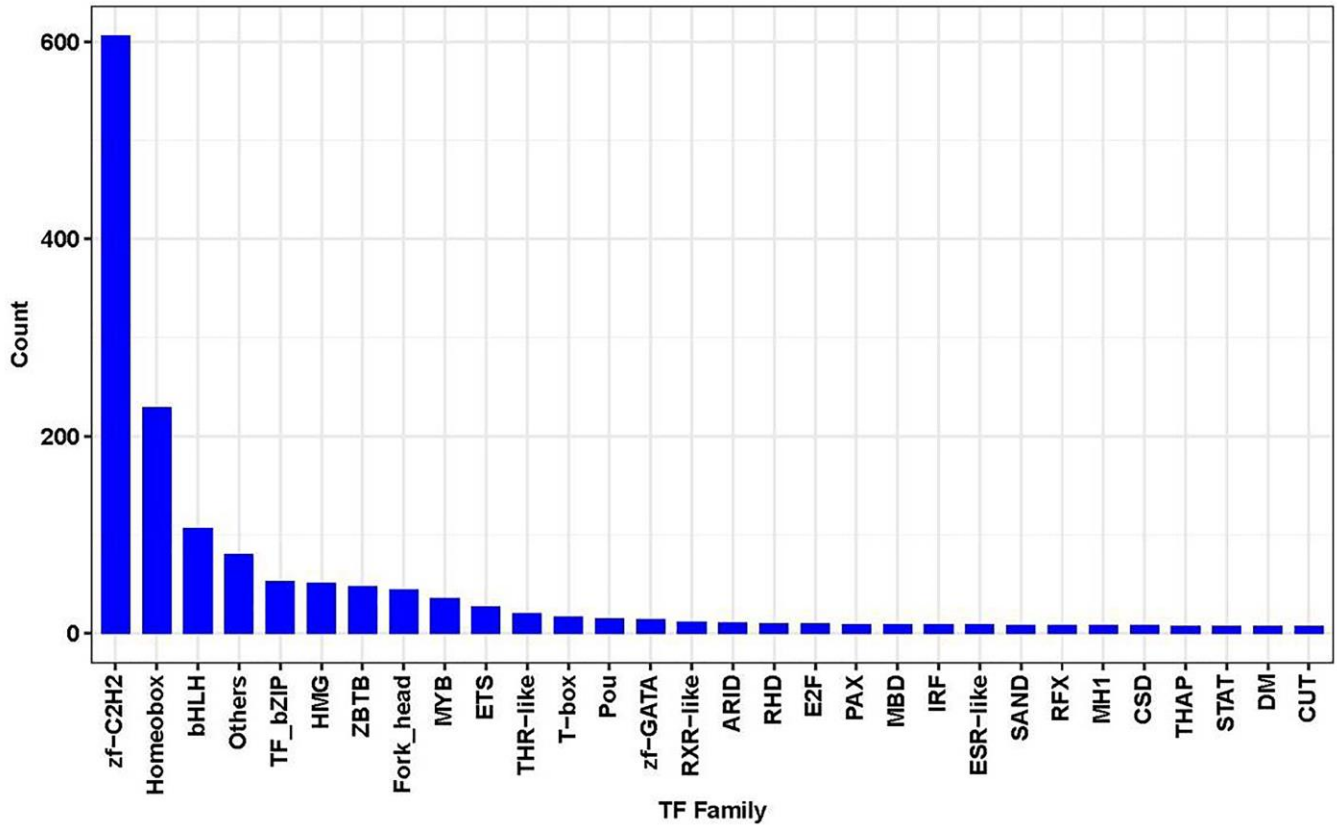

Figure 6. Results of TF family gene counts. This chart showed the top significantly enriched TF family (X-axis) with the gene count (Y-axis).

Protein-protein interaction network construction in response to *Pias1* overexpression

In addition, a regulatory network of protein-protein interaction in response to the overexpression of *Pias1* in HT-22 cells was constructed (Figure 8). A total of 43 encoding DEGs were involved in this network. Our findings provide evidence for a direct link between PIAS1 and NR3C2. The *Nr3c2* gene encodes the mineralocorticoid receptor (MLR), which plays important roles in blood pressure control and fluid balance by regulating the amount of sodium in the body

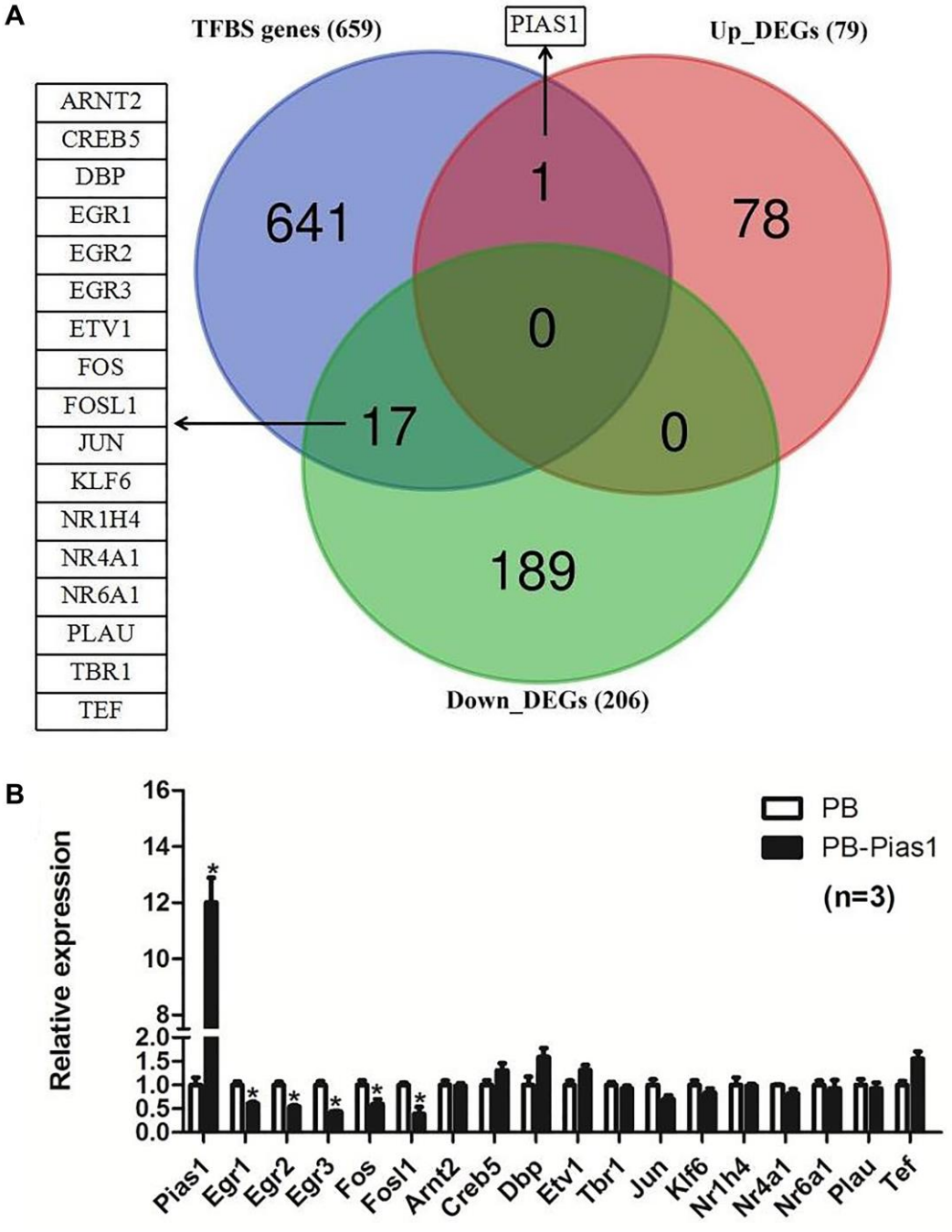

Figure 7. Overlapping targets between transcription factor binding site genes of STAT1 and DEGs of PIAS1. (**A**) Transcription factor binding site (TFBS) prediction for STAT1 was performed and compared with the differentially expressed genes of RNA-seq results in PIAS1 overexpression cell lines. As a result, 18 overlapping genes, including one upregulated gene PIAS1 and 17 downregulated genes, were identified as the target genes. (**B**) The regulation of the overlapping genes was evaluated by qPCR. As a result, five genes, including early growth response 1 (*Egr1*), early growth response 2 (*Egr2*), early growth response 3 (*Egr3*), FBJ osteosarcoma oncogene (*Fos*) and fos-like antigen 1 (*Fosl1*), were significantly downregulated in PIAS1-overexpressing cells.

[36]. MLR has been determined to be involved in guiding spatial and stimulus-response learning in mice [37]. In humans, PIAS1 was reported to repress MLR ligand-dependent transcriptional activity through the interaction with the N-terminal domain of MLR through SUMO modifications [38]. The upstream gene was *Agt* (angiotensinogen), acting as a serpin peptidase inhibitor, which is important for the renin-angiotensin system (RAS) [39]. The pathophysiology and morphology of RAS in mice brains may be influenced by the knockout of neurolysin [40]. The cell death and production of reactive oxygen species (ROS) in the mouse brain may be restrained by the ACE2/Ang-(1-7)/Mas pathway activation with angiotensin II overproduction [41].

In conclusion, the regulatory mechanism of the *Pias1* gene in MHNC was elucidated in our functional genomics study by using RNA-seq. The overexpression of *Pias1* was determined to repress several AD markers expressions and to accelerate the proliferation of HT-22

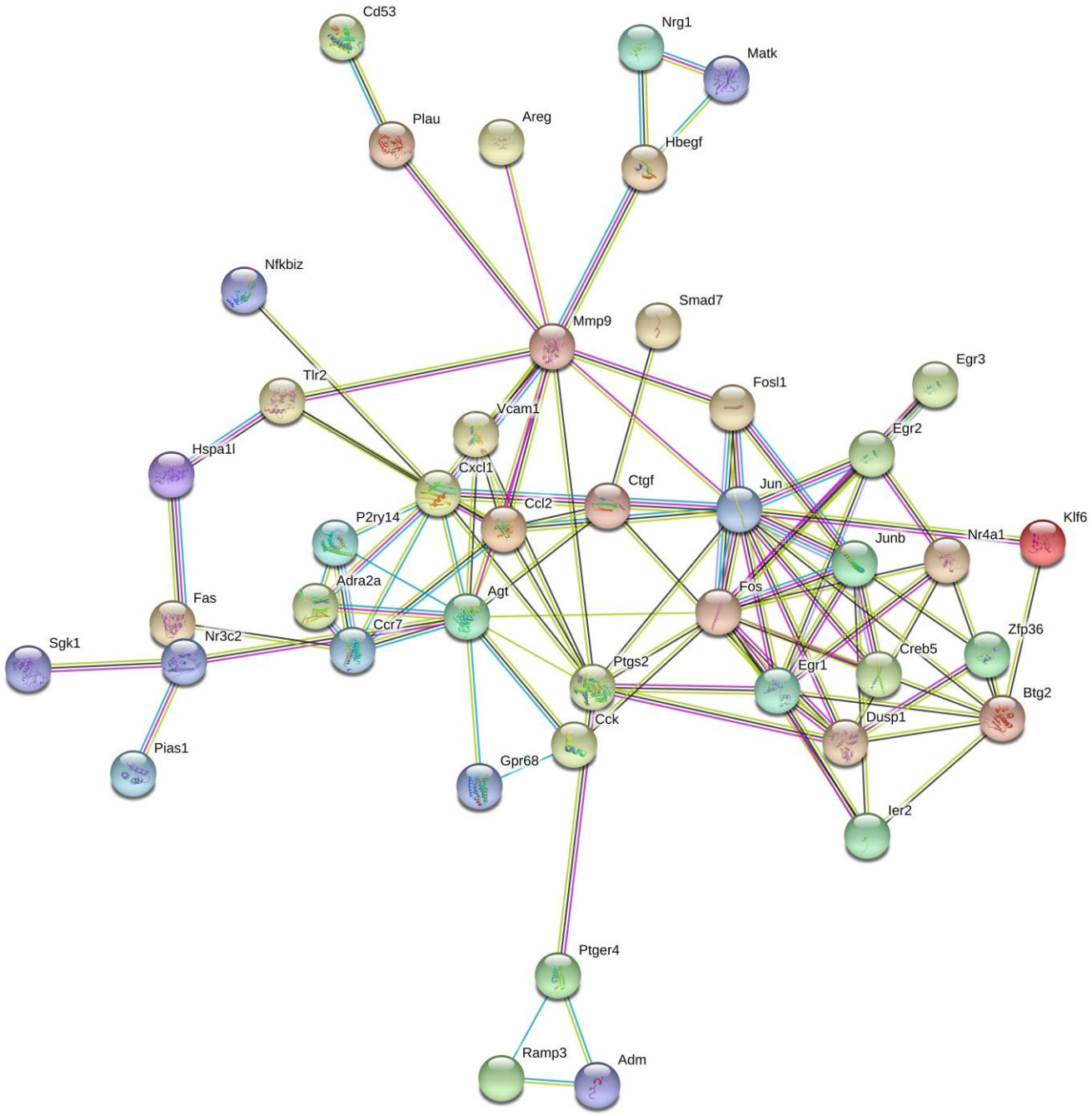

Figure 8. Protein-protein interaction network associated with *Pias1* overexpression in HT-22 cells. The protein-protein interaction network was reconstructed by the top 43 DEG proteins in response to *Pias1* overexpression in HT-22 cells, such as Nr3c2.

cells. There were 79 upregulated genes and 206 downregulated genes, significantly identified in *Pias1*/+ cells compared to WT cells through the DEGs analysis. Functional enrichment analysis and TFBS prediction results showed that basic region leucine zipper domain-containing transcription factors including zinc finger C2H2 (zf-C2H2), homeobox and basic/helix-loop-helix (bHLH) in several signaling pathways were significantly involved in PIAS1 regulation in MHNCs. We also reconstructed the regulatory network under PIAS1 overexpression with 43 coding proteins, especially NR3C2, which directly interacted with PIAS1.

MATERIALS AND METHODS

PIAS1 gene expression in AD brains

The expression profiles of the PIAS1 gene in four brain regions in humans, including the frontal pole (including 111 AD patients and 76 controls), the inferior frontal gyrus (including 90 AD patients and 64 controls), the parahippocampal gyrus (including 90 AD patients and 68 controls), and the superior temporal gyrus (including 102 AD patients and 65 controls), were sourced from the database Synapse [42, 43]. Next, we tested the differences of PIAS1 gene expressions between AD patients and controls in each brain region. The Wilcoxon signed-rank test was used to analyze the significance.

Cell culture and gene overexpression

For the *in vitro* experiments, we employed one subclone of the mouse hippocampal HT-22 cells, which were purchased from CHI SCIENTIFIC, Shanghai, China. HT-22 cells were maintained in Dulbecco's modified Eagle's medium (DMEM, Sigma) supplemented with 10% fetal bovine serum (FBS, Gibco BRL, USA), MEM nonessential amino acids (Invitrogen), L-glutamine (Invitrogen), penicillin and streptomycin (Sigma), 0.01 mM β-mercaptoethanol (Invitrogen) and 1000 units/ml LIF (Millipore). Based on the database of GenBank, the reference sequence of Mus musculus *Pias1* (NM_019663.3) was obtained. The gene insertion and transfection were performed by using a vector of PiggyBac (PB) in addition with 2 μg transposase using LTX (Invitrogen). The cell selection was continued for seven days by using 2 μg/ml puromycin.

Cell proliferation assay

The experiments of Cell Counting Kit-8 (CCK-8) was performed to assess the proliferation of HT-22 cells based on the manufacturer's instructions. Cell culture *in vitro* was conducted in 96-well plates with 100 μl DMEM supplemented with 10% FBS. The cell density was set as 1000 cells per well. Three time points (24 hours, 36 hours, 48 hours and 60 hours after transfection) were selected for observation. The cell viability was determined based on the absorbance at 450 nm by using a microplate reader (Spectramax i3 platform, MD, Austria). All of the experiments were performed in three biological repeats.

Experimental grouping and RNA sequencing

Two groups of the control (named as PB-1 and PB-2) and two groups of the treatment (named as Pias1-1 and Pias1-2) were selected in the following experiments. Total RNA in each group was extracted by using TRIzol, the cDNA was synthesized from 1 μg of total RNA by using reverse transcriptase (Takara). The preparation of the cDNA library and RNA sequencing (RNA-Seq) utilizing the platform of Illumina HiSeq 3000 were performed by Personal Biotechnology (Shanghai, China). On average, 6 G reads per sequencing sample were obtained. Based on the general process of data analysis, our sequencing data was analyzed [44]. The data preprocessing and quality control were performed by using the softwares of Trim Galore and FastQC. Using STAR software, the preprocessing sequence alignment was performed based on the mouse reference genome sequences (Mus_musculus.GRCm.38.83) downloaded from the database of Ensembl. Cufflinks software was used to perform transcript assembly and Cuffcompare software was employed to parsimoniously merge the assembled transfrags. The DEGs analysis was performed by using the package of DESeq2 under R software [45]. The significance threshold is defined as $p < 0.05$ and $|\log2 FC| > 1$. The functional enrichment analysis was performed by GSEA to identify the significantly enriched KEGG pathways and GO terms, including biological process (BP), molecular function (MF) and cellular component (CC) [46]. The significance cutoff was chosen as FDR < 0.05. The web server SMART version 8.0 was employed to identify and annotate the specific domains of the target proteins [47]. The protein-protein interaction network was reconstructed by using the tool of STRING version 11.0 [48].

Transcription factor binding site (TFBS) prediction

The promoter sequences, including 3000 bases upstream and 50 bases downstream of STAT1, were downloaded from the UCSC Genome Browser [49]. The animal transcription factor database AnimalTFDB 3.0 was employed to predict the transcription factor binding sites (TFBS), as well as predicting the associated TFs and their family information [50]. Next, we compared the above TFBS genes with the DEGs identified by RNA-seq data analysis to explore the overlapping genes that were affected by overexpressing PIAS1.

Table 4. The primers of target genes for qPCR.

Genes	Forward	Reverse
Arnt2	GTTCCAGGACATGCTACCCAT	ATGCCCAGGTCAGCAAAGTC
Creb5	GTTGAAAAACGAGGTGGCCC	AGGGCTGCTCTCTGGACTTA
Dbp	CAAGCCCAAAGAACCGGC	AGCGGCGCAAAAAGACTCG
Etv1	TGCAAAAGGTGGCTGGAGAA	ATGTCCGTCTTCAGCAGTGG
Fosl1	GTTCCACCTTGTGCCAAGCA	GGACTGTACTGGGGATAGGC
Jun	CAAAACTCCGAGCTGGCATC	TGCGTTAGCATGAGTTGGCA
Klf6	CTTGAAAGCACATCAGCGCA	TCTTGCAAAACGCCACTCAC
Nr1h4	GAGATGGGGATGTTGGCTGA	CAGCGTGCTGCTTCACATTT
Nr4a1	CCGGTGACGTGCAACAATTT	TCCCCTCACCGGGTTTAGAT
Nr6a1	CATCCAGTAGGTCTGTGGAACT	TGGTGAGTGGCCAGAATAGC
Plau	TAAAATGCTGTGTGCTGCGG	CCGGGCTTGTTTTTCTCTGC
Tef	GAAGCTGATGGAGAACCCCC	GAAGGACTCGCCATCGTAGG
Egr1	AGTGATGAACGCAAGAGGCA	TAGCCACTGGGGATGGGTAA
Egr2	ATCGAAAGCCGTTTCCCTGT	GCGGATTATAAGGGGTGGCA
Egr3	GATGGCTACAGAGAATGTGATGG	TTGGAAGGAGAGTCGAAAGCG
Fos	TCGCATCAAGGCCATCATTG	TACGCACGTAGACCAGGATC
Tbr1	GGATTTACGAGCAGGCCAAG	CTATGTCCTTGGCGCAGTTC
Fabp4 (p15)	GGGATGGAAAGTCGACCACA	CTTGTGGAAGTCACGCCTTT
Stmn1 (p18)	TTCTCAGCCCTCGGTCAAAA	CTGCTTCAAGACTTCCGCCT
Cdkn1a (p21)	AGTACTTCCTCTGCCCTGCT	GAATCTTCAGGCCGCTCAGA
Cdkn1b (p27)	TCGCAAAACAAAAGGGCCAA	TTACGTCTGGCGTCGAAGG
Trp53 (p53)	GCCCATGCTACAGAGGAGTC	TCAGGCCCCACTTTCTTGAC
Coro1a (p57)	CTACCTCTGTGGCAAGGGTG	CCATACCACGCTGAGACTCC
Rpl19	GACGGAAGGGCAGGCATATG	TGTGGATGTGCTCCATGAGG

Quantitative polymerase chain reaction (qPCR)

The primers of target genes were designed by using Primer 5.0. The experiments of qPCR were performed using TransStart Tip Green qPCR SuperMix (Takara). The normalization of each transcript expression was performed by using the 2−ΔΔCt method with the reference gene ribosomal protein L19 (Rpl19). The sequences of above primers are shown in Table 4.

Immunofluorescence staining

Cells were fixed in 4% paraformaldehyde (PFA fixative) for 20 mins at room temperature, washed in PBS three times, permeabilized for 1 h at 37°C in PBS-T solution(PBS containing 5% BSA and 0.2% Triton X-100), rinsed with PBS three times and incubated overnight at 4°C with the following primary antibodies: APP (25524-1-AP, Proteintech, 1:500). In dark, Alexa Fluor 594 (Invitrogen, 1:1000)-conjugated secondary antibody was used at 1:1000 and nuclei were stained with Hoechst (Invitrogen, 1:5000).

Western blotting

Cells were lysed in pre-cooled RIPA cell buffer (P0013B, Beyotime Biotechnology, China) containing protease inhibitor (DI111-02, TRANSGEN BIOTECH, China, 1: 100). Proteins were separated by a 10% SDS-PAGE gel at RT and electrotransferred to a PVDF membrane in ice-bath. Probing was performed with specific primary antibodies and HRP-conjugated secondary antibodies. The primary antibodies used were PIAS1 (ab2474-1, Abcam, 1:1000) and GAPDH (SC-8035, Santa Cruz, 1:2000). Proteins were detected in super ECL detection reagent (No.180-501, Tanon, China).

AUTHOR CONTRIBUTIONS

DL and KH conceived and designed the study. JZ, JL, LGL, YC, SY, QB and RP performed the experiments and data analysis. DL and KH wrote the paper. DL, JZ, JL and KH reviewed and edited the manuscript. All authors read and approved the manuscript.

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CONFLICTS OF INTEREST

The authors declare that they have no competing interests.

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SUPPLEMENTARY MATERIALS

Supplementary Tables

Please browse Full Text version to see the data of Supplementary Tables 1 and 2.

Supplementary Table 1. Full information of DEGs in response to the overexpression of *Pias1* in the hippocampal neuronal HT22 cells.

Supplementary Table 2. Full information of significantly enriched GO terms in response to the overexpression of *Pias1* in HT22 cells.

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Original Article

β-catenin regulates myocardial ischemia/reperfusion injury following heterotopic heart transplantation in mice by modulating PTEN pathways

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Abstract: Ischemia reperfusion (I/R) injury, an inevitable event accompanying heart transplantation, is the primary factor leading to organ failure and graft rejection. In order to prevent I/R injury, we established murine heart transplantation model with I/R and cell culture system to determine whether β-catenin is a mediate factor in preventing I/R injury in heart transplantation. After successfully established heterotopic heart transplantation mice model, the I/R injury was induced, and two dynamic temporal were studied during different I/R phases. With the increase of ischemia and reperfusion time, heart damage was more severe. In the initial study, we observed that β-catenin was significantly decreased, while ROCK1 and PTEN increased during the perfusion phase from day 0 to day 1, and remain the same level until 3 days later. The similar pattern that β-catenin was down-regulated while ROCK1 and PTEN were up-regulated was also observed in the dynamic temporal ischemia study. To further investigate the role of β-catenin signaling in I/R injury *in vitro*, β-catenin over-expressing plasmid was transfected into HL-1 cells, a cardiac cell line. We noted that β-catenin over-expressing cardiomyocytes showed decreased ROCK1/PTEN expression both at mRNA and protein levels. In addition, cobalt dichloride ($CoCl_2$) -induced oxidative stress model was further established to mimic cardiac I/R injury. We observed that $CoCl_2$-induced activation of ROCK1/PTEN signaling pathway were attenuated by transient transfection of a β-catenin over-expressing plasmid. Taken together, our results suggest that cardiac transplant induced IR injury is closely associated with the down-regulation of β-catenin and up-regulation of ROCK1 and PTEN expression.

Keywords: Heart transplantation, Ischemia reperfusion (I/R) injury, β-catenin, ROCK/PTEN

Introduction

Cardiac transplantation is the preferred treatment for patients with end-stage heart disease, i.e, advanced heart failure [1-3]. It offers a significant improvement of life quality and the most beneficial survival of heart failure patients. However, in cardiac transplantation, the big challenge we have to encounter with is Ischemia/reperfusion (IR) injury [4, 5]. As after transplantation, cardiac grafts inevitably suffer from ischemia when separated from the donor. After revascularization, blood flow in post-ischemic hearts could activate a sequence of events that aggravates cardiac injury, i,e., ischemia-reperfusion injury (IRI), and contributes to a high rate of morbidity. Hence, it is critical important to minimize IR injury.

Recently, a robust of studies mainly on molecular level, regarding the important genes, gene networks, key signaling cascades, participating in I/R injury have been reported, and modulation of these pathway by endogenous factors or

pharmacological agents has been postulated as a relevant way to prevent I/R injury [6-9]. However, the molecular mechanism of heart transplantation induced I/R injury, involving activation of multiple and interlocked signaling pathways remains exclusive. Given the limited efficacy of these current therapeutic strategies, it is of critical importance to gather information about the status of key cardiac signaling pathways at early stages after I/R, to identify potential new targets for interventional therapies.

β-catenin pathway is an important regulator of cell development, regeneration, and carcinogenesis [10]. It has been shown to possess multiple regulatory functions during the inflammatory response [11, 12]. There are a lot of literature suggests that β-catenin and its related pathways PTEN/ROCK participates in the regulatory mechanism of IR injury events, such as: liver, lung, cerebral, etc. Kamo et al. (2013) found that β-catenin directly inhibits NF-κB activation, resulting in suppression of pro-inflammatory gene programs that otherwise drive the hepatocellular damage in IR-stressed liver forward [13]. Ke et al. (2013) demonstrated that β-catenin inhibits PTEN and promotes PI3K/Akt pathway, which acts as a novel regulator of innate and adaptive immune responses in the mechanism of liver IRI [14]. Zhou et al. (2019) found that PTEN/β-catenin signaling regulates the immune response and thus alleviates the inflammation of lung injury [15]. Li et al. (2019) found that Wnt/β-catenin pathway may play an important role in the protection against cerebral ischemia/reperfusion injury in rats [16]. However, currently, few reports have been published on the molecular mechanism of β-catenin events in cardiac I/R injury. The cross talk between β-catenin and other signal pathways, pivotal in the mechanism of cardiac IR injury, remains to be elucidated.

The heterotopic heart transplantation in rodents, like mice, rat, is a widely accepted animal model as which is mainly used to investigate transplantation biology [17-23]. Heterotopic abdominal mice heart transplantation has been extensively used to investigate ischemic-reperfusion injury, immunological consequences during heart transplantations [22].

Due to our initial experience on the model establishment of heterotopic heart transplantation, herein, we carried out studies to investigate the molecular mechanisms of β-catenin regulation involved in cardiac I/R injury. In the present work, firstly, using an *in vivo* mice model, we found that the β-catenin participates in myocardial ischemia/reperfusion injury following heterotopic heart transplantation in mice by modulating PTEN regulation.

Materials and methods

Experimental animals

The male C57BL/6 Mice, 8-10 weeks of age, with mean weight of 24-27 g, were maintained in a specific pathogen-free facility of Xiamen University. The mice were kept in individual stainless steel cages in a room with controlled temperature (23 ± 3°C) and humidity (55 ± 15%) on a 12-hour light/dark cycle, and provided with standard rodent food and free access to water prior to the experiments. The animals were fasted overnight before the experiments, but were given free access to water. All procedures involving mice were conducted in accordance with the guidelines of the Animal Care and Committees of Xiamen University and complied with the National Research Council's Guide for the Humane Care and Use of Laboratory Animals.

Cell line and cell culture

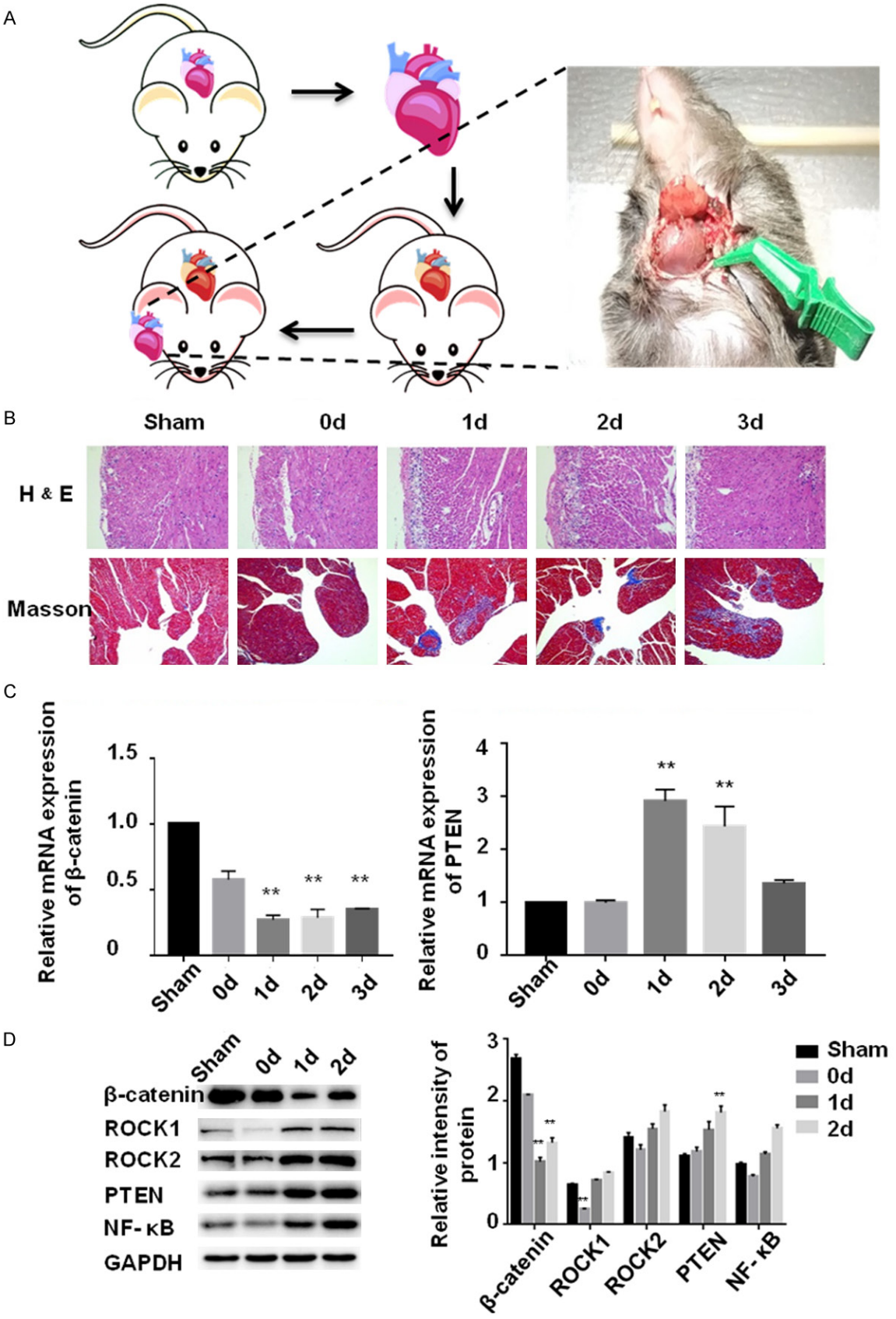
The murine cardiac HL-1 cell lines were originally purchased from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China). Cells were cultured and maintained in 2 mL DMEM (50:50), supplemented with 10% fetal bovine serum (FBS), penicillin/streptomycin mixture (100 U/ml) (Gibco, Grand Island, NY). Cell lines were incubated at 37°C with 100% humidity in 5% CO_2.

Experimental design

The experimental design is schematically illustrated in **Figure 1A**.

Heterotopic heart transplantation procedure

Heterotopic heart transplantation procedure was demonstrated as previously described [24]. The rats were placed on the white corkboard and their limbs were fixed with tape. The rats were anaesthetized using 1% pentobarbital. Finding the right external jugular vein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ROCK2
PTEN
NF-κB
GAPDH
Relative intensity of protein
Sham
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1d
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Figure 1. Ischemia/reperfusion injury following heterotopic heart transplantation in mice. A. Schematic illustration of mice heterotopic heart transplantation in the study. Donor hearts were harvested from C57BL/6 mice (n=3, each) and preserved at 4°C for 24 h, followed by a heterotopic syngeneic transplantation. B. Histology analysis. The upper panel is HE staining. After transplantation, heart grafts were harvested for H&E staining, at different perfusion time points. Endocardium infiltrated with neutrophils. (day 0) Cardiomyocyte damage (day 1). Epicardium thickening (day 2). Trichrome staining for fibrosis (day 3). The lower panel is Mason staining. Representative images from 3 independent grafts per group. Images were taken at ×200 magnification. C. Real-time PCR was used to detect the mRNA expression of β-catenin (Left) and PTEN (Right) in the mice underwent different period of perfusion varying from day 0 to days 3: sham-operated mice, I/R mice with perfusion at day 0 (samples were obtained immediately when transplantation was done), I/R mice with perfusion at day 1, I/R mice with perfusion at day 2, and I/R mice with perfusion at day 3, (n=3/group). (Left) Compared with shame group, mRNA expression of β-catenin was decreased during the perfusion phase from day 0 to day 1, and remain the same level until 3 days after perfusion. (Right) Compared with shame group, mRNA expression of PTEN was significantly increased during the perfusion phase from day 0 to day 3 ($P<0.01$ both on day 1 and day 2). The PTEN level was significantly increased at 1 day of reperfusion ($P<0.01$), but then decreased at the IR 2 days compared with IR 1 day of reperfusion time point, however, when compared with controls, it was still significantly up-regulated ($P<0.01$). Columns, mean of three independent experiments; bars, SD (n=3). *, $P<0.05$, **, $P<0.01$. D. Western blot analysis was used to detect the expression of β-catenin, ROCK1, ROCK2, PTEN and NF-KB (Left) in the mice underwent different period of perfusion at 0, 1 and 2 days time point. (Right Pannel) Band intensity was expressed compared with total GAPDH. β-catenin protein accumulation was decreased at IR 1 day and 2 days reperfusion time points. ROCK1, ROCK2, and NF-KB protein levels were decreased at 0 day of reperfusion, but increased at 1 day and 2 days reperfusion compared with sham control.

(REJV), the tributaries of the REJV were cut off. The right common carotid artery (RCA) was exposed and external carotid arteries. The RCA was occluded by vessel clip and tied, then incised between the clip and the tie. The RCA was then overturn and ligature of 9-0 silk. The REJV was prepared in the same way. The donor was intravenously injected with 1 mL of 4°C heparinized saline (100 U/mL). The thorax was opened and the anterior chest wall was pulled in a cranial direction. After removal of thymus and pericardium, we performed a second reverse perfusion through the thoracic aorta to completely drain the remaining blood from the donor heart. Then we cut around the ascending aorta (AO) and the pulmonary artery (PA), just above the PA bifurcation. All blood vessels, except AO and PA, are connected to the posterior surface of the heart. The donor heart was placed in the right side of the neck of the recipient immediately. The aorta of the donor heart is connected to the common carotid artery of the recipient heart; Connect the pulmonary artery of the donor heart to the external jugular vein of the recipient heart. The heart rapidly reperfusion and gradually developed a sinus rhythm.

Myocardial I/R model

Myocardial I/R model mice were subjected to myocardial I/R as previously described [15]. Briefly, to induce ischemia, mice were first intraperitoneally anesthetized with sodium 1% pentobarbital (50 mg/kg, purchased from Sigma, St. Louis, USA). Surgical sutures were cut to expose the heart and right common carotid artery (RCA). The right common carotid artery (RCA) was sutured with 7-0 silk for 10 min before release for reperfusion. At the 0, 1, 2, 3 day of reperfusion, the hearts were obtained for further investigations.

For ischemia study, the same procedures were administered to mice exposed to ischemia for 5 min, 10 min, and 15 min, followed by 1 day of reperfusion. At the end of the procedures, samples were immediately harvested. Blood samples were collected by cardiac puncture and centrifuged at 3000 g for 10 minutes to abstract serum, which was stored at -80°C and to determine biochemical parameters. Tissues were snap frozen in liquid nitrogen for gene expression detection or fixed in 4% buffered formalin for histopathological evaluation.

In vitro ischemia reperfusion model

An in vitro $CoCl_2$-induced apoptosis model was established to mimic cardiac IRI. In the present study, the $CoCl_2$ induced cell model was established to mimic the cardiac IRI in vitro. HL-1 cells were plated in a 6 well plate (80,000 cells/well) and cultured in DMEM medium supplemented with 10% fetal bovine serum and 100 U/ml penicillin and streptomycin overnight. DMEM culture medium was replaced by deoxygenized PBS and HL-1 cells were treated with $CoCl_2$ (500-1000 μM) for 12 h, 24 h and 36 h, respectively. OD was detected in HL-1 cells treated with different concentrations (500,

600, 700, 800, 900, and 1000 μM). The optimal concentration was 600 μM at 24 h was selected for the further study, Prior to cell collection, some cells were treated with $CoCl_2$ (600 μM) for 24 h. The cells were then lysed and processed for protein analysis or Real-time PCR detection. Cells were seeded into 6-well plates at 80% confluency and transfected with the indicated plasmids using Lipofectamine 2000 (Invitrogen, United States) for 72 h. The cells were then lysed and processed for protein analysis or Real-time PCR detection.

Real-time PCR

Total RNA was extracted from cells and heart tissues using Trizol (TransGen BiotechChina) according to the manufacturer's instructions. One μg of total RNA was used to synthesize cDNA using reverse transcriptase kit (YEASEN, China). Gene expression of β-catenin, ROCK1, ROCK2, PTEN, NF-κB, and GAPDH were detected by qPCR using the SybGreen method. Primer sequences used for PCR were listed: β-catenin F: 5'-ACTGGAGCTCTCCACATCC-3' and R: 5'-GTGGCTCCCTCAGCTTCAAT-3', ROCK1 F: 5'-AACGCTCCGAGACACTGTAG-3' and R: 5'-ACTTTCCTGCAAGCTTTTATCCAC-3,' ROCK2 F: 5'-CATCAACGTGGAGAGTCTGCTG-3' and R: 5'-TCTGTGATGCCTTATGACGAAC-3', PTEN F: 5'-CTGCAGAAAGACTTGAAGGTGT-3' and R: 5'-CTAGCTGTGGTGGGTTATGGT-3', NF-κB F: 5'-GCTATGTGTGTGAAGGCCCA-3' and R: 5'-TTGCAAATTTTGACCTGTGGGT-3', GAPDH F: 5'-CAGGAGAGTGTTTCCTCGTCC-3' and R: 5'-TTCCCATTCTCGGCCTTGAC-3'. qPCR was conducted in a 10 μL PCR with 1× SybrGreen mixture (Bio-Rad), 2.5 μM primers, and 1 μL of cDNA, with the following thermal profiling: an initial activation step was carried with at 95°C for 5 mins, followed by 40 cycles of: 95°C for 10 s, 60°C for 30 s, and 72°C for 20 s. Data were quantified automatically using the Light Cycler 480 (Roche, Switzerland) and the relative expression was determined using the $2^{-\Delta\Delta Ct}$ method.

Western blotting analysis

Total cell protein was abstracted from cardiac tissue, or HL-1 cells after transfection. The concentration of the protein was measured by BCA protein assay kit (Solarbio, Beijing, China), following manufacturer's instruction. Samples were electrophoresed by using 10% SDS-PAGE. The protein was then transferred onto a PVDF (polyvinylidene fluoride) membrane (Bio-Rad, USA). After blocking in 5% dry milk for 4 hours, the membranes were incubated with incubated with primary antibodies: GAPDH, (Product# 60004-1-Ig, Proteintech), β-catenin (Product# 66379-1-Ig, Proteintech), PTEN (Product# 60300-1-Ig, Proteintech), ROCK1 (Product# 66782-1-Ig, Proteintech), ROCK2 (Product# 66633-1-Ig, Proteintech), P65 (Product# 10745-1-AP, Proteintech) The membranes were then incubated with HRP-conjugated secondary antibodies: HRP-conjugated Affinpure Goat Anti-Rabbit IgG (H+L) (SA00001-2, Proteintech), HRP-conjugated Affinpure Goat Anti-Mouse IgG (H+L) (SA00001-1, Proteintech). Specific protein bands were visualized using an enhanced chemiluminescence (ECL) kit and then processed using Image J for quantification. Autoradiograms were quantified by densitometry (Quantity One software; Bio-Rad). GAPDH was used as internal reference.

Histology

After reperfusion, tissues from the donor mice hearts were fixed in 4% paraformaldehyde (Solarbio, Beijing) for 14 h at 4°C, embedded in paraffin. Formalin-fixed paraffin-embedded heart tissues were then cut in 5 μm sections. Serial slides were used for hematoxylin and eosin (H&E) staining and Masson staining according to standard procedures.

The pathological features were carefully examined under 200× magnification. Images were analyzed by an independent investigator in blind manner. H&E stained slides were evaluated for the area of necrosis and inflammatory cells infiltration. The area of necrosis and inflammatory cells infiltration was determined as previously described and expressed as a percentage of the whole section.

Cardiac fibrosis was assessed using Masson Trichrome staining as previously described by us (Chu et al. 2011). 4 μm thick sections of cardiac tissue were sectioned and for each section and measurement of fibrosis, 10 fields were chosen at random and imaged under light microscopy using the Olympus BH2 microscope at ×40 magnification. Fibrosis was then quantified using ImagePro Plus software (Adept Electronic Solutions Pty Ltd, Moorabin, Australia).

Cell viability assay

Cell viability was analyzed by using Cell Counting Kit-8 (CCK-8) (TransGene, Beijing, China), followed by manufacturer's instructions. Briefly, the cells were seeded into the 96-well plate. 10 μL of Cell Counting Kit-8 solution with 100 μL MES were then added into each well. Followed by incubation at 37°C for 1 h, cells were subjected to detect the OD450 values by using the SpectraMicroplate Reader (BIOTEC). Results were showed as percentage of the vehicle control levels set at 100%. Each treatment was performed in triplicate.

Serum analysis

Blood samples were obtained from mice with different ischemia time. Serum was collected and stored at -80°C until analysis was performed. Serum levels of cytokines, including proinflammatory cytokines (IL-1β, IL-6, TNF-α) and anti-inflammatory cytokines (IL-10) and TGF-β, were measured using enzyme-linked immunosorbent assays (ELISAs) performed according to the manufacturer's instructions.

Statistical analysis

The data was presented as mean ± SD. The students test and one-way ANOVA were used to conduct the comparison of the different protein, mRNA, luciferase reporter and miRNA expression levels. Statistical analyses were performed by SPSS 16.0 software. A *P*-value less than 0.05 was considered a statistically significant difference.

Results

Ischemia/reperfusion injury following heterotopic heart transplantation in mice

We first examined the cardiac IR injury in the heart tissue underwent different reperfusion time point by histology analysis. We found that in the sham group, the cardiac cells were arranged neatly, and no bleeding or inflammatory cell infiltration was observed, however, 0d group showed obvious bleeding. Either on 1 d, 2 d and 3 d after reperfusion, biopsies revealed that significant infiltrations of inflammatory (mononuclear cells) and edema of the intercellular substance in myocardial tissue, in comparison with biopsies from sham groups (HE staining, **Figure 1B**). Masson Trichrome staining demonstrated that cardiac perivascular fibrosis in biopsies of mice with 1 d, 2 d and 3 d after reperfusion was significantly greater than in sham controls and mice with 0 day of ischemia reperfusion (**Figure 1B**).

Since PTEN-mediated β-catenin signaling regulates innate immune responses in mouse liver ischemia/reperfusion injury [13], we were keen to determine the effect of PTEN-mediated β-catenin signaling in heart ischemia/reperfusion injury.

Cardiac ischemia/reperfusion induces rapid and time-dependent degradation of β-catenin

Real-time PCR was used to detect the mRNA expression of β-catenin in mice underwent different period of perfusion varying from day 0 to days 3: sham-operated mice, I/R mice with perfusion at day 0 (samples were obtained immediately when transplantation was done), I/R mice with perfusion at day 1, I/R mice with perfusion at day 2, and I/R mice with perfusion at day 3 (n=3/group). Compared with shame group, β-catenin mRNA expression was decreased during the perfusion phase from day 0 to day 1, and remained the same level until 3 days after perfusion (**Figure 1C**). We investigated whether such β-catenin reduction was related to changes in other key cardiac signaling pathways.

Compared with shame group, mRNA expression of PTEN was significantly increased during the perfusion phase from day 0 to day 3 ($P<0.01$ both on day 1 and day 2). The PTEN level was significantly increased at 1 day of reperfusion ($P<0.01$), but then decreased at the IR 2 days compared with IR 1 day of reperfusion time point, however, when compared with controls, it was still significantly up-regulated ($P<0.01$) (**Figure 1C**). We further investigated whether the decreased β-catenin or increased PTEN transcript levels detected by real-time PCR correlated to decreased or increased levels of the corresponding protein, as well as what is the event of its upstream (ROCK1/ROCK2) or downstream of PTEN pathways during IR injury. By using Western Blot analysis, β-catenin, ROCK1, ROCK2, PTEN, and NF-KB protein levels were detected. Results showed that β-catenin protein accumulation was decreased at IR 1 day and 2 days reperfusion time points.

ROCK1, ROCK2, and NF-κB protein levels were decreased at 0 day of reperfusion, but increased at 1 day and 2 days reperfusion compared with sham control (**Figures 1D** and S1).

Cardiac transplant induced heart IR injury is closely associated with the down-regulation of β-catenin and up-regulation of ROCK1 and PTEN expression

However, whether β-catenin levels are altered at early time points of cardiac transplantation induced myocardial ischemia, the molecular mechanisms involved and the impact of such changes in β-catenin related signaling networks has not been fully investigated. In our study, when heart was removed from the donor, it was subjected to 5 min, 10 min, and 15 min of cold ischemia followed by 1 day of subsequent reperfusion. To our knowledge, the pattern of β-catenin levels at such initial times of I/R has not been previously described.

In order to evaluate the injury of cardiac myocytes during reischemia reperfusion, HE and Masson staining were performed. According to the results of HE staining, myocardial cell structure was intact in the sham operation group, and no bleeding or inflammatory cell infiltration was observed. However, with the prolonged period of ischemia, myocardial tissue begun to be surrounded by thick layer of inflammation, and the cells were disordered and degraded. When at 15 min time point, myocardial tissue showed obvious bleeding and necrosis. The results of Masson staining also showed that the degree of myocardial fibrosis increased greatly when ischemia time was prolonged (**Figure 2A**).

Real-time PCR was used to detect the mRNA expression of β-catenin, ROCK1, ROCK2, PTEN, and NF-κB in mice subjected to different ischemic time point (sham, 5 min, 10 min and 15 min after ischemia). Compared with shame group, mRNA expression of β-catenin underwent a decreased pattern with the ischemic time point increased. However, there were no significant changes observed for ROCK1 mRNA expression during the whole period. For ROCK2, in general, there was a decline trend although it underwent an up-regulation at 10 min. It is obvious that mRNA expression of β-catenin was significantly increased. A similar pattern of concomitant up-modulation of NF-κB levels in early I/R was observed in mice hearts, although a little difference at 15 min time point (**Figure 2B**). Western blot analysis showed that except for a decreased β-catenin protein levels were observed, others all increased with the ischemic time increased (**Figures 2C**, **2D** and S2).

The pattern of inflammatory factors including proinflammatory cytokines (IL-1β, IL-6, TNF-α) and anti-inflammatory cytokines (IL-10) and TGF-β were evaluated in the blood serum of mice with different time point of ischemia (Sham, 5 min, 10 min, and 15 min). A decrease pattern of IL-6 was observed during the whole time course. We found that serum levels of IL-1β and TNF-α proinflammatory cytokines began to increase at 5 min time point, although decrease and remain the base level at 15 min. These cytokines are crucial in tissue injury and inflammation [25-27]. However, no significant difference was observed between each group. For IL-10, at 5 min time point, the expression of IL-10 increased significantly, and peaked at 10 min, compared to other time points. However, the whole time course witnessed a decreased pattern of TGF-β levels (**Figure 2E**).

The over-expression of β-catenin inhibited the expression of PTEN and ROCK

Since it is well known that EGFR mutations usually lead to the abnormal expression of itself and promote cellular proliferation in lung cancer, we investigated the potential fact whether β-catenin over-expression on ROCK/PTEN signal transduction exists in cardiomyocyte. HL-1 cell lines were used in the present study, β-catenin plasmid DNA with different dosage (1, 2, 3, 4, 5 ug), and plasmid with blank were transfected into cells. 4 μg were selected as the optimal dosage for β-catenin plasmid, as at such a dosage (**Figure 3A**).

We found a significant increase in the expression of β-catenin mRNA and protein was observed after transfection with β-catenin plasmid DNA in HL-1 cells, when compared with PB-IP and control group. However, we also noted that β-catenin over-expressing cardiomyocytes showed decreased ROCK1, ROCK2, PTEN, and NF-KB mRNA expression. Significant down-regulation of PTEN both at mRNA and protein levels were observed in β-catenin over-expressing HL-1 cells ($P<0.01$), when compared with others (**Figures 3B**, **3C** and S3). Our *in vitro*

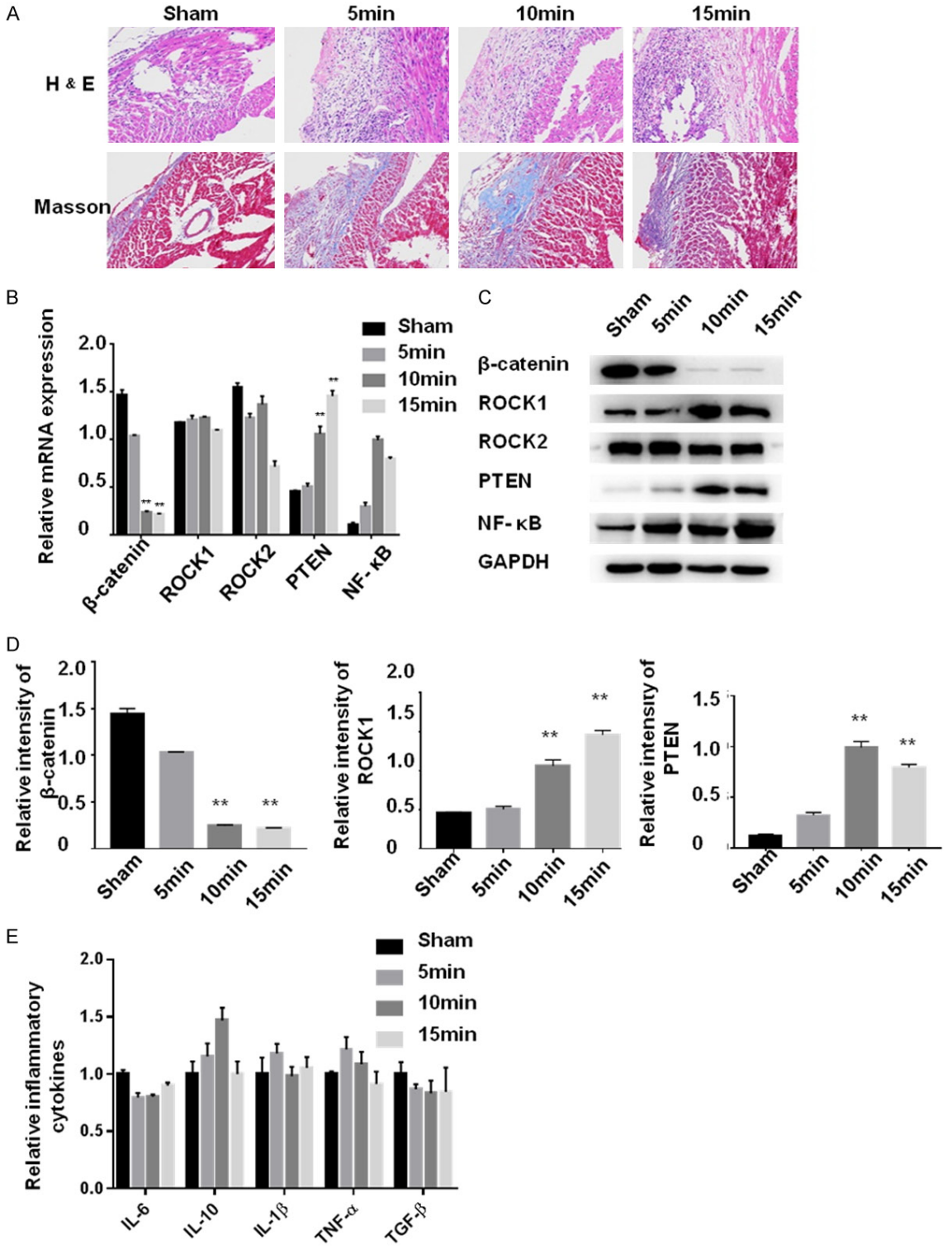

Figure 2. Cardiac ischemia/reperfusion induces rapid and time-dependent degration of β-catenin. A. Histology analysis. The upper panel is HE staining. After transplantation, heart grafts were harvested for H&E staining, at different ischemia time points, 5 min, 10 min and 15 min. The lower panel is Mason staining. Representative images from 3 independent grafts per group. Images were taken at ×200 magnification. B. Real-time PCR was used to detect the mRNA expression of β-catenin, ROCK1, ROCK2, PTEN, and NF-κB in the mice exposed to different period of ischemic at 5 min, 10 min, and 15 min: sham-operated mice, I/R mice with ischemic for 5 min, I/R mice with ischemic for 10 min, and I/R mice with ischemic for 15 min (n=3/group). Compared with shame group, mRNA expres-

sion of β-catenin underwent a decreased pattern with the ischemic time point increased. However, there were no significant changes observed for ROCK1 mRNA expression during the whole period. For ROCK2, in general, it was a decline trend although underwent an up-regulation at 10 min. It is obvious that mRNA expression of β-catenin was significantly increased. Similar trend was observed at NF-KB, although a little difference at 15 min time point. Columns, mean of three independent experiments; bars, SD (n=3). *, $P<0.05$, **, $P<0.01$. C. Western blot analysis was used to detect the expression of β-catenin, ROCK1, ROCK2, PTEN and NF-κB in the mice with cold ischemia at 5 min, 10 min and 15 min time point. GAPDH is used as a loading control. D. β-catenin protein accumulation was decreased at 5 min, 10 min and 15 min time points. PTEN and NF-KB protein levels were decreased at 0 day of reperfusion, but increased at 1 day and 2 days reperfusion compared with sham control. (Right Panel) Band intensity was expressed compared with total GAPDH. E. ELISA was used to detect the serum pattern of inflammatory factors including proinflammatory cytokines (IL-1β, IL-6, TNF-α), anti-inflammatory cytokines (IL-10) and TGF-β mice with different time point of ischemia (Sham, 5 min, 10 min, and 15 min). Columns, mean of three independent experiments; bars, SD (n=3). *, $P<0.05$, **, $P<0.01$.

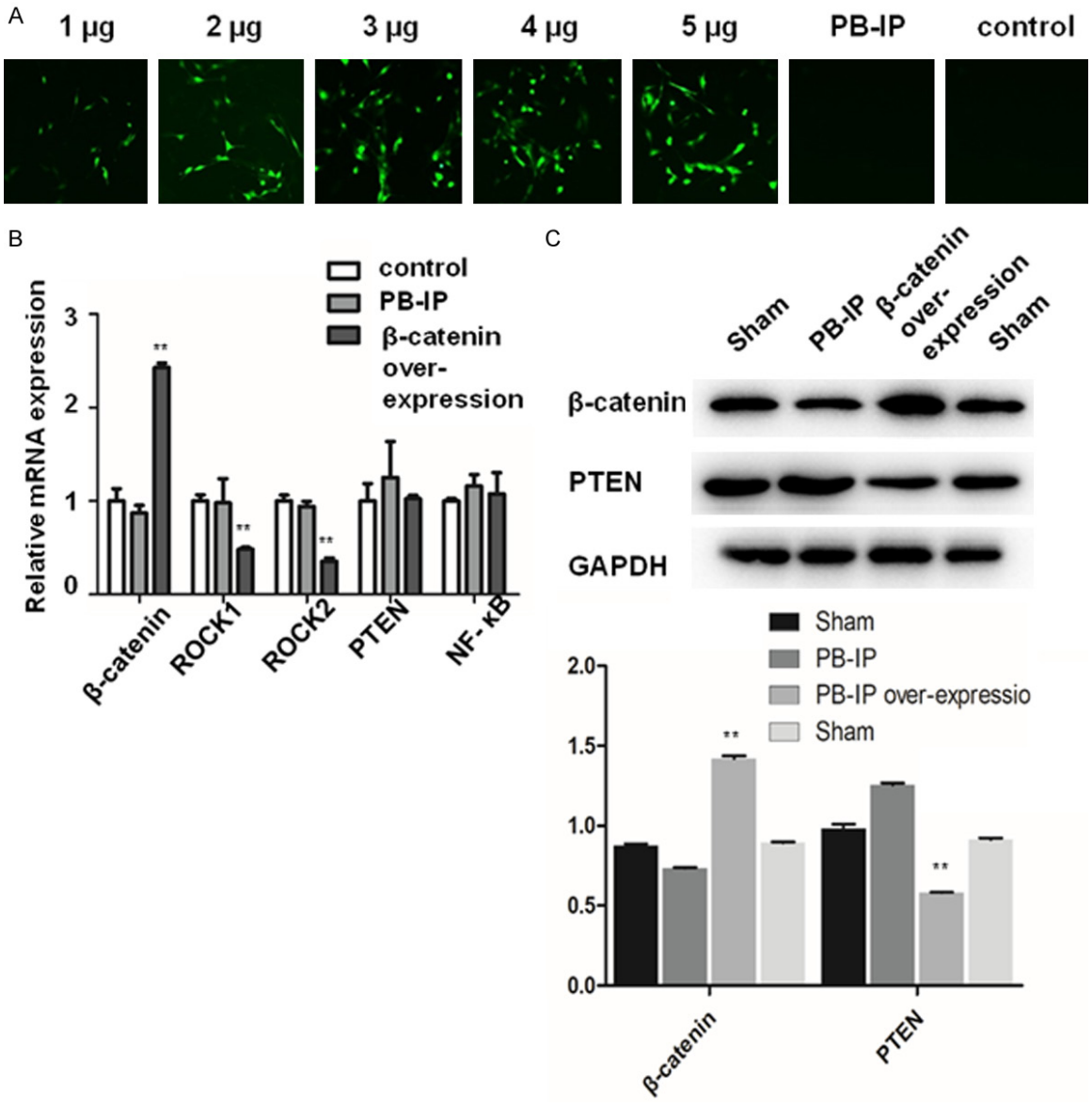

Figure 3. Over-expression of β-catenin inhibited the expression of PTEN and ROCK in HL-1 cells. A. β-catenin plasmid DNA with different dosage (1, 2, 3, 4, 5 ug), and black plasmid were transfected into HL-1 cells. 4 ug were chose and cells were divided into three groups: Control, PB-IP and β-catenin over-expression group. Cell density was monitored after treatment. Typical pictures are shown. Original magnification, ×100. B. The results of the real-time PCR. Real-time PCR was used to detect the mRNA expression of β-catenin, ROCK1, ROCK2, PTEN, and NF-κB in the three

groups. Columns, mean of three independent experiments; bars, SD (n=3). *, $P<0.05$, **, $P<0.01$. C. Western blot analysis. Western blot was used to determine the protein expression of β-catenin, and PTEN in the three groups, which were detected significantly altered for the mRNA level. GAPDH is used as a loading control.

study suggests that β-catenin could inhibit ROCK, and PTEN pathways.

$CoCl_2$-induced activation of ROCK1/PTEN signaling pathway was attenuated by transient transfection of a β-catenin over-expressing plasmid

To further investigate the role of β-catenin and its related signaling in injury in *vitro*, we utilized $CoCl_2$-mediated oxidant injury as an *in vitro* model of heart ischemic reperfusion injury, since $CoCl_2$ was often used as hypoxia mimic reagent and is able to establish hepatic I/R injury *in vitro* [28]. Here, HL-1 cells were treated with $CoCl_2$ (500-1000 μM) for 12 h, 24 h and 36 h, respectively. OD was detected in HL-1 cells treated with different concentrations (500, 600, 700, 800, 900, and 1000 μM). We revealed that the OD value in cells treated with 600 μM at 24 h time point was significantly lower than that in other groups (**Figure 4A**), the optimal concentration was 600 μM at 24 h and it was therefore selected for the further study. Cells were selected into the four groups: Control group, $CoCl_2$ group (HL-1 cells treated with $CoCl_2$), $CoCl_2$+PB-IP group (HL-1 cells with blank plasmid transfection and treated with $CoCl_2$), and β-catenin+$CoCl_2$ group (Cells transfected with β-catenin over-expressing plasmid and treated with $CoCl_2$). Real-time PCR and Western Blot analysis were used to detect ROCK1, ROCK2, PTEN, and NF-kB levels. Compared with controls, $CoCl_2$ treated cells showed significantly increased ROCK1, ROCK2, PTEN, and NF-κB levels (**Figures 4B**, **4C** and S4). However, in β-catenin+$CoCl_2$ group, we demonstrated that $CoCl_2$-induced activation of ROCK1/PTEN signaling pathway was attenuated by transfection of a β-catenin over-expressing plasmid. **Figure 4D** displayed schematic illustration of the role of β-catenin and its related PTEN/ROCK pathways during myocardial Ischemia/Reperfusion injury following heterotopic heart transplantation in mice.

Discussion

Ischemia and reperfusion injury (IRI) is a major pathogenesis to cause heart dysfunction and failure following heart transplantation, which is a key cause of death and disability. Prompt restoration of blood flow to the ischemic area is crucial for reducing infarct size and mortality, but it can itself trigger additional myocardial damage, termed reperfusion injury. Despite advances in the protocols allowing rapid and effective reperfusion in myocardial ischemia patients, therapies adequately targeting reperfusion injury remain elusive. Therefore, it is crucial to gather information about the status of key cardiac signaling pathways at early stages after I/R to identify potential new targets for intervention. Experiments on myocardial ischemia/reperfusion model demonstrated that β-catenin regulates IR injury, but the underlying mechanism in heart IR remains unknown. Even less is known about the effects of ischemia/reperfusion in the heart transplantation models.

In the present study, we performed systematic analysis of β-catenin and ROCK/PTEN signal pathways during ischemia and reperfusion (with different time points) using heterotopic heart transplantation mice model. We have investigated (a) β-catenin and the activations of ROCK/PTEN signal pathways during reperfusion, as during heart transplantation, difference may exist between the gene responses evident in initial, intermediate, and prolonged reperfusion times when following an acute ischemic insult. (b) β-catenin and its related signal pathways during early ischemia time, since early ischemia may decrease the ischemia-reperfusion (IR) injury than prolonged ischemia, which is related to the survival rate after transplantation (c) the relationship between β-catenin and ROCK/PTEN signal pathways *in vitro* (d) the role of β-catenin playing in the process of I/R *in vitro* cell model.

β-catenin is a very relevant signaling hub in cardiac physiopathology [29]. In our study, firstly, we found that during transplantation, cardiac ischemia/reperfusion induces a rapid and time-dependent degradation of β-catenin either in reperfusion time point. Ke et al. (2013) reported that β-catenin could regulate immune responses in hepatic IR injury and is a novel

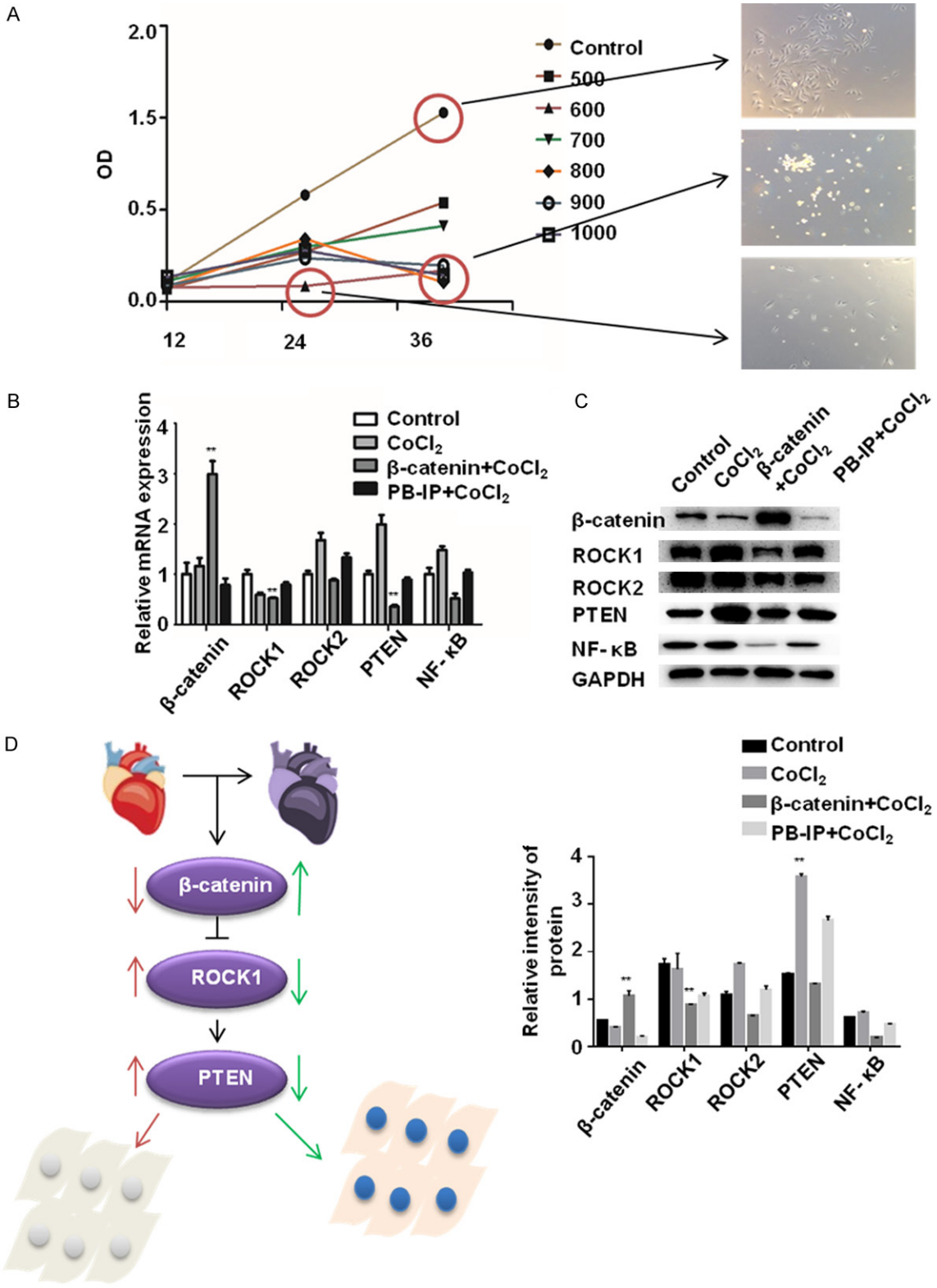

Figure 4. $CoCl_2$-induced activation of ROCK1/PTEN signaling pathway were attenuated by β-catenin over-expression. A. HL-1 cells were treated with $CoCl_2$ (500-1000 μM) for 12 h, 24 h and 36 h, respectively. OD was detected in HL-1 cells treated with different concentrations (500, 600, 700, 800, 900, and 1000 μM). Data are expressed as the OD value. Data shown are means ± SD (n=3). B. The results of the real-time PCR. Real-time PCR was used to detect the mRNA expression of β-catenin, ROCK1, ROCK2, PTEN, and NF-κB in the four groups: Control group, $CoCl_2$ group (HL-1 cells treated with $CoCl_2$), $CoCl_2$ +PB-IP group (HL-1 cells with black plasmid transfection and treated with

$CoCl_2$), and β-catenin+$CoCl_2$ group (Cells transfected with β-catenin over-expressing plasmid and treated with $CoCl_2$). C. Western blot analysis. Western blot was used to determine the protein expression of β-catenin, ROCK1, ROCK2, PTEN, and NF-KB in the four groups (Upper panel). Band intensity was expressed compared with total GAPDH (Down Panel). D. Schematic illustration of the role of β-catenin and its related PTEN/ROCK pathways during myocardial Ischemia/Reperfusion injury following heterotopic heart transplantation in mice.

regulator of innate and adaptive immune responses in the mechanism of liver IRI [14]. Others found that β-catenin is up-regulated in the long-term of patients and experimental models of heart failure. However, in our study, β-catenin mRNA expression was decreased during the perfusion phase from day 0 to day 1, and it remains the same level until 3 days after perfusion, which suggests a time dependent expression of β-catenin in heart I/R injury.

PTEN, a tumor suppressor gene, is an essential regulator of cell proliferation, apoptosis, differentiation, migration, etc. Recently, studies have found that PTEN plays an important role in myocardial remodeling, cardiac hypertrophy, myocardial fibrosis and myocardial ischemia reperfusion injury [30, 31]. Ruan et al. (2009) found that PTEN may play a role in ischemia/reperfusion injury by inhibiting antiapoptotic survival signals [32]. Inhibiting PTEN may serve as a potential approach to exert cardiac protection against ischemia reperfusion injury. In our study, we found that PTEN was significantly up-regulated with the prolonged reperfusion time, suggested that activation of PTEN plays an important role in the IR injury during heart transplantation, and its activation is positively associated with reperfusion time.

ROCK belongs to the family of serine/threonine kinases and includes two isoforms: ROCK1 and ROCK2. ROCK plays important roles in the regulation of platelet activation. ROCK is a major regulator of cell contractility, motility, and proliferation. Many of these ROCK-mediated processes in endothelial cells, vascular smooth muscle cells, pericytes, astrocytes, glia, neurons, leukocytes, and platelets are important in stroke pathophysiology, and the inhibition of such processes could improve stroke outcome. It has been demonstrated that ROCK participate in the event of heart I/R injury [33, 34]. ROCK1 promotes migration and invasion of non-small-cell lung cancer cells through the PTEN/PI3K/FAK pathway [35].

In our initial study designed to detect the molecular mechanism of β-catenin and its related signal pathways in mice underwent different period of reperfusion (0, 1, 2, and 3 days), we found that with the reperfusion time increased, an inhibition of β-catenin pattern occurred in transplanted induced IR injury, while ROCK1 and PTEN increased during the reperfusion phase from day 0 to day 1, and remain the same level until 3 days after reperfusion.

Since early ischemia time may decrease the ischemia and reperfusion injury than prolonged ischemia time [36, 37], the role of β-catenin in I/R dynamic injury during heart transplantation was conducted. When hearts were removed from the donor, a robust mouse was subjected to 5, 10, and 15 min of cold ischemia followed by 1 day of subsequent reperfusion. The molecular mechanism on β-catenin and ROCK/PTEN pathway in I/R injury following heart transplantation were also investigated. The time point selected in our study is different from previous studies which investigated IR injury in murine heart transplantation utilized ischemia time varying from 18 to 24 hours [38]. We found that mRNA expression of β-catenin underwent a decreased pattern with the ischemic time point increased. However, PTEN mRNA expression was significantly activated during the whole period. For ROCK2, in general, it has a decline trend although underwent an up-regulation at 10 min. It is obvious that mRNA expression of β-catenin was significantly increased. The above data suggest that cardiac transplant induced heart IR injury is closely associated with the down-regulation of β-catenin and up-regulation of ROCK1 and PTEN expression.

Subsequently, we poured our attention to analyze the relationship of β-catenin and ROCK/PTEN pathways. We found that β-catenin over-expressing cardiomyocytes showed decreased ROCK1/PTEN expression both at mRNA and protein levels. In addition, cobalt dichloride ($CoCl_2$)-induced oxidative stress model was further established to mimic cardiac I/R injury. We found that $CoCl_2$-induced activation of ROCK1/PTEN signaling pathway were attenuated by

transient transfection of a β-catenin over-expressing plasmid. It is, therefore, concluded that over-expression of β-catenin, might play a protective role in the process I/R injury, or at least participate in the process of transplantation induced heart I/R injury, by inhibit ROCK/PTEN pathways.

In summary, to our knowledge, our study is the first time to investigate the protective role of β-catenin and its related pathways in transplanted induced I/R injury. It suggests that over-expression of β-catenin seems to ameliorate transplanted induced I/R injury by inhibiting ROCK1/PTEN signaling pathway. However, more studies, like *in vivo* studies on over-expression of β-catenin to detect improved cardiac function, modulated oxidative injury, repressed inflammatory damage, should be carried out to verify this hypothesis.

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Disclosure of conflict of interest

None.

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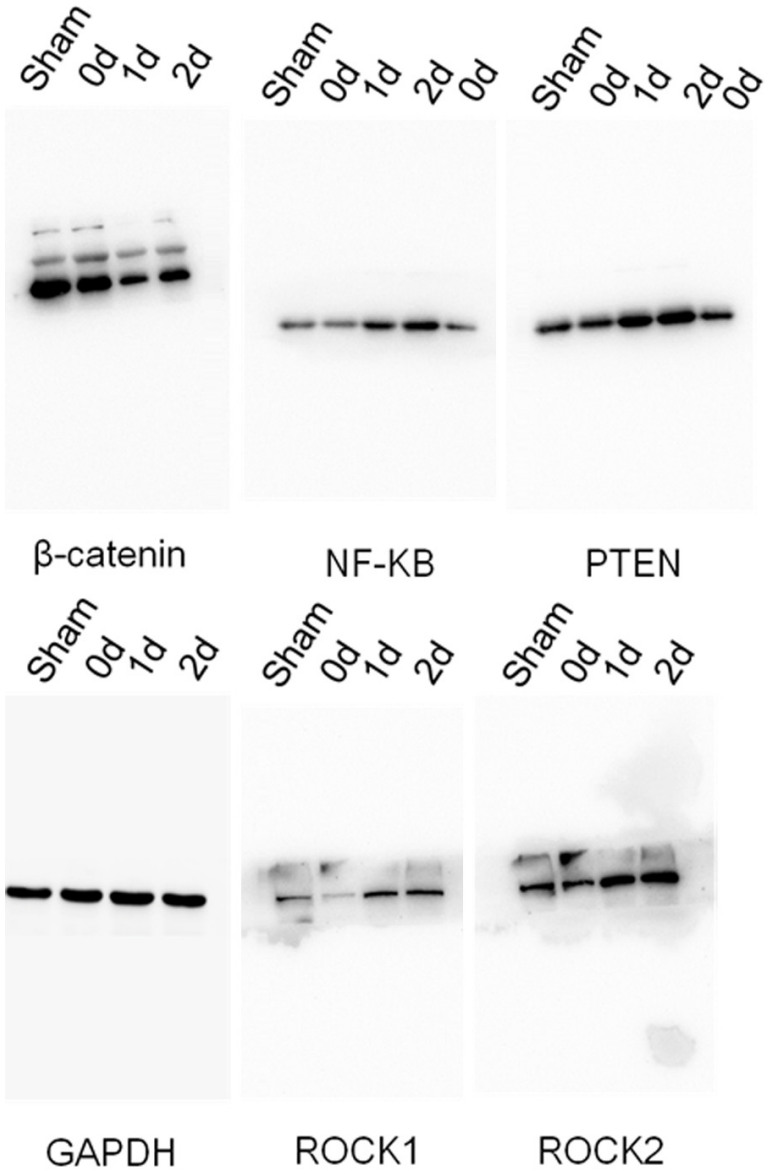

Figure S1. Original western images for relevant western blots of **Figure 1D**.

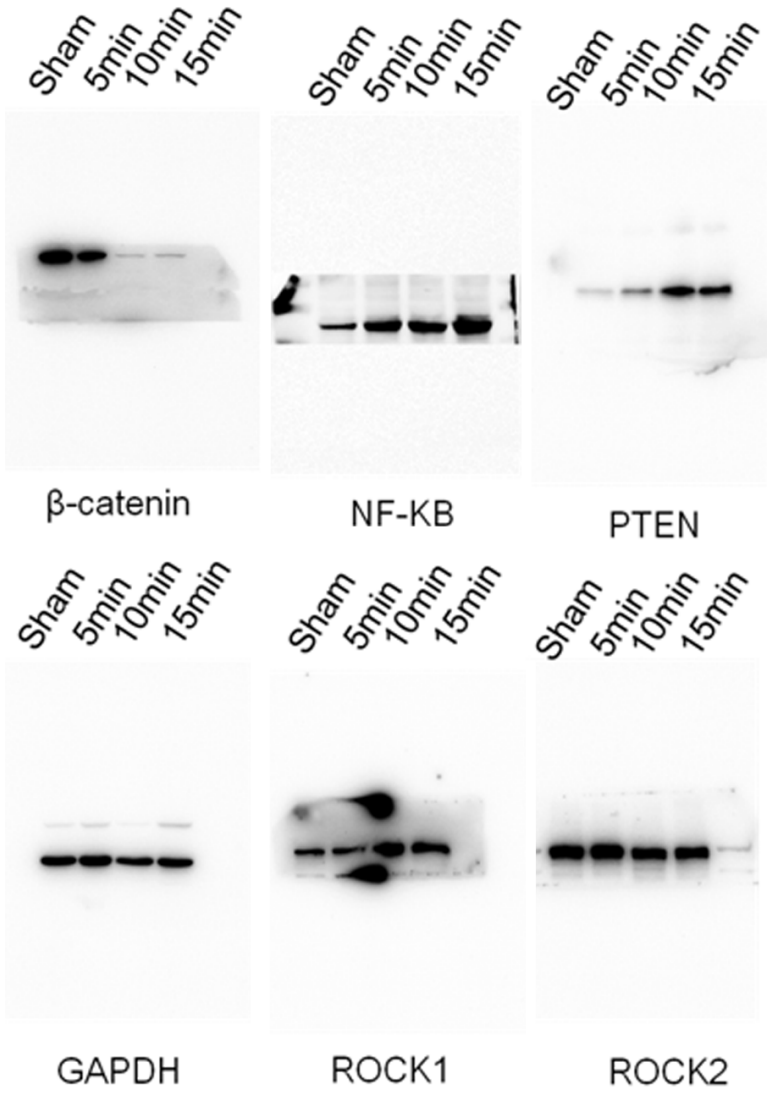

Figure S2. Original western images for relevant western blots of **Figure 2C**.

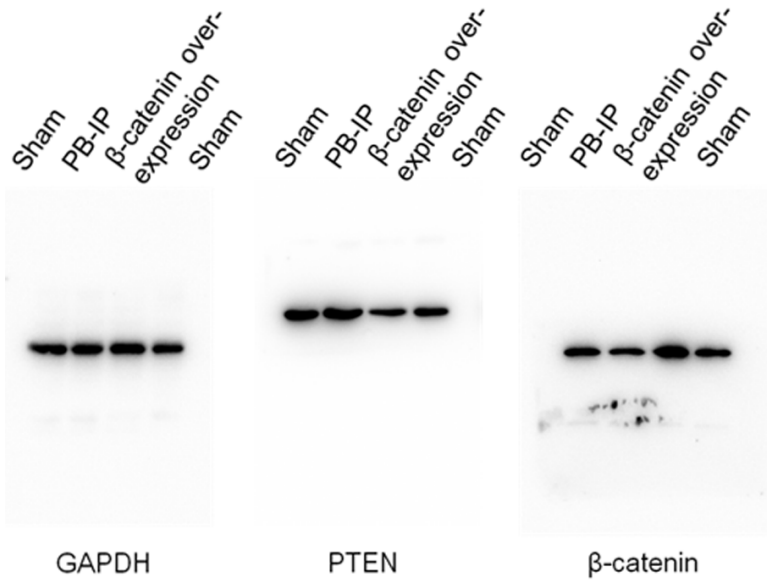

Figure S3. Original western images for relevant western blots of **Figure 3C**.

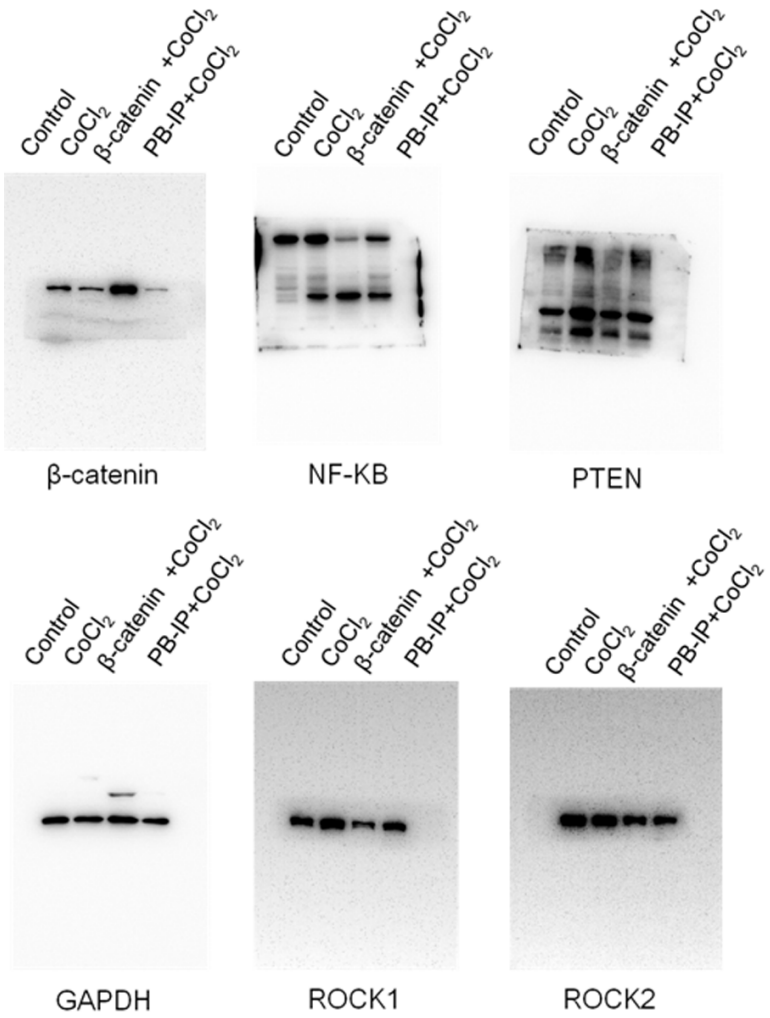

Figure S4. Original western images for relevant western blots of **Figure 4C**.

Research Paper

Transcriptome integration analysis and specific diagnosis model construction for Hodgkin's lymphoma, diffuse large B-cell lymphoma, and mantle cell lymphoma

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ABSTRACT

Transcriptome differences between Hodgkin's lymphoma (HL), diffuse large B-cell lymphoma (DLBCL), and mantle cell lymphoma (MCL), which are all derived from B cell, remained unclear. This study aimed to construct lymphoma-specific diagnostic models by screening lymphoma marker genes. Transcriptome data of HL, DLBCL, and MCL were obtained from public databases. Lymphoma marker genes were screened by comparing cases and controls as well as the intergroup differences among lymphomas. A total of 9 HL marker genes, 7 DLBCL marker genes, and 4 MCL marker genes were screened in this study. Most HL marker genes were upregulated, whereas DLBCL and MCL marker genes were downregulated compared to controls. The optimal HL-specific diagnostic model contains one marker gene (MYH2) with an AUC of 0.901. The optimal DLBCL-specific diagnostic model contains 7 marker genes (LIPF, CCDC144B, PRO2964, PHF1, SFTPA2, NTS, and HP) with an AUC of 0.951. The optimal MCL-specific diagnostic model contains 3 marker genes (IGLV3-19, IGKV4-1, and PRB3) with an AUC of 0.843. The present study reveals the transcriptome data-based differences between HL, DLBCL, and MCL, when combined with other clinical markers, may help the clinical diagnosis and prognosis.

INTRODUCTION

Lymphoma is a malignant tumor originating from the lymphoid hematopoietic system and is mainly divided into two categories: Hodgkin's lymphoma (HL) and non-Hodgkin's lymphoma (NHL). Lymphoma is considered to be a chemosensitive tumor and the risk of lymphoma increases significantly with age [1]. Weakened organismal functions, defects in cellular and tissue homeostasis, immune deficiency, and multiple genetic alterations such as increased DNA damage in cells were correlated with aging. These risk factors are also the main causes of many cancers such as lymphoma [2]. Studies have reported that many anti-aging measures are also helpful in the treatment of lymphoma [2]. Enhancing the expression of anti-aging genes can be an effective way to inhibit lymphoma, a recent study showed that enforced expression of Klotho could significantly induce cell apoptosis and inhibit tumor growth in diffuse large B-cell lymphoma (DLBCL) [3]. Furthermore, downregulated telomere-binding genes (TRF1, TRF2, and POT1) lead to complex chromosomal aberrations, alternative lengthening of telomeres, and induced the progression of HL [4].

According to the WHO classification of lymphoid neoplasias (2016 version), more than 40 types of lymphoma are recognized, with clinical behaviors spanning from remarkably indolent to profoundly aggressive [5]. There are many subtypes of NHL and the most common of which is DLBCL. HL and DLBCL are B-cell-derived lymphomas with high incidence [6]. Mantle cell lymphoma (MCL) is a rare type of B-cell lymphoma that is still incurable, accounting for about 3-6% of all NHL because of its high malignant aggressiveness [7]. There are large differences in histological classification, pathological diagnostic markers, clinical treatment, and prognostic status among HL, DLBCL, and MCL [7–9]. The prognosis of the three B cell-derived lymphoma subtypes is quite different [7, 9, 10]. Especially the prognosis of MCL patients is very poor, and many therapy methods have not achieved the expected outcomes [11].

The different types of lymphoma or lymphoma subtypes can be distinguished by gene expression profiling [12]. Several genes can be used as diagnostic markers for specific types of lymphoma. The ligands of the tumor necrosis factor (TNF) family (APRIL and BAFF) showed high specificity and sensitivity in the diagnosis of central nervous system lymphoma [13]. The high expression of FOXP-1 in pediatric-type follicular lymphoma can also distinguish it from follicular hyperplasia [14]. Furthermore, high-throughput T cell receptor (TCR) gene sequencing technology facilitates the detection of early-stage cutaneous T-cell lymphoma [15]. However, most of the previous lymphoma diagnostic or prognostic models did not consider the heterogeneity among different tumors or subtypes [16–20]. Due to a large number of lymphoma subtypes, some genes may show consistent differential expression in multiple lymphoma subtypes that may interfere with the diagnostic specificity. Furthermore, complex gene-gene interactions also affect the accuracy of tumor diagnosis [21] and prognostic status [22]. Therefore, a diagnostic model with excellent performance should also be robust and hardly affected by gene-gene interactions.

The identification of tumor subtypes contributes to the effective treatment of the disease and prolongs the survival of cancer patients. Using gene expression characteristics to screen specific molecular markers is an effective approach to distinguish tumor subtypes. Through the identification of subtype-specific genes and constructing corresponding models, researchers can accurately perform subtype-specific diagnosis and prognostic evaluation for various tumor patients [23–25]. Currently, the diagnosis of HL, DLBCL, and MCL is mainly based on the morphology and the different combinations of CD surface antigens [26]. The clinical application of genotyping differences among these lymphomas is still limited, and there is lacking an effective molecular diagnostic model. Therefore, this study aims to screen for subtype-specific marker genes and constructed lymphoma-specific diagnostic models, and then explore the related biological functions and prognostic status of these specific molecular markers.

RESULTS

Differentially expressed genes overview

There were more than 3000 differentially expressed genes (DEGs) in the tumor samples compared to the controls in each type of lymphoma, and most of the genes were upregulated (Figure 1A). Most of these differentially expressed genes are upregulated, and a few genes are downregulated. This result is consistent with previous reports that the number of up-regulated genes in different lymphomas is much greater than down-regulated genes [27–29]. Regarding the intergroup comparisons, DLBCL showed a large difference compared with the other two lymphomas, whereas only a small difference was detected between HL and MCL (Figure 1B). There were 67, 369, and 59 specific DEGs in HL vs. control, DLBCL vs. control, and MCL vs. control, respectively (Figure 1C). Furthermore, the results of the intergroup comparisons suggest that there were 182 intergroup difference genes (IDGs) shared by HL vs. DLBCL and HL vs. MCL,

1145 IDGs shared by DLBCL vs. HL and DLBCL vs. MCL, and 186 IDGs shared by MCL vs. HL and MCL vs. DLBCL (Figure 1D). According to the screening criteria for lymphoma-specific genes (defined as the intersection of specific differentially expressed genes in lymphoma samples compared to controls and the common intergroup differentially expressed genes in different lymphoma groups), we identified 20 HL-specific genes (Figure 1E), 88 DLBCL-specific genes (Figure 1F) and 8 MCL-specific genes (Figure 1G). The GO enrichment results showed that the HL specific genes are mainly involved in muscle functions, differentiation, and development functions; the DLBCL specific genes are mainly involved in proliferation, development, and neuromodulation functions; and the MCL specific genes are mainly involved in multiple immune-related functions (Supplementary Table 1). Interestingly, most of the HL-specific genes were upregulated, whereas more than 90% of the DLBCL-specific genes were downregulated. Half of the MCL-specific genes were upregulated while the other half were downregulated.

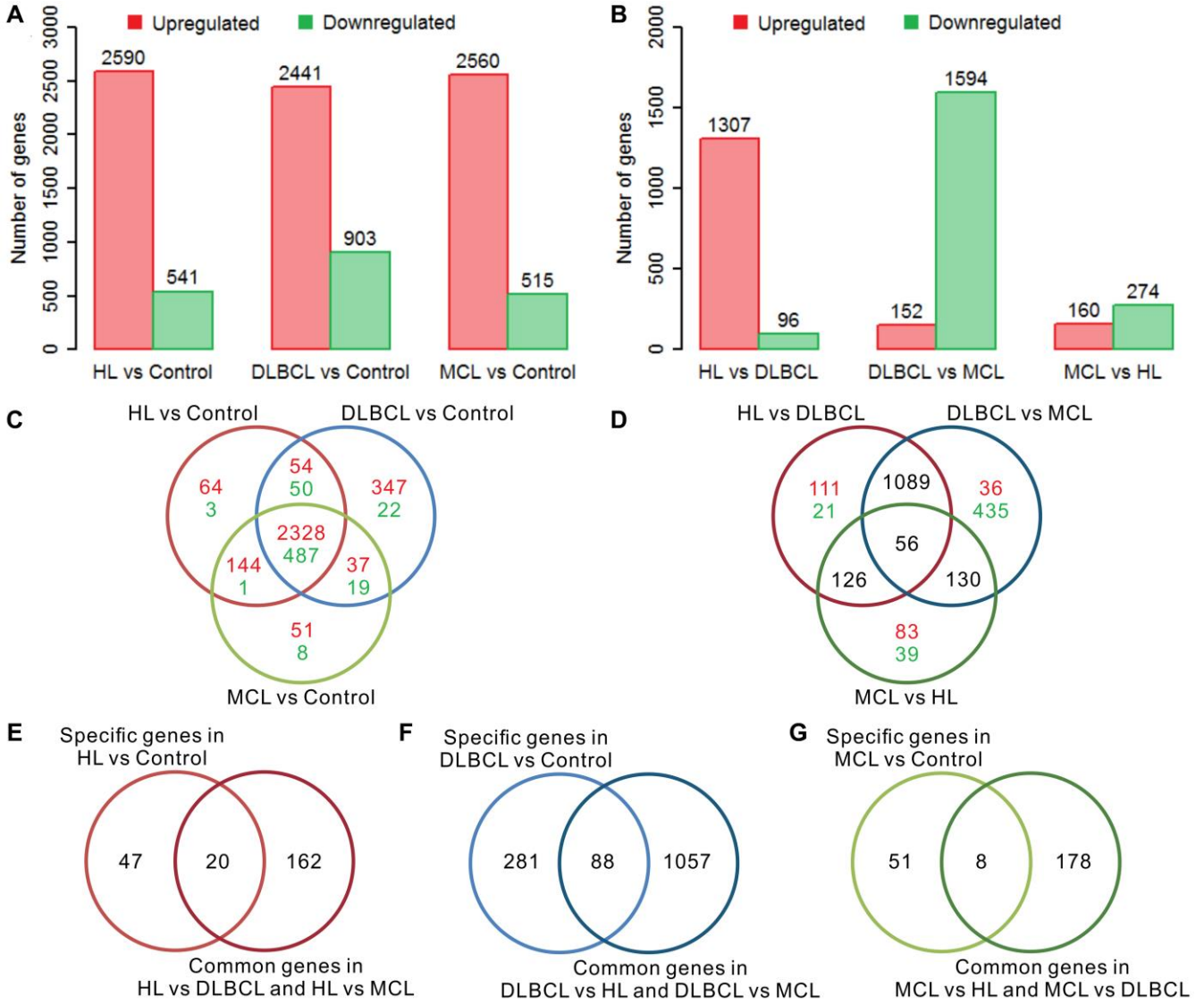

Figure 1. Differential gene expression analysis of three lymphomas. (**A**) The number of differentially expressed genes (DEGs) in lymphoma samples compared to controls. (**B**) The number of intergroup difference genes (IDGs) in three types of lymphoma. (**C**) Venn diagram of DEGs in lymphomas compared to controls. The red color indicates the number of upregulated genes and the green color indicates the number of downregulated genes. The expression trends of these genes are consistent in different types of lymphoma compared with controls. (**D**) Venn diagram of the IDGs between the lymphoma groups. The red color indicates the number of upregulated genes and the green color indicates the number of downregulated genes. (**E**) Venn diagram of HL-specific DEGs and HL common IDGs. (**F**) Venn diagram of DLBCL-specific DEGs and DLBCL common IDGs. (**G**) Venn diagram of MCL-specific DEGs and MCL common IDGs. The red bar indicates the upregulated genes, and the green bar indicates the downregulated genes. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.

Expression and function of the lymphoma marker genes

There were 20 lymphoma-specific genes (9 HL marker genes, 7 DLBCL marker genes, and 4 MCL marker genes) with a mean absolute value of intergroup fold-change high than 0.5 that were defined as lymphoma marker genes (Figure 2A). Among these genes, IL9, SFTPA2, and IGLV3-19 showed the highest specificity in HL, DLBCL, and MCL, respectively. The GO enrichment results showed that these marker genes were mainly involved in the regulation of various immune response and metabolic processes (Supplementary Table 2). Gene-gene interaction analysis proved that most marker genes were independently correlated with lymphoma status (Supplementary Table 3). The high expression of MYH2 increased HL risk whereas the high expression of LIPF and IGLV3-19 reduced DLBCL and MCL risk. The functional interaction network shows that most of the HL marker genes showed coexpression relationships with each other (Figure 2B). DLBCL and MCL marker genes showed multiple interaction relationships with other genes (Figure 2C and 2D). The enrichment results suggest that HL marker genes are mainly involved in actin- and cytoskeleton-related functions (Figure 2E), DLBCL marker genes are mainly involved in chromatin modification and regulation processes (Figure 2F), and MCL marker genes correlate with organismal homeostasis (Figure 2G). The prognostic analysis shows that IL9 and CRNN correlated with the International Prognostic Score (IPS) in HL (Supplementary Figure 1). Furthermore, low expression of CCDC144B and PHF1 and high expression of HP, LIPF, and SFTPA2 correlate with poor overall survival and progression-free survival in DLBCL (Supplementary Figure 2).

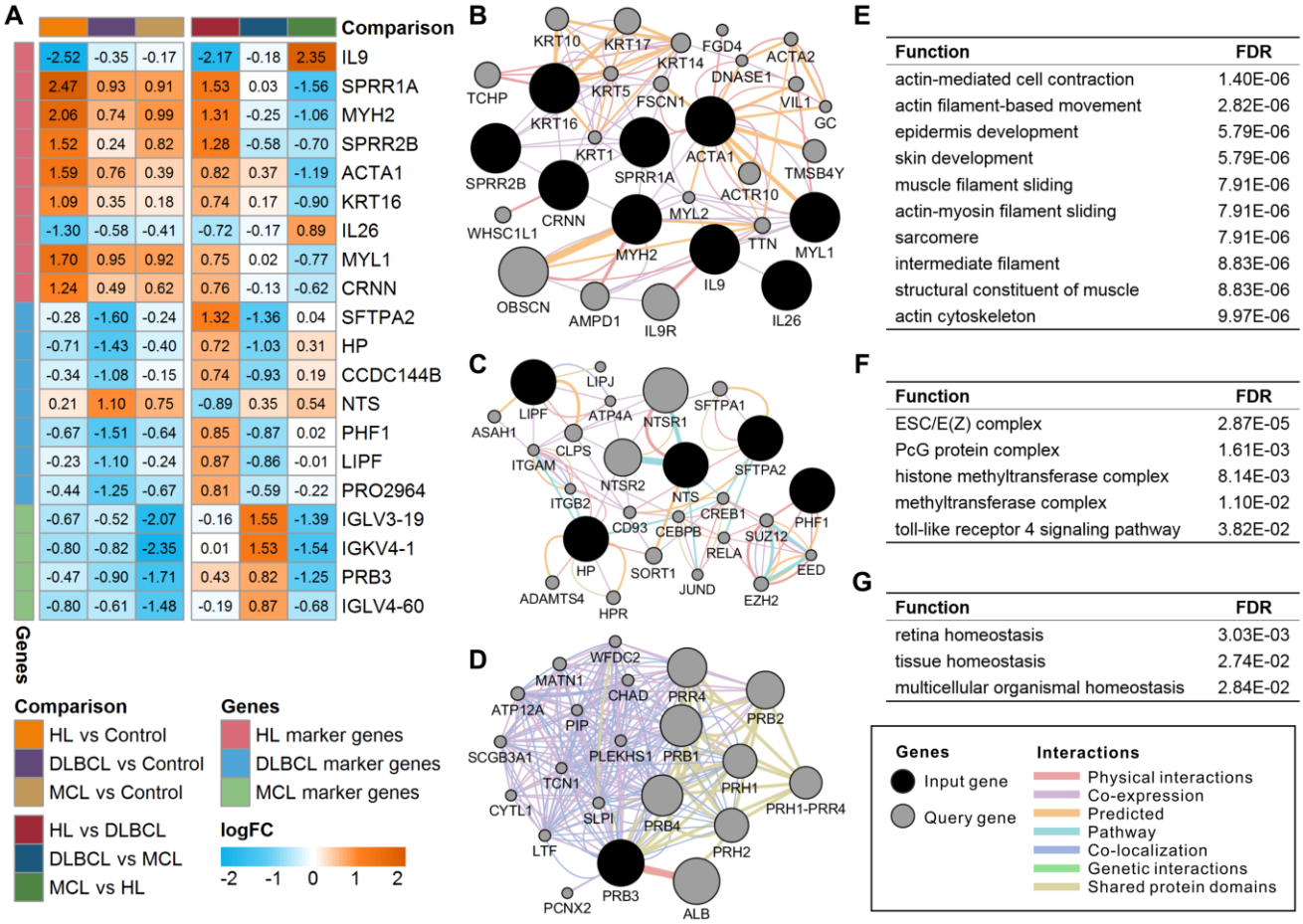

Figure 2. Expression and functional interaction network of lymphoma marker genes. (**A**) Log2 transformed the fold-change (logFC) of lymphoma marker genes in different comparisons. The orange color indicates the logFC of the gene > 0, and the cyan color indicates the logFC of the gene < 0. (**B**) Functional interaction network of HL marker genes. (**C**) Functional interaction network of DLBCL marker genes. No records of CCDC144B or PRO2964 were found in the GeneMANIA database. (**D**) Functional interaction network of MCL marker genes. No records of IGLV3-19, IGKV4-1, or IGLV4-60 were found in the GeneMANIA database. (**E**) Enriched functions of HL marker genes and query genes. (**F**) Enriched functions of DLBCL marker genes and query genes. (**G**) Enriched functions of MCL marker genes and query genes. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.

Single-gene prediction model

A logistic regression model showed that all these marker genes could significantly separate the lymphoma samples from the controls (Figure 3A–3C). The odds ratios of these marker genes are relatively consistent with the expression difference between lymphomas and controls. The results of the ROC analysis of these marker genes are shown in Figure 3D. The ideal classification effect of marker genes is that they have high sensitivity and specificity in the specific type of lymphoma (the AUC value is close to 1) and have a random effect for the other two lymphomas (the AUC value is close to 0.5). For the HL marker genes, MYH2 showed the highest AUC of 0.901 in HL, a low AUC in DLBCL, and an AUC close to 0.5 in MCL; therefore, it can be used as the optimal model in the single-gene prediction model in HL (Figure 3E). LIPF showed the highest AUC of 0.875 in DLBCL and low AUCs in HL and MCL and is considered to be the optimal single-gene prediction model in DLBCL (Figure 3F). However, IGLV3-19 had the highest AUC in MCL marker genes but only showed a general prediction effect (Figure 3G).

Multigene prediction model

The optimal model in HL is MYH2 (Figure 4A and 4G), which had the highest AUC compared with the remaining gene combination models (Figure 4D). The optimal model in DLBCL is the combination of 7 marker genes, including LIPF, CCDC144B, PRO2964, PHF1, SFTPA2, NTS, and HP (Figure 4B and 4G), which had the highest AUC of 0.951 (Figure 4E). The optimal model

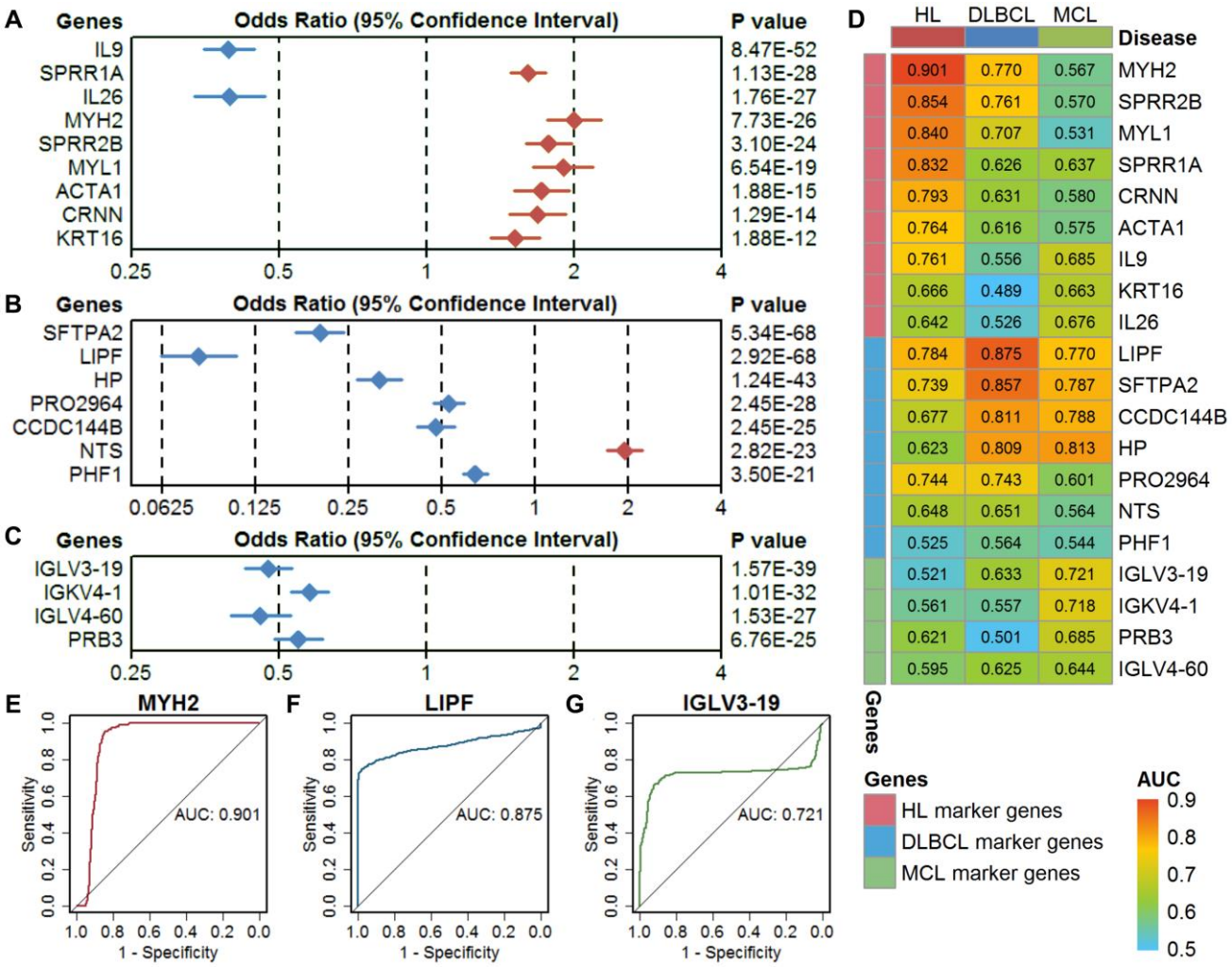

Figure 3. Evaluation of single-gene models in three types of lymphoma. (**A**–**C**) The classification performance of HL marker genes, DLBCL marker genes, and MCL marker genes using a univariate logistic regression model. The diamond shape indicates the odds ratio (OR), and the line indicates the 95% confidence interval (CI). The red color indicates OR > 1, and the blue color indicates OR < 1. (**D**) The area under the curve (AUC) of the marker genes in three types of lymphoma. (**E**–**G**) Receiver operating characteristic (ROC) curves of the optimal single-gene model in HL (MYH2), DLBCL (LIPF), and MCL (IGLV3-19). HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.

in MCL is the combination of 3 marker genes, including IGLV3-19, IGKV4-1, and PRB3 (Figure 4C and 4G), which had the highest AUC of 0.843 (Figure 4F). These three optimal models all show high specificity in a certain type of lymphoma and showed relatively poor specificity for the other two types of lymphomas (Supplementary Figure 3). Considering the analyzed gene expression data derived from samples including not only lymphoma cells, but also stroma. We screened the data derived from isolated lymphoma cells and normal B cells (Table 1) and analyzed the expression of the lymphoma marker genes between cases and controls. Despite the small sample size, most marker genes still showed the differential expression consistent with the overall analysis (Supplementary Figure 4). Furthermore, the optimal diagnostic models of these genes showed high prediction accuracy in the data derived from isolated lymphoma cells (Supplementary Figure 5). The dataset of GSE132929 including multiple types of lymphomas (no HL or controls) was used to verify the predictive performance of the above optimal models. Previous studies suggested that it is difficult to distinguish Burkitt's lymphoma (BURL) and DLBCL [30, 31], the DLBCL optimal model showed a high AUC of 0.843 in the validation set with removed BURK (Supplementary Figure 6).

DISCUSSION

Accurate and effective diagnosis is critical to the appropriate treatment of lymphoma. Although many new techniques are used for the diagnosis of lymphoma, such as immunohistochemical tests, flow cytometry, cytogenetic, and other molecular biology techniques [32], the most effective diagnostic strategy is still tissue biopsy [33]. Given that some genes show extremely high mutation frequencies in certain types of lymphoma, using a single-gene mutation or a combination of mutations may accurately diagnose few types of lymphomas [34]. However, the diagnosis of most lymphomas using genetic mutations may not achieve the desired accuracy. Previous transcriptome studies have revealed that there are a large number of abnormally expressed genes in different types of lymphomas compared with normal tissues [29, 35–37]. These highly differential genes may be used as diagnostic and prognostic markers for lymphomas [38]. The difference in clinical treatment and prognosis of B-cell-derived lymphoma is correlated to its molecular heterogeneity. In this study, lymphoma marker genes and specific diagnostic models are proposed, which are helpful to improve the diagnosis accuracy of HL, DLBCL, and MCL. These results indicate that there are certain differences at the molecular level among HL,

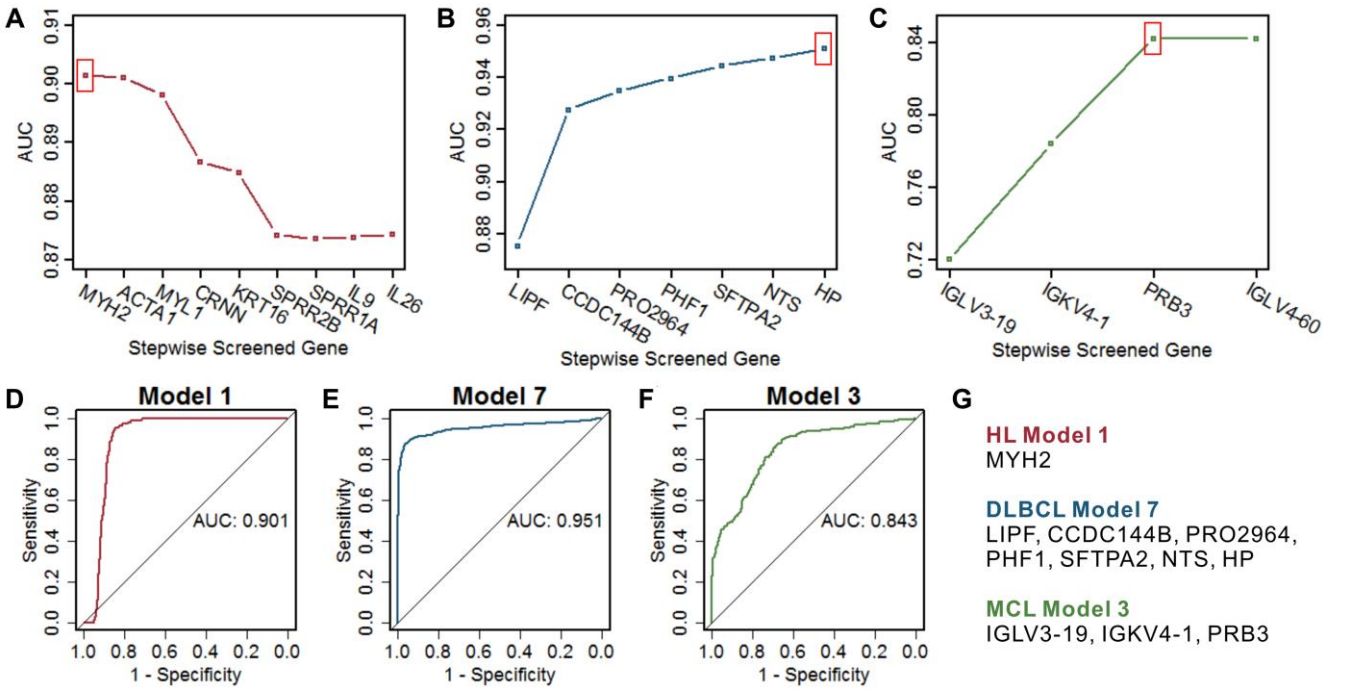

Figure 4. Screening of the optimal multigene prediction model for three lymphomas. (**A**–**C**) Stepwise screened multigene prediction models in HL, DLBCL, and MCL. From left to right on the x-axis (stepwise screened genes), each additional gene corresponds to a model [for example, in (A), MYH2 represents model 1, which contains one gene of MYH2, ACTA1 represents model 2, which contains two genes including MYH2 and ACTA1]. The red box shows the optimal model for each type of lymphoma. (**D**–**F**) ROC curves of the screened optimal models for each type of lymphoma. (**G**) Genes in the screened optimal models for three lymphomas. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.

Table 1. Information on the datasets of three types of lymphoma.

GEO ID	Contributor	Samples	Sample type	Platform
Hodgkin's lymphoma (HL)				
GSE7788[1]	Van Loo P, 2007	10 cases 1 control	lymph nodes	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE12453[2]	Brune V, 2008	17 cases 25 controls	isolated lymphoma cells (case) isolated normal B cells (control)	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE13996	Chetaille B. 2008	64 cases	lymph nodes	Affymetrix HG-U133A 2.0 Array (GPL571)
GSE17920	Steidl C, 2009	130 cases	lymph nodes	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE47044	Hartmann S, 2013	19 cases 5 controls	isolated lymphoma cells (case) isolated normal B cells (control)	Affymetrix Human Gene 1.0 ST Array (GPL6244)
Diffuse large B-cell lymphoma (DLBCL)				
GSE12453[2]	Brune V, 2008	11 cases	isolated lymphoma cells	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE31312	Li Y, 2011	498 cases	lymphoma tissue	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE56315	Bødker JS, 2014	89 cases 33 controls	lymphoma tissue (case) isolated normal B cells (control)	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE64555	Linton K, 2014	40 cases	lymphoma tissue	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE69053	Sha C, 2015	212 cases	lymphoma tissue	Illumina HumanRef-8 WG-DASL v3.0 (GPL8432) Illumina HumanHT-12 WG-DASL V4.0 (GPL14951)
GSE86613	Bødker JS, 2016	41 cases	lymphoma tissue	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
Mantle cell lymphoma (MCL)				
GSE21452	Staudt LM, 2010	64 cases	lymph nodes	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE36000	Jares P, 2012	38 cases	isolated lymphoma cells	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE70910	Liu D, 2015	55 cases	lymph nodes, peripheral blood	Affymetrix HG-U133 Plus 2.0 Array (GPL570)
GSE93291	Staudt LM, 2017	59 cases	lymph nodes	Affymetrix HG-U133 Plus 2.0 Array (GPL570)

[1]The one control sample in this study was the mixed five control samples.
[2]This dataset included 17 HL samples, 11 DLBCL samples, and 25 controls. The number of control samples was shown in the HL group and was not repeated in the DLBCL group.

DLBCL, and MCL, which provides some insights for the molecular diagnosis and prognosis assessment of these three types of lymphomas.

Gene expression profiling has broad application prospects in tumor diagnosis [39], and numerous novel biomarkers have been identified in the most common types of B-cell, T-cell, and NK-cell lymphomas [40]. Multiple diagnostic models based on the combined effects of tumor biomarkers have been developed and show high prediction accuracy. A previous report constructed two logistic regression models based on mammography features and demographic data; both of these models showed high accuracy for breast cancer diagnosis [41]. A logistic regression model integrating transcriptome and clinical data also showed high diagnostic accuracy in lung cancer [42]. Furthermore, using machine learning methods to construct tumor diagnostic models is also an effective strategy. Diagnostic models based on support vector machines and their derived methods by feature extraction of transcriptome data exhibited high prediction accuracy in multiple cancer datasets [43, 44]. The sample size is an important factor affecting the accuracy of the diagnostic model [45]. In this study, the sample size of DLBCL is relatively large, while the sample size of HL and MCL is relatively small, and the final multigene diagnosis models also showed the highest diagnostic accuracy for DLBCL. In future work, a larger sample size can be used to develop more accurate tumor-specific diagnosis

or prognosis models. With the expansion of the sample size, it is expected to be further upgraded to a personalized prediction model.

The screened HL marker genes were mainly involved in actin- and cytoskeleton-related functions. Multiple studies showed that the actin cytoskeleton plays a crucial role in aging and apoptosis [46], and the dysfunction of the actin cytoskeleton correlated to many age-related diseases, such as cancer [47]. Actin polymerization and actin-myosin interactions directly drive the movement and migration of lymphocytes [48]. A proteomics study showed that several upregulated proteins were involved in the regulation of the cytoskeleton and/or cell migration in HL [49]. Inhibited cytoskeleton-related proteins promoted the differentiation of Hodgkin's and Reed-Sternberg (H/RS) cells toward terminal B-cells in HL cell lines [50]. DLBCL marker genes were mostly enriched in chromatin modification and regulation processes. Increased variations in chromatin modification were correlated with aging [51], studies showed that these epigenetic factors can also induce tumorigenesis [52]. Mutations in chromatin modification-related genes are correlated to gene expression profiles and clinical outcomes in DLBCL [53]. These genes can be used as signatures for evaluating the effect of medical treatments on DLBCL [54, 55]. Several immunoglobulin (Ig) subunit genes were chosen as MCL marker genes in this study. Alterations in IgG glycosylation patterns have been observed in aging and various cancers [56]. The regulation and modification of Ig are essential to maintain immune homeostasis *in vivo* [56]. The imbalanced Ig heavy and light chain stereotypy was found in MCL [57].

The above reports indicated that the marker genes screened in this study correlated with the specific biological changes of different lymphomas. Furthermore, these marker genes also showed high diagnostic accuracy in other tumors and correlated with tumorigenesis and prognosis. A previous study showed that MYH2 was correlated with multiple prognostic factors in lymph-node-negative primary breast cancer [58]. Downregulated IL26 promotes anaplastic large cell lymphoma cell growth and survival [59]. Gastric lipase (LIPF) is highly expressed in the normal stomach and showed significantly low expression in gastric adenocarcinoma, suggesting that it can be used as a diagnostic and prognostic indicator for gastric cancer [60, 61]. Low expression of PRB3 was found to be associated with tumor recurrence in prolactinomas [62] and salivary gland acinic cell carcinoma [63]. Besides, there are several representative markers associated with each subtype, such as CD15, CD30, CD45, and PD-L1 for HL [10], CD5, MYC, BCL2, and BCL6 for DLBCL [9], and CCND1, CD5, and SOX11 for MCL [7]. However, most of these genes did not meet the differential expression screening criteria. Therefore, these genes are not included in the specific markers screened in this study.

In conclusion, screening for tumor-specific biomarkers requires the rigorous consideration of differences between tumor and normal cells as well as the differences among different tumors or subtypes. The present study provides the transcriptome data-based reference markers, which may help the diagnosis of HL, DLBCL, and MCL when combined with other clinical markers. As there are multiple subtypes of lymphoma according to the WHO classification, whether the currently obtained marker genes can be used to diagnose other types of lymphomas requires further research. One potential shortcoming in this study is the sample size of different lymphoma datasets varies largely, especially for the control group is relatively low. The present study provides a molecular diagnostic method, a reference for tumor diagnosis with a subtle difference to clarify tumor subtypes.

MATERIALS AND METHODS

Lymphoma dataset collection

Transcriptome datasets of HL, DLBCL, and MCL were downloaded from the Gene Expression Omnibus (GEO) database (https://www.ncbi.nlm.nih.gov/geo/). The dataset selection criteria were as follows: (1) all datasets were genome-wide; (2) the number of samples in each dataset must be ≥ 10; (3) all samples were non-cell-line samples; and (4) complete microarray data (raw or normalized) were available. If a dataset contained any of the following items, it was excluded: (1) the number of samples was less than 3 for cases or controls; (2) the samples were treated with drugs or other agents; and (3) serious RNA degradation or the number of detected genes was too small. Based on the above criteria, 14 datasets were chosen for the integrated analysis (GSE12453, GSE13996, GSE17920, GSE21452, GSE31312, GSE36000, GSE47044, GSE56315, GSE64555, GSE69053, GSE70910, GSE7788, GSE86613 and GSE93291). The sample type of most data is lymphoma tissue, only a small part of the data derived from isolated lymphoma cells, the details of these datasets are provided in Table 1. In total, the collected datasets contained 240 HL samples, 891 DLBCL samples, 216 MCL samples, and 64 healthy samples.

Data preprocessing

R statistical software v3.3.3 (https://www.r-project.org/) was used to perform data preprocessing. Because these

datasets contain different microarray platforms, they were grouped into 15 batches according to the study and platform. Each batch contained only one study and one platform (Supplementary Table 4). Gene annotation, integration, and renormalization of the 15 batches were carried out using custom-designed Python code. The method and scripts are detailed in our previous publications [64, 65]. Because there were missing values for genes in a few samples, the mean expression value of these genes in the whole sample was used to replace the missing data. Fortunately, the missing values had little effect on the data (Supplementary Figure 7). After global renormalization, the distribution of gene expression values across all studies had a consistent range (Supplementary Figure 8). Heatmap in the pheatmap package in R was used to show all gene expression profiles in the integrated and the global renormalized datasets. The method of unsupervised clustering was chosen as "ward.D". There was a strong batch effect in the integrated datasets, and this batch effect has been mostly eliminated in the global renormalized datasets (Supplementary Figure 9).

Differential expression analysis

Differential gene expression analysis was performed using the empirical Bayesian algorithm in the limma package in R [66]. Up- and downregulated genes were defined as a log2 transformed fold-change (logFC) ≥ 1 or ≤ -1 for lymphoma samples compared with controls. Because the difference between lymphoma groups was smaller than the difference between lymphoma samples and controls, the fold-change cutoff was set as 1.2. A false discovery rate (FDR)-corrected *P*-value ≤ 0.05 was considered significant.

Screening of lymphoma marker genes

Lymphoma-specific genes were defined as the intersection of specific differentially expressed genes in lymphoma samples compared to controls and the common intergroup differentially expressed genes in different lymphoma groups. For example, the filtered HL-specific genes are differentially expressed between HL vs. control with no difference in DLBCL vs. control or MCL vs. control and are differentially expressed between HL vs. DLBCL and HL vs. MCL. To ensure that the screened marker genes have a relatively large differential expression compared to other types of lymphoma, the lymphoma marker genes were defined as lymphoma-specific genes with a mean absolute value of intergroup fold-change ≥ 0.5.

Gene-gene interaction analysis

Considering that the gene-gene interactions between the screened marker genes and other genes may affect prediction accuracy, Pearson correlation analysis was used to calculate the correlation coefficient between each marker gene and all other genes. An FDR corrected *P*-value ≤ 0.05 was considered significantly correlated. A multiple logistic regression model was used to analyze the effect of each marker gene on the corresponding lymphoma. The top 10 significantly correlated genes (filtered by significance) were used as covariates for model correction.

GO enrichment analysis

The information on human genes and related GO biological functions were downloaded from the QuickGO database (http://www.ebi.ac.uk/QuickGO-Beta/). GO enrichment analysis was performed using a hypergeometric test and the formula shown in a previous report [67]. An FDR corrected *P*-value ≤ 0.05 was considered significantly enriched.

Functional interaction analysis

The GeneMANIA application [68] in Cytoscape v3.4.0 was used to perform functional interaction analysis of marker genes in three types of lymphoma. The interaction networks were built with the default parameter settings. The application predicts 20 query genes that are correlated to the input genes and generate a functional association network based on their relationships. The functional enrichment results of the genes in the network were automatically generated, and an FDR-corrected *P*-value ≤ 0.05 was considered significantly enriched.

Prognostic analysis

Two datasets (GSE17920 and GSE31312) had prognostic information. The GSE17920 dataset (HL) contains multiple prognostic indicators but no survival data, and the GSE31312 dataset (DLBCL) contains complete overall and progression-free survival information. The difference in HL marker genes regarding prognostic indicators was determined using Student's *t*-test. Survival analysis was conducted using the survival package in R. The effects of DLBCL marker genes on overall and progression-free survival were assessed using Kaplan–Meier survival curves.

Single-gene and multigene prediction models

The single-gene prediction model and the multigene prediction model were built using the lymphoma marker genes. A univariate logistic regression model was used to calculate the odds ratios of the screened lymphoma marker genes in each type of lymphoma. For the single-gene prediction models, the specified type of lymphoma

was classified as "case", whereas the healthy samples and the other two types of lymphomas were classified as "control". The receiver operating characteristic (ROC) curve and the area under the curve (AUC) of the single marker genes were calculated using the pROC package in R. The model with the largest AUC was defined as the optimal model. A stepwise modeling strategy was used to screen the optimal multigene combination models for each type of lymphoma. First, a gene with the largest AUC was selected. Then, we used a multivariate logistic regression model to generate the combined effect of the selected gene and each of the remaining genes. Next, we selected the best two-gene model with the highest AUC and repeated the previous steps. Finally, we selected the optimal model with the highest AUC in each multigene combination model.

AUTHOR CONTRIBUTIONS

DL, JFH, FL, and WXL designed the study. WXL, SXD, SQA, JL, LGL, YX, and HY, collected the data. WXL, SXD, SQA, TTS, JW, LXF, XLZ, WQL, JFH, and DL designed the method and analyzed the data. WXL, SXD, SQA, TTS, HY, LT, FL, JFH, and DL wrote the manuscript. WXL, FL, JFH, and DL supervised the research, interpreted the data, and revised the manuscript. All authors read and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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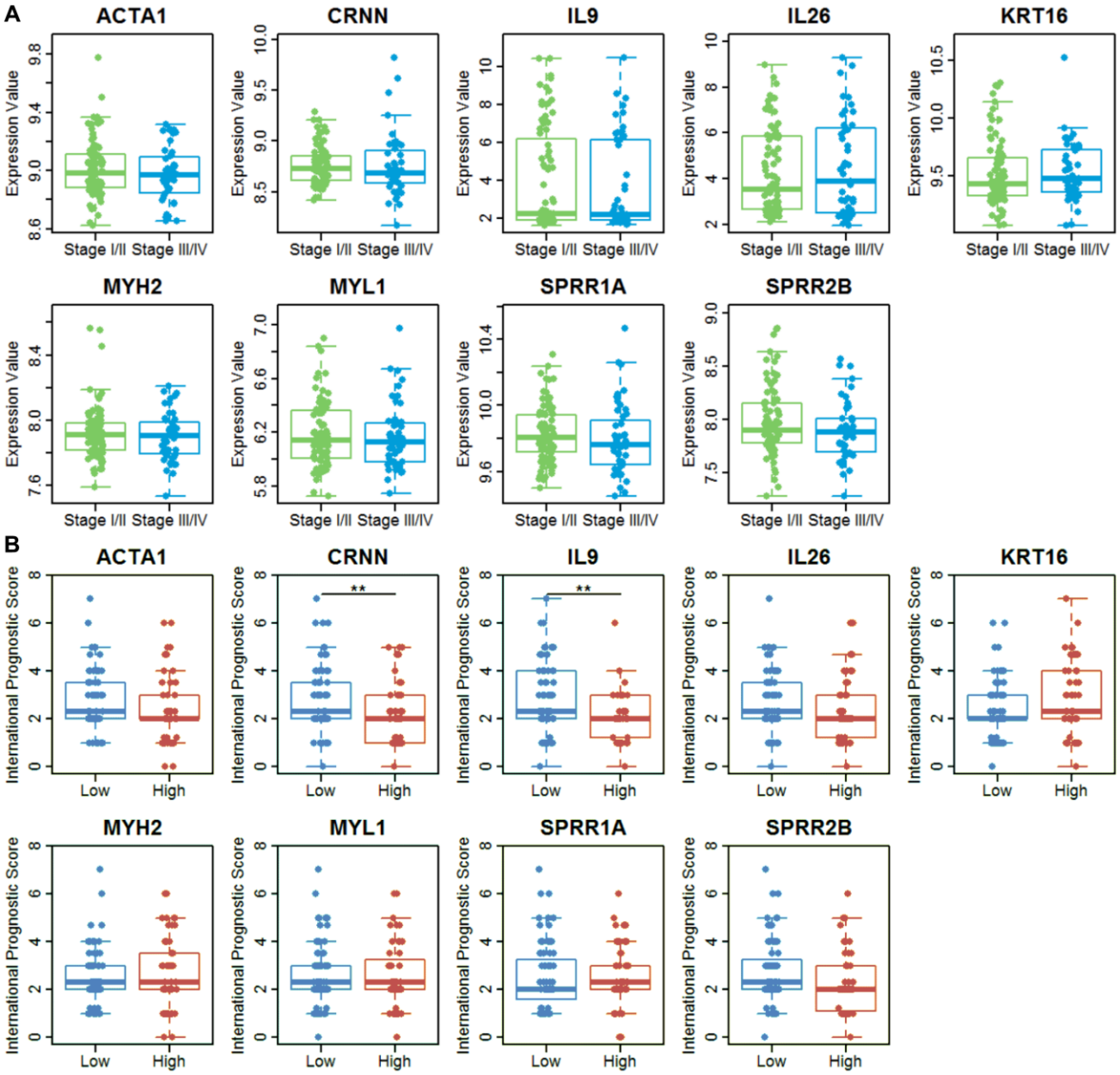
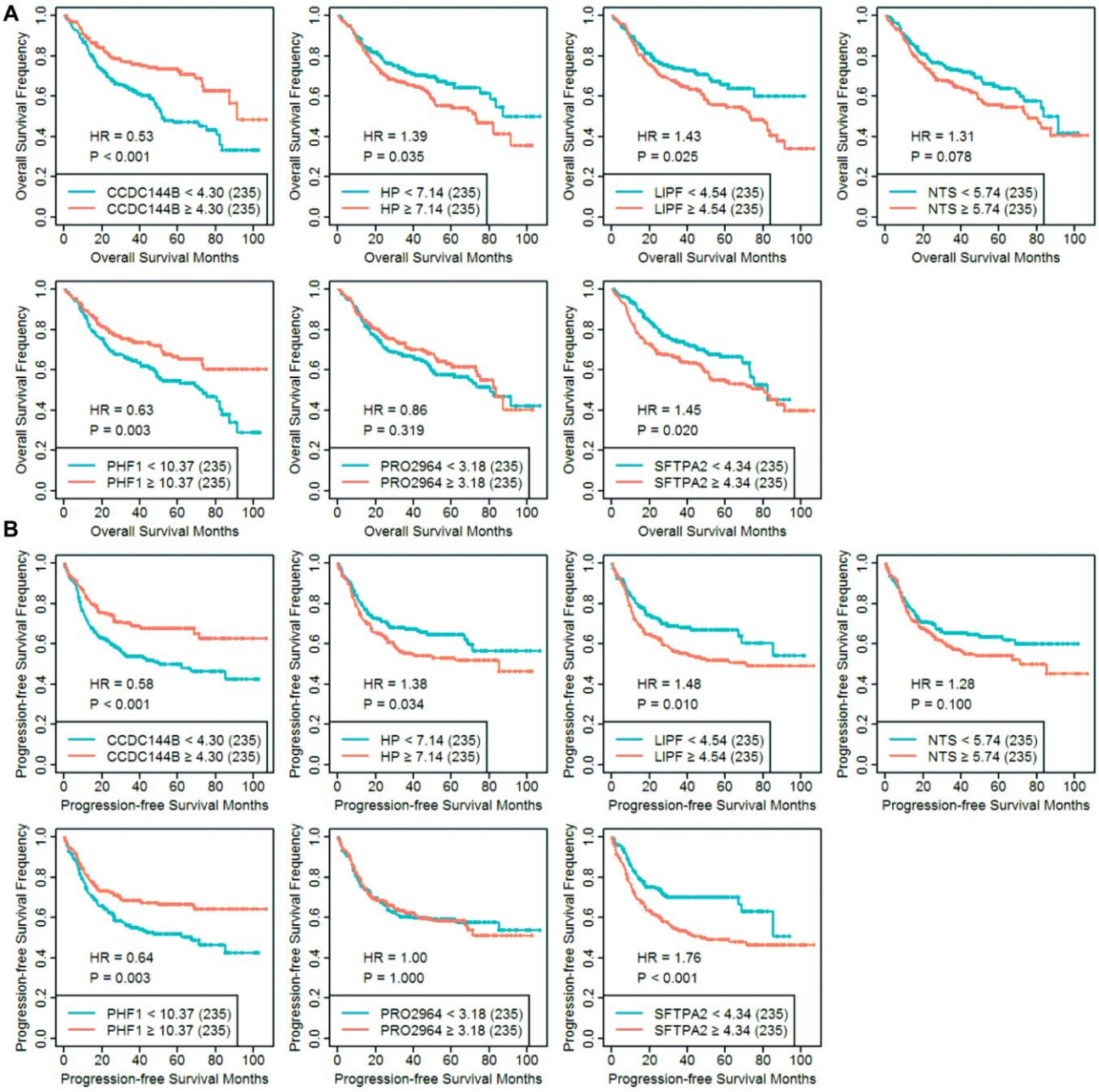
SUPPLEMENTARY MATERIALS

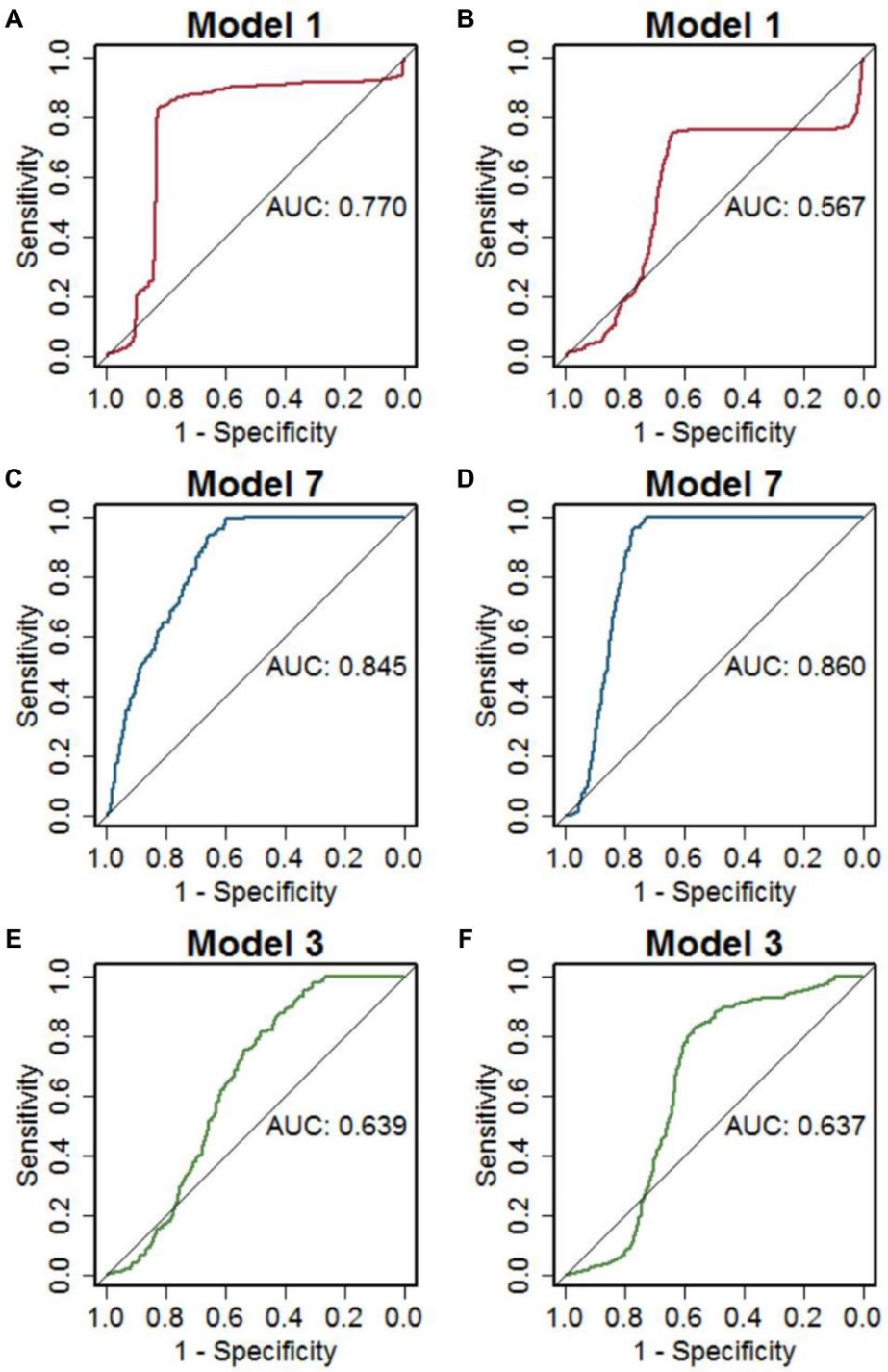
Supplementary Figures



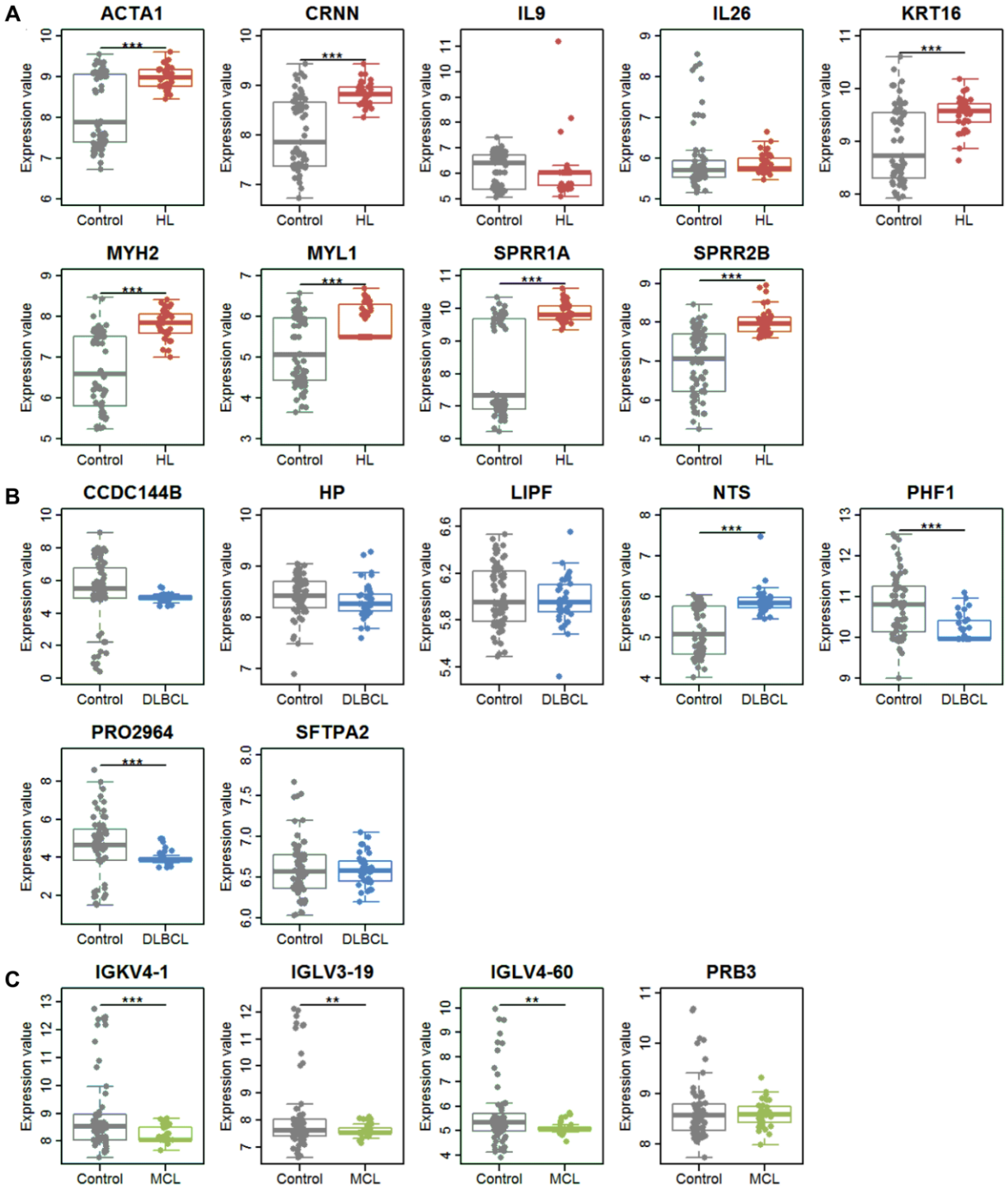
Supplementary Figure 1. Correlation of HL marker genes and prognostic indicators. (**A**) Correlation between HL marker genes and stage. (**B**) Correlation between HL marker genes and International Prognostic Score. The genes were divided into high and low groups based on the median expression value. HL, Hodgkin's lymphoma. Significance: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.



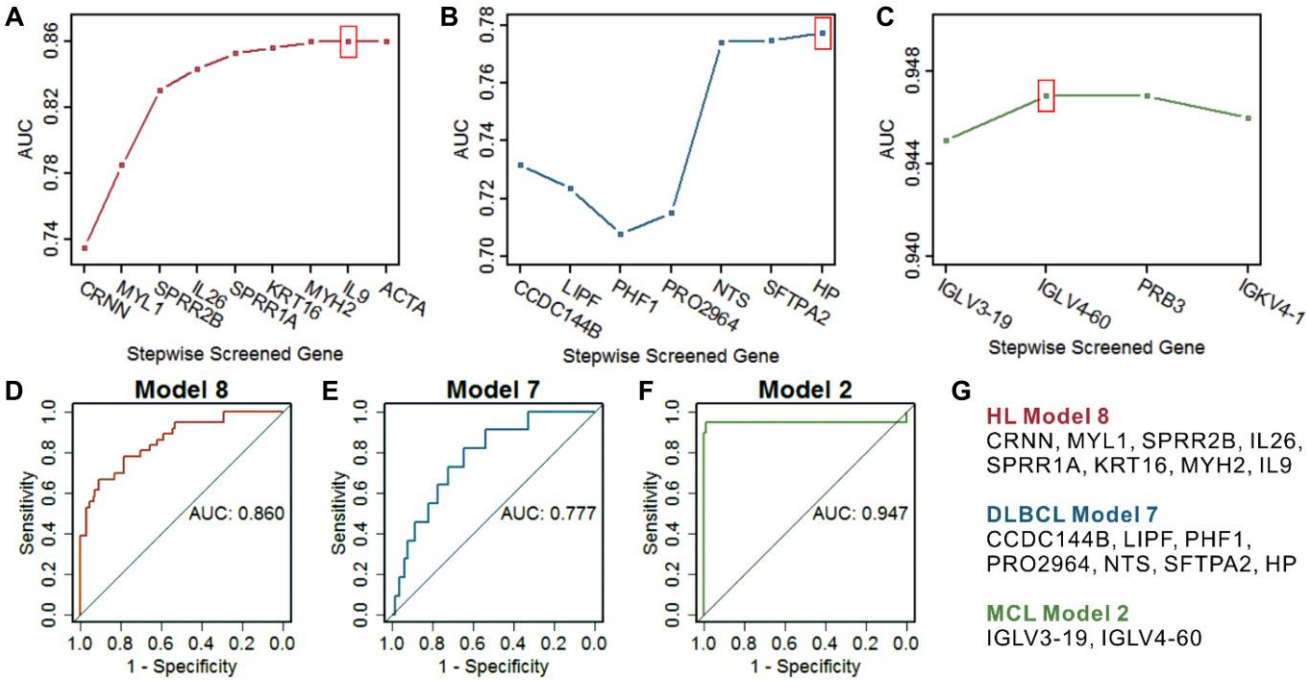
Supplementary Figure 2. Effect of DLBCL marker genes on patient overall survival (**A**) and progression-free survival (**B**). The genes were divided into high and low groups based on the median expression value.



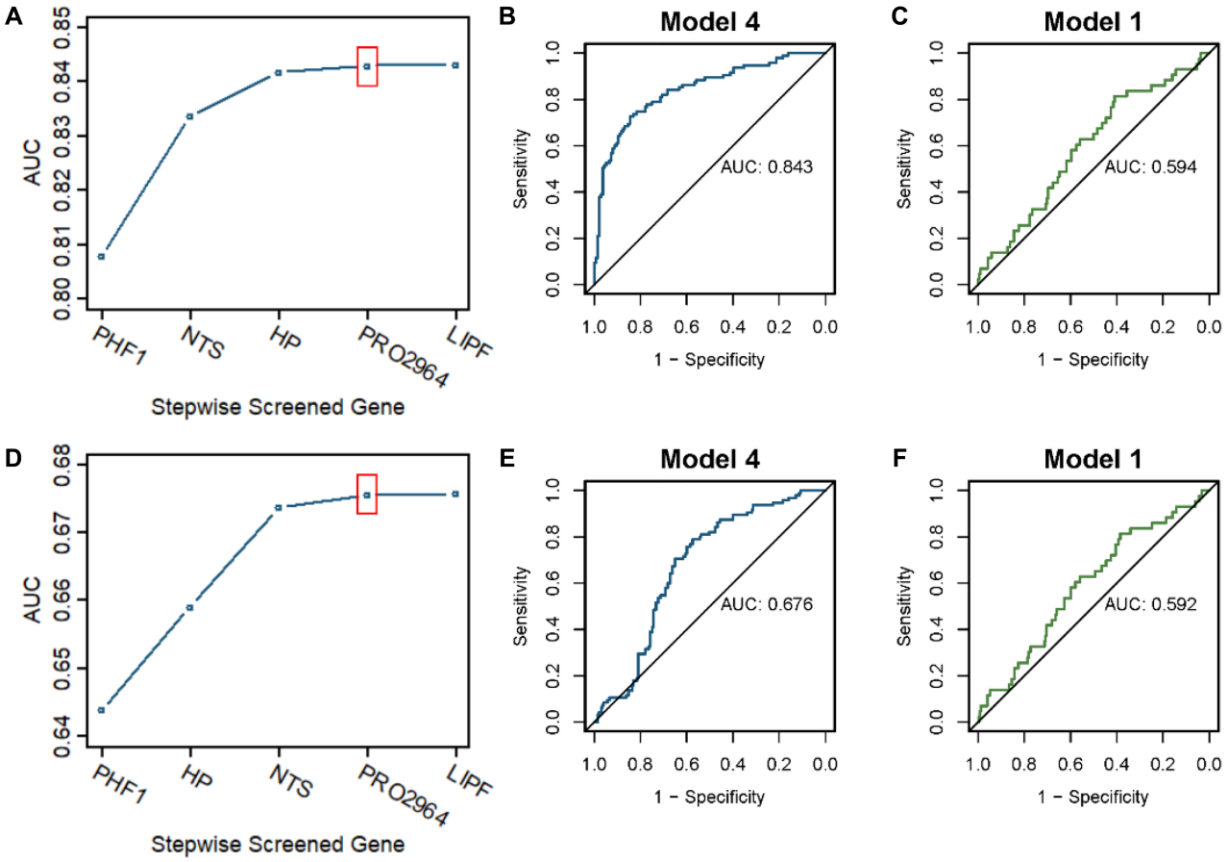
Supplementary Figure 3. The marker genes showed relatively poor specificity for the other two types of lymphomas (corresponding to the optimal model in Figure 4). (**A**) The classification performance of HL marker genes on DLBCL. (**B**) The classification performance of HL marker genes on MCL. (**C**) The classification performance of DLBCL marker genes on HL. (**D**) The classification performance of DLBCL marker genes on MCL. (**E**) The classification performance of MCL marker genes on HL. (**F**) The classification performance of MCL marker genes on DLBCL. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.



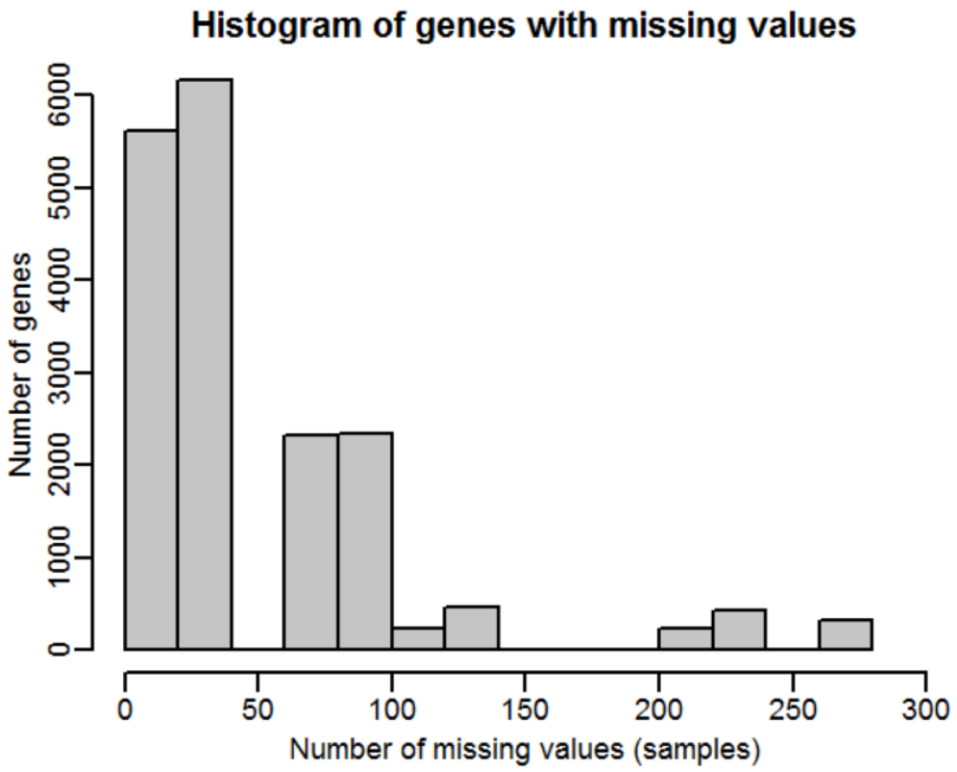
Supplementary Figure 4. Expression of lymphoma marker genes in the data derived from isolated lymphoma cells and normal B cells. The corresponding datasets see in Table 1. (**A**) Expression of HL marker genes between HL samples and controls. (**B**) Expression of DLBCL marker genes between DLBCL samples and controls. (**C**) Expression of MCL marker genes between MCL samples and controls. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma. Significance: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.



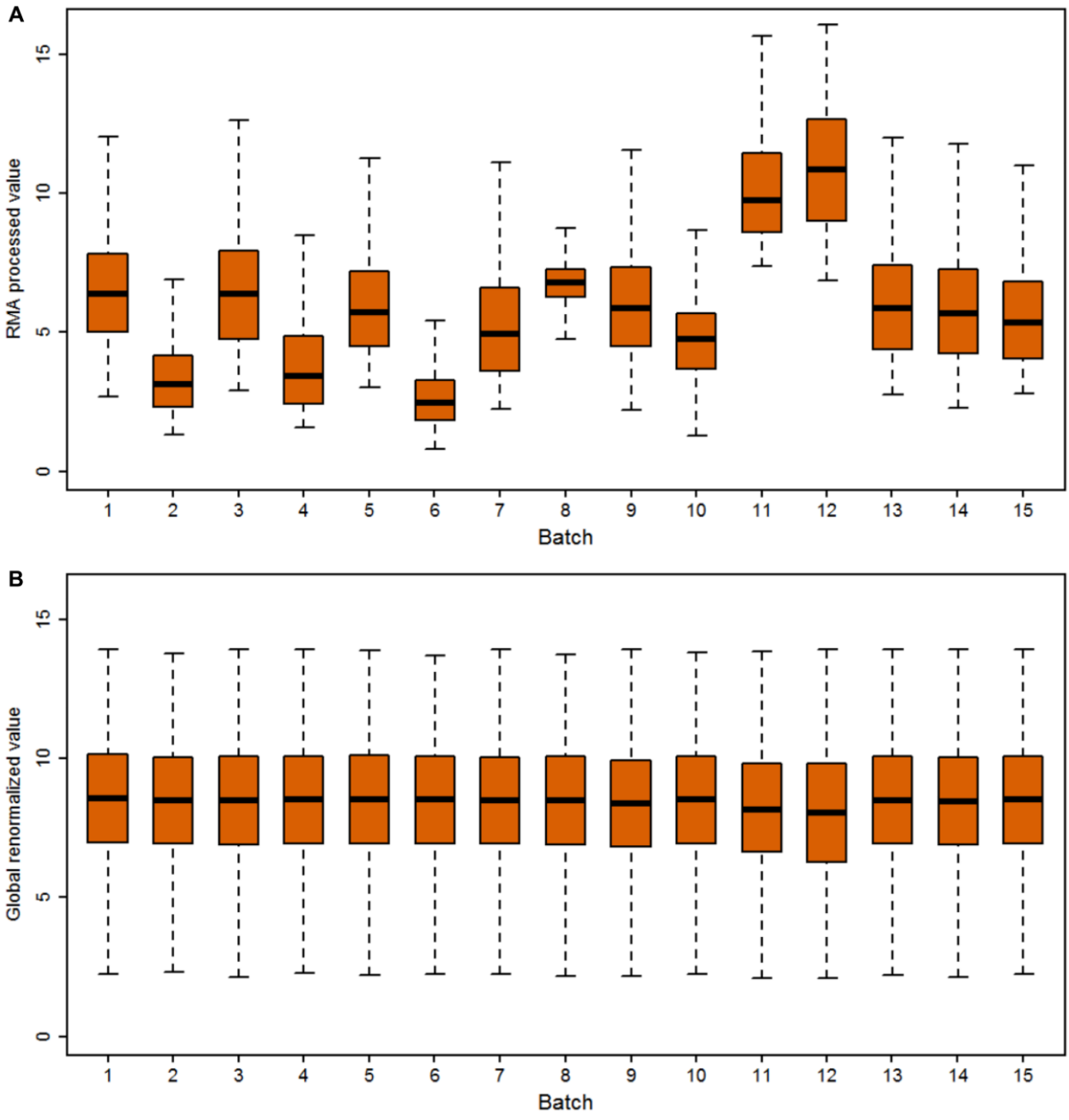
Supplementary Figure 5. Screening of the optimal multigene prediction model for three lymphomas using the data derived from isolated lymphoma cells. The corresponding datasets see in Table 1. (**A**–**C**) Stepwise screened multigene prediction models in HL, DLBCL and MCL. From left to right on the x-axis (stepwise screened genes), each additional gene corresponds to a model [for example, in (A), CRNN represents model 1, which contains one gene of CRNN, MYL1 represents model 2, which contains two genes including CRNN and MYL1]. The red box shows the optimal model for each type of lymphoma. (**D**–**F**) ROC curves of the screened optimal models for each type of lymphoma. (**G**) Genes in the screened optimal models for three lymphomas. HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.



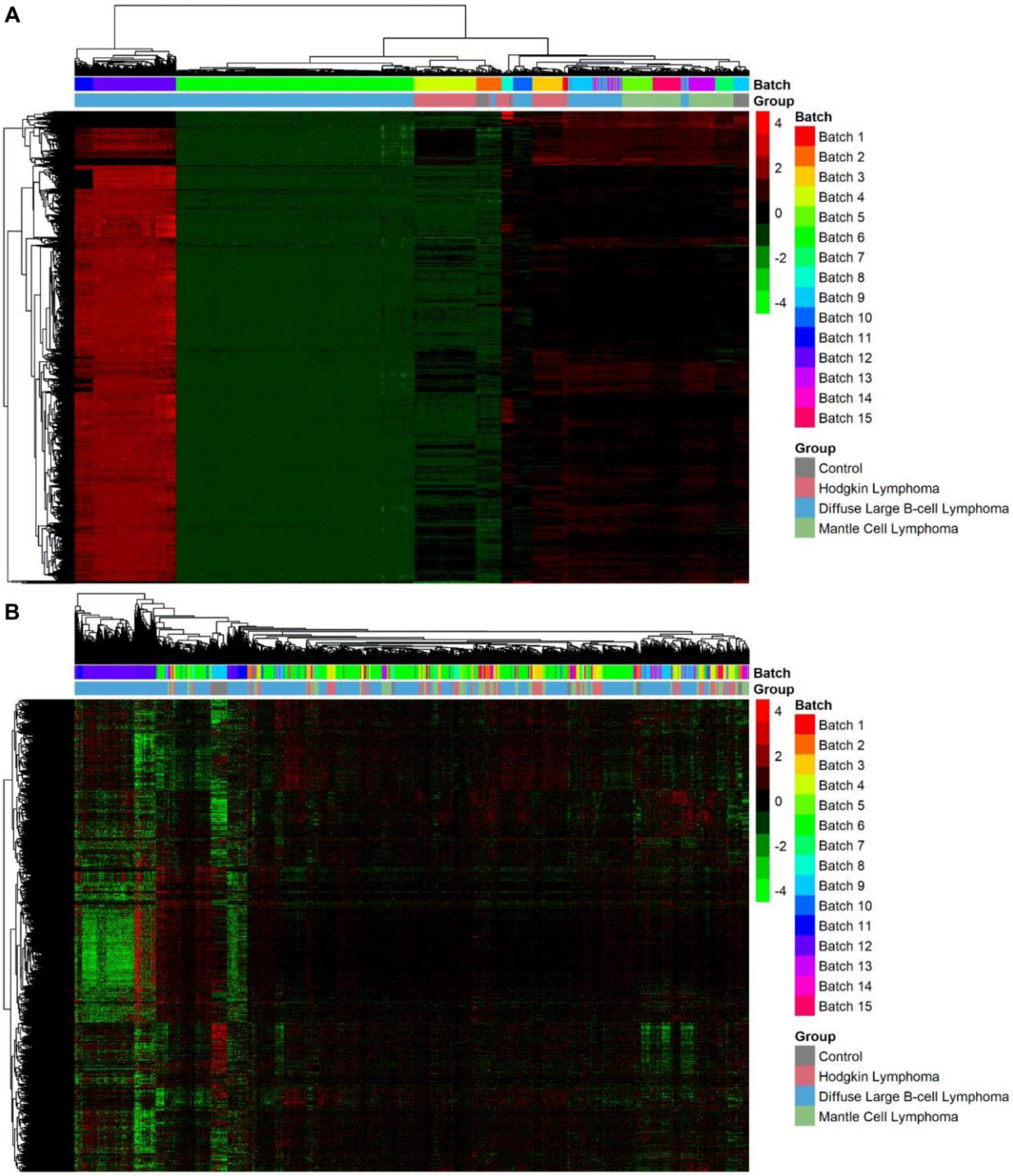
Supplementary Figure 6. The prediction performance of the lymphoma marker genes in the validation dataset of GSE132929. The dataset including Burkitt's lymphoma (BURK), diffuse large B-cell lymphoma (DLBCL), double hit lymphoma (DHL), follicular lymphoma (FL), mantle cell lymphoma (MCL), medial zone lymphoma (MZL), and other high-grade B-cell lymphomas (no Hodgkin's lymphoma (HL) or controls). There were only 5 DLBCL marker genes and 1 MCL marker gene (PRB3) in the GSE132929 dataset. (**A**–**C**) The validation dataset does not include BURK. (**D**–**F**) The validation dataset includes BURK. (**A**, **D**) Stepwise screened multigene prediction models in DLBCL. From left to right on the x-axis (stepwise screened genes), each additional gene corresponds to a model [for example, in (A), PHF1 represents model 1, which contains one gene of PHF1, NTS represents model 2, which contains two genes including PHF1 and NTS]. The red box shows the optimal model for each type of lymphoma. (**B**, **E**) ROC curves of the screened optimal model for DLBCL. (**C**, **F**) ROC curves of the PRB3 model for MCL.



Supplementary Figure 7. Histogram of genes with missing values. The x-coordinate indicates how many samples have missing values. This study collected 1411 samples, and the total number of genes was 18116. The figure shows that most of the samples' expression data are relatively complete (80% of samples have no missing values, 13% of samples have less than 3% missing values, and only 7% of samples have more than 10% missing values).



Supplementary Figure 8. The distribution of the RMA-processed gene expression values (**A**) and the global renormalized gene expression values (**B**) of the lymphoma datasets. Details of these batches see Supplementary Table 4. There was a relatively large deviation in the distribution of gene expression values across these batches in the RMA-processed gene expression values. The distribution of gene expression values across these batches had a consistent range in the global renormalized gene expression values.



Supplementary Figure 9. Heatmap of the gene expression profiles in the integrated (**A**) and the global renormalized (**B**) lymphoma datasets. All gene expression values were z-score converted. There was a strong batch effect in the integrated datasets whereas only a weak batch effect in the global renormalized datasets.

SUPPLEMENTARY MATERIALS

Supplementary Tables

Supplementary Table 1. Enriched GO biological processes of lymphoma specific genes.[1]

ID	Term	FDR	Genes in GOBP
Hodgkin's lymphoma			
GO:0070268	cornification	7.43E–08	KRT24, SPRR2B, SPRR1A, KRT16, PI3
GO:0018149	peptide cross-linking	2.71E–06	SPRR2B, SPRR1A, PI3
GO:0006936	muscle contraction	4.10E–06	MYH2, ACTA1, MYL1, MYOT
GO:0030049	muscle filament sliding	5.09E–06	MYH2, ACTA1, MYL1
GO:0030216	keratinocyte differentiation	1.15E–05	SPRR2B, SPRR1A, KRT16
GO:0008544	epidermis development	2.00E–05	SPRR2B, SPRR1A, KRT16
GO:2000648	positive regulation of stem cell proliferation	2.64E–04	SOX11
GO:0071305	cellular response to vitamin D	2.64E–04	PHEX
GO:0010226	response to lithium ion	2.64E–04	ACTA1
GO:0060174	limb bud formation	3.16E–04	SOX11
Diffuse large B-cell lymphoma			
GO:0021846	cell proliferation in forebrain	1.24E–05	FGF8, LHX5
GO:0007512	adult heart development	3.71E–05	MYH6, MYH7
GO:0014898	cardiac muscle hypertrophy in response to stress	4.62E–05	MYH6, MYH7
GO:0019226	transmission of nerve impulse	6.86E–05	CACNG7, CNTNAP2
GO:2000311	regulation of AMPA receptor activity	9.70E–05	CACNG7, SHANK1
GO:0071625	vocalization behavior	9.70E–05	CNTNAP2, SHANK1
GO:0002026	regulation of the force of heart contraction	1.14E–04	MYH6, MYH7
GO:0035176	social behavior	1.38E–04	CNTNAP2, TH, SHANK1
GO:0006941	striated muscle contraction	1.53E–04	MYH6, MYH7
GO:0006936	muscle contraction	1.68E–04	MYH13, GALR2, MYH6, MYH7
Mantle cell lymphoma			
GO:0002377	immunoglobulin production	4.62E–07	IGLV4-60, IGKV4-1
GO:0006956	complement activation	3.92E–06	IGKV4-1, IGLV3-19
GO:0006958	complement activation, classical pathway	6.29E–06	IGKV4-1, IGLV3-19
GO:0030449	regulation of complement activation	7.87E–06	IGKV4-1, IGLV3-19
GO:0038096	Fc-gamma receptor signaling pathway involved in phagocytosis	1.75E–05	IGKV4-1, IGLV3-19
GO:0048333	mesodermal cell differentiation	3.59E–05	INHBA
GO:0071372	cellular response to follicle-stimulating hormone stimulus	3.59E–05	INHBA
GO:2001241	positive regulation of extrinsic apoptotic signaling pathway in absence of ligand	4.13E–05	INHBA
GO:0061029	eyelid development in camera-type eye	4.70E–05	INHBA
GO:0071397	cellular response to cholesterol	5.28E–05	INHBA

[1]All GO biological process enrichment results are sorted by FDR and showed the top 10 enriched results for each lymphoma.

Supplementary Table 2. Enriched GO biological processes of lymphoma marker genes.[1]

ID	Term	FDR	Genes in GOBP
Hodgkin's lymphoma			
GO:0045407	positive regulation of interleukin-5 biosynthetic process	<1.00E–16	IL9
GO:0030049	muscle filament sliding	1.77E–07	ACTA1, MYH2, MYL1
GO:0030216	keratinocyte differentiation	4.02E–07	KRT16, SPRR1A, SPRR2B
GO:0008544	epidermis development	9.04E–07	KRT16, SPRR1A, SPRR2B
GO:0070268	cornification	2.52E–06	KRT16, SPRR1A, SPRR2B
GO:0006936	muscle contraction	2.56E–06	ACTA1, MYH2, MYL1
GO:0043503	skeletal muscle fiber adaptation	2.56E–06	ACTA1
GO:0009991	response to extracellular stimulus	6.71E–06	ACTA1
GO:0018149	peptide cross-linking	6.71E–06	SPRR1A, SPRR2B
GO:0071417	cellular response to organonitrogen compound	1.17E–05	ACTA1
Diffuse large B-cell lymphoma			
GO:2000296	negative regulation of hydrogen peroxide catabolic process	1.71E–06	HP
GO:0061086	negative regulation of histone H3-K27 methylation	6.15E–06	PHF1
GO:0061087	positive regulation of histone H3-K27 methylation	7.32E–06	PHF1
GO:0051354	negative regulation of oxidoreductase activity	7.32E–06	HP
GO:0050880	regulation of blood vessel size	2.30E–05	NTS
GO:0006108	malate metabolic process	2.56E–05	LIPF
GO:0045814	negative regulation of gene expression, epigenetic	1.19E–04	PHF1
GO:0007585	respiratory gaseous exchange	1.32E–04	SFTPA2
GO:0006953	acute-phase response	1.42E–04	HP
GO:0010942	positive regulation of cell death	1.61E–04	HP
Mantle cell lymphoma			
GO:0002377	immunoglobulin production	2.49E–08	IGKV4-1, IGLV4-60
GO:0006956	complement activation	3.85E–07	IGKV4-1, IGLV3-19
GO:0006958	complement activation, classical pathway	5.55E–07	IGKV4-1, IGLV3-19
GO:0006955	immune response	6.35E–07	IGKV4-1, IGLV3-19, IGLV4-60
GO:0030449	regulation of complement activation	6.35E–07	IGKV4-1, IGLV3-19
GO:0038096	Fc-gamma receptor signaling pathway involved in phagocytosis	1.43E–06	IGKV4-1, IGLV3-19
GO:0050776	regulation of immune response	5.29E–06	IGKV4-1, IGLV3-19
GO:0038095	Fc-epsilon receptor signaling pathway	5.29E–06	IGKV4-1, IGLV3-19
GO:0050900	leukocyte migration	7.66E–06	IGKV4-1, IGLV3-19
GO:0006898	receptor-mediated endocytosis	8.45E–06	IGKV4-1, IGLV3-19

[1]All GO biological process enrichment results are sorted by FDR and showed the top 10 enriched results for each lymphoma.

Supplementary Table 3. Logistic regression between marker genes and lymphomas.

Variable	Crude		Adjusted[1]		Adjusted top 10 significantly correlated genes[1]
	OR (95% CI)	*P*	OR (95% CI)	*P*	
Hodgkin's Lymphoma					
MYH2	1.99 (1.75–2.27)	< 0.001	2.55 (1.97–3.29)	< 0.001	MYL1, ACTA1, KBTBD10, MB, CKM, MYH1, SMPX, CSRP3, MYBPC1, ASB5
SPRR2B	1.77 (1.59–1.98)	< 0.001	1.25 (0.97–1.60)	0.084	SPRR1A, S100A7, SPRR1B, SPRR3, KRT6A, S100A2, KRT6B, CLCA2, CRCT1, CRNN
MYL1	1.89 (1.65–2.18)	< 0.001	1.43 (1.00–2.04)	0.048	KBTBD10, MB, MYBPC1, SMPX, MYH1, MYH2, MYOT, CSRP3, TNNT1, MYL2
SPRR1A	1.61 (1.48–1.74)	< 0.001	1.96 (1.26–3.05)	0.003	KRT13, KRT6A, KRT6B, S100A7, SPRR1B, SPRR2B, SPRR3, S100A2, SPINK5, KRT14
CRNN	1.68 (1.47–1.92)	< 0.001	1.95 (1.41–2.69)	< 0.001	CLCA2, DSG3, TGM3, KRT6B, CRCT1, SPINK7, KRT16, RHCG, TMPRSS11D, KRT78
ACTA1	1.71 (1.50–1.95)	< 0.001	1.51 (1.13–2.03)	0.006	CKM, MYH2, MYL2, CA3, MYH1, MYLPF, TNNC1, TNNC2, MYH7, XIRP2
IL9	0.39 (0.35–0.44)	< 0.001	0.40 (0.33–0.48)	< 0.001	IL13RA2, IL26, AMPH, DHRS2, CCL11, CLC, TFPI2, MFAP2, CYP4Z1, SCG2
KRT16	1.51 (1.35–1.69)	< 0.001	1.78 (1.31–2.43)	< 0.001	KRT6B, LY6D, DSG3, CLCA2, CRNN, RHCG, TGM3, KRT14, SERPINB2, KRT17
IL26	0.39 (0.33–0.47)	< 0.001	1.11 (0.87–1.41)	0.404	IL9, CYP4Z1, CLC, IL13RA2, IL22, DHRS2, COL6A6, CCL26, AMPH, CCL23
Diffuse Large B-cell Lymphoma					
LIPF	0.08 (0.06–0.11)	< 0.001	0.38 (0.25–0.60)	< 0.001	GKN1, CTSE, SFTPA2, OLFM4, DEFA6, KRT20, TMEM183A, MUC7, GKN2, RBP2
SFTPA2	0.20 (0.17–0.24)	< 0.001	0.21 (0.16–0.29)	< 0.001	LIPF, CTSE, GKN1, SLC34A2, TMEM183A, KRT4, MUC7, DEFA6, OLFM4, KRT20
CCDC144B	0.48 (0.41–0.55)	< 0.001	0.46 (0.38–0.54)	< 0.001	TAF13, LOC100287927, ZNF90, GTF2I, PTENP1, PKD2L2, UGGT1, HIST1H4E, PTMS, LOC100509761
HP	0.31 (0.27–0.37)	< 0.001	0.36 (0.30–0.43)	< 0.001	ALB, FGA, AHSG, FGL1, PRG4, FGB, TTR, GC, FGG, C4BPA
PRO2964	0.53 (0.47–0.59)	< 0.001	0.65 (0.57–0.73)	< 0.001	HIST1H2BM, PTENP1, RAB30, HIST1H4B, ZNF814, ST6GAL1, HIST2H2AA3, HIST1H3I, SKIL, LOC100129112
NTS	1.94 (1.70–2.21)	< 0.001	1.85 (1.56–2.19)	< 0.001	CLEC4M, NPY1R, MMRN1, CCL20, MATN2, CETP, CCL21, TFPI, TSPAN7, SDPR
PHF1	0.64 (0.58–0.70)	< 0.001	0.63 (0.58–0.70)	< 0.001	RPSA, RPLP0, RPL35, RPL30, RPS12, RPL3, COX6B1, RPS29, RPLP1, RPS15
Mantle Cell Lymphoma					
IGLV3-19	0.48 (0.43–0.53)	< 0.001	0.16 (0.12–0.20)	< 0.001	IGLJ3, IGLV4-60, IGLV2-23, IGLV1-40, IGKV3-20, IGKV1-5, LOC100508797, IGKV1D-8, LOC100287927, IGLV6-57
IGKV4-1	0.58 (0.53–0.63)	< 0.001	0.29 (0.23–0.37)	< 0.001	IGKV1-5, IGLV2-23, IGLJ3, IGKV3-20, IGLV1-40, IGKC, SDC1, IGKV1D-8, IGLV3-19, IGLV4-60
PRB3	0.55 (0.49–0.61)	< 0.001	0.19 (0.13–0.26)	< 0.001	PRB4, PRB1, C20orf70, HTN1, CRISP3, AZGP1, LPO, CST5, PIP, TCN1
IGLV4-60	0.46 (0.40–0.53)	< 0.001	0.28 (0.20–0.39)	< 0.001	IGLV1-40, IGKV3-20, IGLJ3, LOC100287927, IGKV1-5, SDC1, IGLV3-19, IGLV6-57, IGKV1D-8, IGLV2-23

[1]Pearson correlation analysis was used to calculate the correlation between marker genes all other genes in the whole samples. A false discovery rate (FDR) corrected *P*-value ≤ 0.05 was considered as significant correlated. The top 10 significant correlated based on FDR *P*-value were used for covariates correction in logistic regression analysis.

Supplementary Table 4. Grouping the lymphoma datasets for data integration and global renormalization.

GEO ID	Category	Platform	Batch
GSE7788	HL, Control	GPL570	1
GSE12453	HL, DLBCL, Control	GPL570	2
GSE13996	HL	GPL571	3
GSE17920	HL	GPL570	4
GSE21452	MCL	GPL570	5
GSE31312	DLBCL	GPL570	6
GSE36000	MCL	GPL570	7
GSE47044	HL, Control	GPL6244	8
GSE56315	DLBCL, Control	GPL570	9
GSE64555	DLBCL	GPL570	10
GSE69053	DLBCL	GPL8432	11
GSE69053	DLBCL	GPL14951	12
GSE70910	MCL	GPL570	13
GSE86613	DLBCL	GPL570	14
GSE93291	MCL	GPL570	15

Abbreviations: HL, Hodgkin's lymphoma; DLBCL, diffuse large B-cell lymphoma; MCL, mantle cell lymphoma.

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Original Article

Combination of cetuximab with met inhibitor in control of cetuximab-resistant oral squamous cell carcinoma

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Abstract: *Objective:* To investigate the underlying molecular mechanisms contributing to oral squamous cell carcinoma (OSCC) cell resistance to the epidermal growth factor receptor (EGFR) inhibitor. *Materials and methods:* OSCC cell lines HSC-2 and HSC-3 were assessed *in vitro* for drug treatment, cell viability, and gene expression and the online gene expression in OSCC tissues was analyzed for association with OSCC prognosis. *Results:* HSC-2 and HSC-3 cells expressed high EGFR levels, but hepatocyte growth factor (HGF) treatment induced cetuximab resistance, whereas the Met inhibitor PHA-665752 as well as Met siRNA was able to restore OSCC cell sensitivity to cetuximab. HGF treatment induced tumor cells to express p-Akt and p-ERK1/2. In contrast, the activity of Akt and ERK1/2 was suppressed by treatment with PHA-665752, Met siRNA, or their combination. Furthermore, Met was highly expressed in OSCC tissues and associated with a poor patient survival, while Met/HGF-activated Akt also was associated with a poor patient survival. *Conclusions:* This study demonstrates that Met/HGF expression results in OSCC resistance to cetuximab and tumor recurrence after cetuximab therapy; thus, inhibition of Met/HGF activity could restore OSCC sensitivity to cetuximab.

Keywords: Oral squamous cell carcinoma, cetuximab resistance, HGF, Met, Akt, ERK1/2

Introduction

Head and neck cancer, which occurs in the oral cavity, nose, throat, larynx, sinuses, salivary glands, nasopharynx, or hypopharynx, significantly affects the quality of life of patients [1]. Oral squamous cell carcinoma (OSCC) is a commonly diagnosed head and neck squamous cell carcinoma (HNSCC). The risk factors include tobacco smoking and alcohol consumption [2] as well as excessive consumption of processed meats and red meat [3], human papillomavirus infection [4], and Frequently chewing betel nuts [5]. OSCC management is usually surgery, chemoradiotherapy, targeted therapy, or photodynamic therapy [6, 7]. Early-stage OSCC has a favorable prognosis after treatment, but drug resistance, disease recurrence, and metastasis result in overall 5-year survival rates ≤ 50% [8]. Targeted therapy is a hot topic for the treatment of OSCC, and clinical drugs used frequently include epidermal growth factor receptor (EGFR) inhibitors, such as cetuximab, bevacizumab, and erlotinib, which have shown improvement of OSCC patient survival [9, 10]. Indeed, EGFR is a transmembrane receptor tyrosine kinase that is highly overexpressed in head and neck, lung, and breast cancers [11, 12]. EGFR promotes cancer development by increasing cell growth, migration, and survival [13]. Cetuximab is a chimeric monoclonal antibody that can bind to EGFR and in turn inhibit EGFR tyrosine kinase activity to suppress EGFR-positive cancer progression [14]. In the treatment of HNSCC, cetuximab has gained much attention as a novel therapeutic strategy and was approved by the European Medicines Agency in 2004 and the US Food and Drug

Administration (FDA) in 2006 [15]. Unfortunately, long-term cetuximab treatment results in HNSCC drug resistance, even in tumors with high EGFR expression [16, 17]. Thus, a better understanding of the underlying molecular events of this drug resistance is critical to restore the sensitivity of cancer cells to cetuximab [18-20]. To date, there is increasing evidence implicating that receptor tyrosine kinases play a pivotal role in regulating cetuximab resistance in colon and lung cancers [21, 22], and the combined treatment with receptor tyrosine kinase inhibitors may overcome cetuximab resistance [22].

Hepatocyte growth factor (HGF) plays a crucial role in cell motility, growth, and morphogenesis through binding to hepatocyte growth factor receptor (also known as Met) to activate the receptor tyrosine kinase cascade, which is related to cancer development [23-25]. HGF is highly expressed in OSCC, and the HGF/Met signaling pathway has been shown to induce OSCC cell migration, invasion, and metastasis through lamellipodia and filopodia formation [26] and the destruction of E-cadherin [27]. The cross-talk of Met with other signaling proteins, like EGFR, vascular endothelial growth factor receptor, and Wnt, also has been revealed in various cell lines [28-30], indicating the role of HGF/Met in cancer development. In addition, it has been shown that Met activation during cetuximab treatment of recurrent and metastatic HNSCC is associated with poor outcomes [31].

In this study, we investigated the underlying molecular events contributing to OSCC cell resistance to cetuximab using an *in vitro* OSCC cell line model. Our data provide novel insight into a therapeutic strategy to restore cetuximab sensitivity of OSCC.

Materials and methods

Cell culture

The human OSCC cell lines HSC-2 and HSC-3 (Immuno-Biological Laboratories Co. Shanghai, China) were maintained in Dulbecco's modified Eagle medium (DMEM; Gibco, Gaithersburg, MD, USA) supplemented with 10% fetal bovine serum (Gibco), 100 units/mL penicillin, and 100 μg/mL streptomycin (Gibco) in a humidified incubator with 5% CO_2 at 37°C.

Cell viability assay

Cell viability was measured by using the 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) method. In brief, 2×10^3 tumor cells/well were seeded into 96-well plates and grown for 24 h and then treated with various concentrations of cetuximab (Merck, Germany), PHA-665752 (Selleck Chemicals, TX, USA), HGF (R&D Systems, Minneapolis, MN, USA), and a goat anti-human HGF neutralizing antibody, normal goat IgG (R&D Systems) for 72 h. At the end of each experiment, 50 μL of MTT solution (2 mg/mL; Sigma Chemicals, St. Louis, MO, USA) was added to each well of the cell culture plate, and the cells were incubated for an additional 2 h. The culture medium was replaced with 100 μL of dimethyl sulfoxide. The absorbance rate was measured with a microplate reader (BioTek, Winooski, VT, USA) at a wavelength of 490 nm. The percentage of cell viability was calculated by comparison to untreated controls.

Western blot

A western blot assay was performed, as described previously [32]. Briefly, cells after treatment were lysed in cell lysis buffer, and the protein concentration was measured using a bicinchoninic acid protein assay kit (Beyotime Biotechnology, Shanghai, China). Each total protein sample (30 μg per lane) was separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred onto a polyvinylidene fluoride membrane (Millipore, Billerica, MA, USA). After that, the membranes were blocked by 5% skimmed dry milk solution in phosphate-buffered saline (PBS) at room temperature for 1 h and then incubated with a primary antibody at 4°C overnight (**Table 1**). On the following day, the membranes were washed briefly three times with PBS-Tween 20 and then incubated with the corresponding horseradish peroxidase-conjugated secondary antibody at room temperature for 1 h. The target protein bands were visualized by ECL plus western blotting detection reagent and exposed to X-ray films.

RNA interference (RNAi) assay

For the RNAi assay, Duplexed Stealth RNAi targeting ErbB3 and Met or Stealth RNAi Negative Control Low GC Duplex #3 were purchased from Invitrogen (Carlsbad, CA, USA). OSCC cells

Table 1. Antibodies used for Western blot

Antibody	Cat #	Source	Dilution
Anti-EGFR	R&D Systems, AF231	Rabbit	1 μg/mL
Anti-phospho-EGFR	Abcam, ab32578	Rabbit	1:1000
Anti-Met	CST, 3127	Mouse	1:1000
Anti-phospho-Met	Abcam, ab47402	Rabbit	1:1000
Anti-Akt	CST, 9272	Rabbit	1:1000
Anti-phospho-Akt	CST, 4060	Rabbit	1:1000
Anti-ErbB3	CST, 4754	Rabbit	1:1000
Anti-phospho-ErbB3	CST, 4791	Rabbit	1:1000
Anti-ERK1/ERK2	R&D Systems, AF1576	Rabbit	0.2 μg/mL
Anti-phospho-ERK1/ERK2	R&D Systems, AF1018	Rabbit	0.1 μg/mL

CST, Cell Signaling Technology (Danvers, MA, USA); Abcam, Cambridge, MA, USA.

were seeded into 24-well plates at a density of 2×10^4 per well in 400 μL of antibiotic-free DMEM and grown overnight. On the next day, the cells were transfected with siRNA (50 μmol) using Lipofectamine 2000 (Invitrogen) for 24 h, according to the manufacturer's instructions, then washed with ice-cold PBS and reseeded into 96-well plates for treatment with cetuximab (1 μg/mL) and/or recombinant human HGF (20 ng/μL) for 72 h, and finally subjected to the MTT assay or another assay. The siRNA sequences targeting ErbB3 and Met were 5'-GGCCAUGAAUGAAUUCUCUACUCUA-3' and 5'-UCCAGAAGAUCAGUUUCCUAAUUCA-3', respectively.

Online gene expression database and survival analysis

Transcriptome data of OSCC tissues were retrieved from the NCBI-GEO database with the series ID of GSE42743 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE42743). R statistical software v3.3.3 (https://www.rproject.org/) was used for data analysis. In brief, we utilized the Robust Multichip Average algorithm in the "oligo" package [33] to normalize the raw data on gene expression and generated the normalized expression matrix. The gene annotation and integration of the expression matrix was conducted by using a custom-designed Python code, according to a previous study [34]. We then removed probes without any gene annotations or those that matched multiple gene symbols. After that, we calculated the average expression value for each gene when there were multiple probe IDs that matched one official gene symbol and made this value to represent the expression intensity of the corresponding gene symbol. Differentially expressed genes were then obtained by using the empirical Bayes method in the "limma" package [35]. The upregulated genes were considered to be those with logarithmic transformed fold-change (log2FC) ≥ 1, and the downregulated genes were considered to be log2FC ≤ -1. A false discovery rate-adjusted P value ≤ 0.05 indicated statistical significance. All survival analyses were conducted using the "survival" package in R. Kaplan-Meier survival curves were used to show the prognostic differences between two groups.

Statistical analysis

All data were expressed as the mean ± standard deviation of triplicate experiments. The two-tailed unpaired Student's t test or the one-way analysis of variance test was used to determine the p values; $P < 0.05$ was considered as a significant difference.

Results

HGF induction of OSCC cell resistance to cetuximab by an increase in akt and ERK1/2 phosphorylation

First, we detected the levels of EGFR and c-Met expression in the OSCC cell lines HSC-2 and HSC-3 using western blot. We found that both cell lines significantly expressed EGFR and MET (**Figure 1A**). We then assessed the effect of cetuximab on OSCC viability using the MTT assay and found that HSC-2 and HSC-3 cell growth was moderately inhibited by cetuximab in a dose-dependent manner (**Figure 2B**). To obtain cetuximab resistance, we cotreated HSC-2 and HSC-3 cells with cetuximab (5 μg/mL) and HGF (20 μg/mL); however, we did not find a significant change in cell viability (**Figure 1B**), indicating that both OSCC cell lines became resistant to cetuximab after HGF treatment. At the protein level, HGF treatment induced Met activity (**Figure 1C**), which could be partially blocked by cetuximab treatment. Specifically, cetuximab treatment downregulated the phosphorylation of Akt and ERK1/2, but it recovered back to the previous levels after the addition of

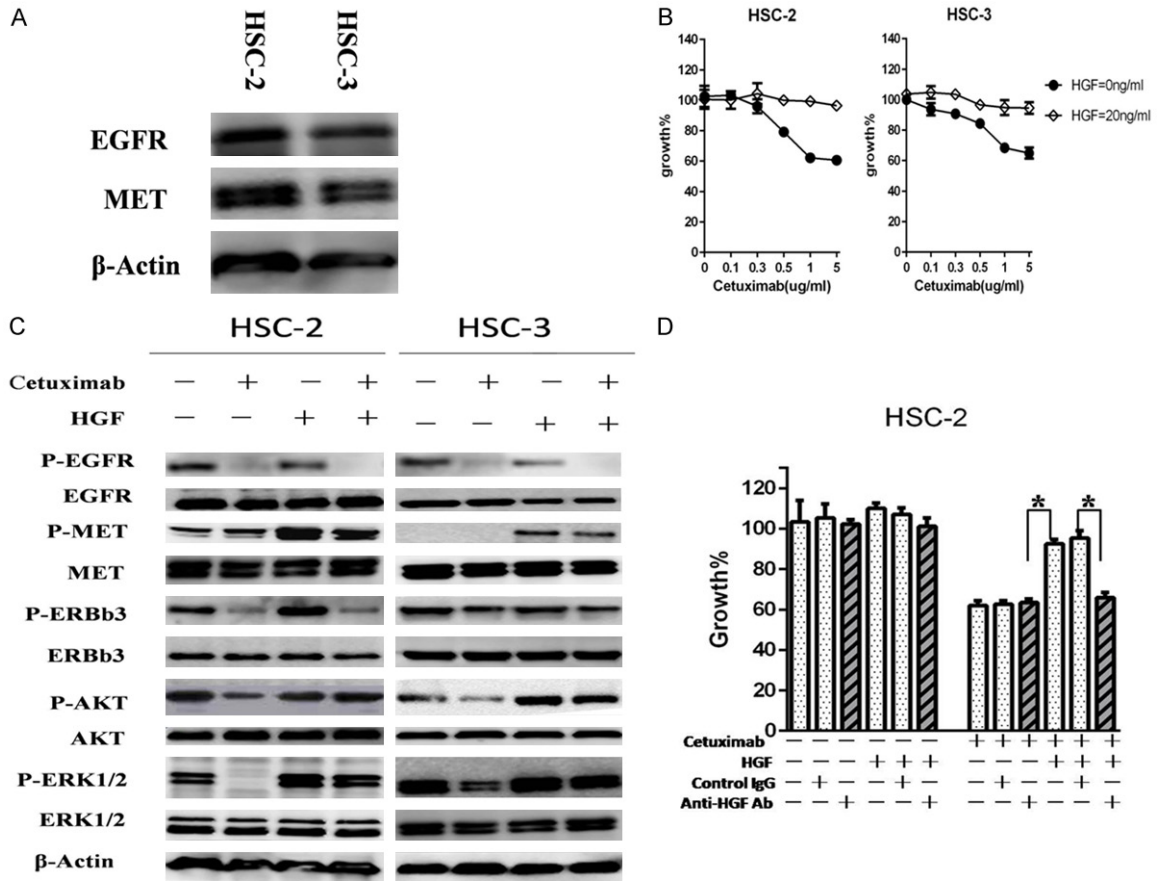

Figure 1. HGF induction of OSCC resistance to cetuximab. A. Western blot. HSC-2 and HSC-3 cells were subjected to protein extraction and showed high protein expression levels of EGFR and MET by western blot. B. The MTT assay. HSC-2 and HSC-3 cells were treated with different doses of cetuximab (0-5 μg/mL) and HGF (20 ng/mL) for 72 h, and the cell proliferation levels were shown by the MTT assay. C. Western blot. HSC-2 and HSC-3 cells were pretreated with control, HGF (20 ng/mL), cetuximab (1 μg/mL), or their combination. D. The MTT assay. HSC-2 cells treated with HGF (20 ng/mL) and the HGF neutralizing antibody or cetuximab (1 μg/mL) for 72 h showed different proliferation abilities by the MTT assay. *$P < 0.05$.

HGF (**Figure 1C**). Under these conditions, p-ErbB3 was dramatically inhibited with cetuximab, even after the addition of HGF, indicating that p-ErbB3 is the target of cetuximab but not involved in in the HGF-induced cetuximab resistance of OSCC cells; whereas the Met, PI3K/Akt, and MAPK pathways may be involved in the HGF-induced cetuximab resistance of OSCC cells. For further confirmation, we pretreated these cells with the HGF neutralizing antibody to block HGF signaling and found that HSC-2 cells regained their sensitivity to cetuximab treatment (**Figure 1D**).

Met inhibitor restoration of cetuximab sensitivity of HGF-treated OSCC cells by decreasing akt and ERK1/2 phosphorylation

The biological effects of HGF are through binding to its receptor, Met; thus, we treated the cells with the Met inhibitor PHA-665752 in the same setting. Our MTT assay data showed that PHA-665752 significantly reduced the tumor cell viability in a dose-dependent manner (**Figure 2A**). Without the stimulus of HGF, PHA-665752 and cetuximab together strongly induced inhibition of cell proliferation, even at a low dose of PHA-665752 (< 1 μM). After stimulus with HGF, the OSCC cells showed cetuximab resistance in low-dose PHA-665752-treated cells, but the cell viability was dramatically reduced after the PHA-665752 dose reached 1 μM, which was similar to that without the HGF stimulus.

To evaluate the mechanism by which HGF rescues OSCC cancer cells from cetuximab-induced inhibition of proliferation, we determined activation of the downstream signaling molecules and found that cetuximab not only

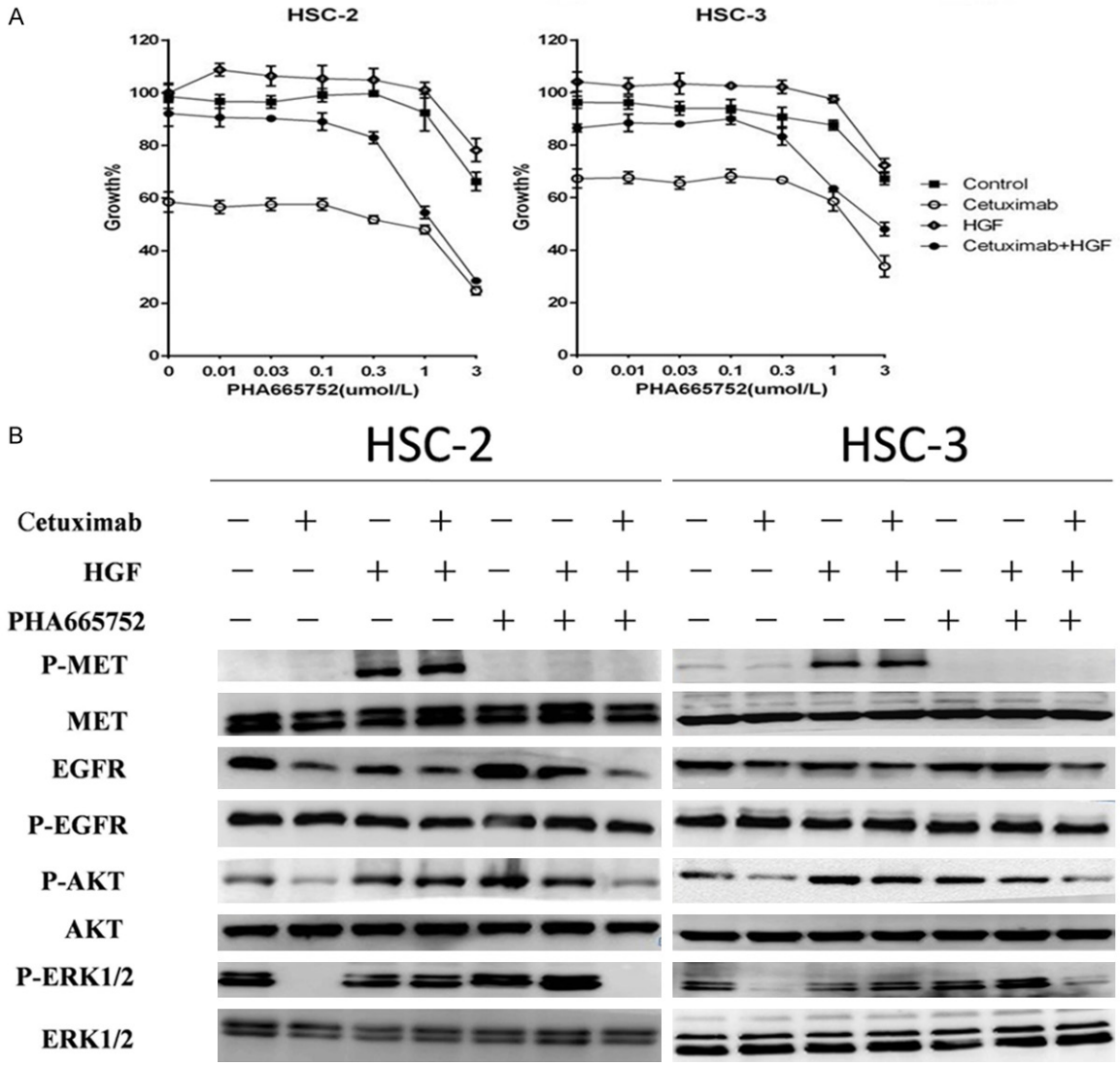

Figure 2. Treatment with the Met inhibitor restored OSCC cell sensitivity to cetuximab in HGF-induced cetuximab-resistant OSCC cells and decreased Akt and ERK1/2 phosphorylation. (A) The MTT assay. HSC-2 and HSC-3 cells were grown and treated with different doses of PHA-665752 or pretreated with or without HGF (20 ng/mL) and cetuximab (1 μg/mL). (B) Western blot. The cells described in (A) were subjected to western blotting.

inhibited EGFR phosphorylation, but also AKT and ERK1/2 activation in HSC-2 and HSC-3 cells. Treating the cells with a combination of cetuximab with HGF led to MET phosphorylation and reactivation of AKT and ERK1/2. However, when treating the cells with a combination of cetuximab with PHA-665752, inhibition of EGFR and MET phosphorylation led to sustained inhibition of the AKT and MAPK pathways in the presence of HGF (**Figure 2B**).

Met siRNA restoration of cetuximab sensitivity of HGF-treated OSCC cells

Next, we assessed whether Met siRNA possesses the same effects as PHA-665752 on HSC-2 cells; it is because cetuximab had even better inhibitory effects on the tumor cells and ErbB3 siRNA was used as a negative control since ErbB3 is not involved in this effect. The western blot data showed that Met siRNA was able to knock down Met expression in OSCC cells, compared to scrambled and ErbB3 siRNA transfection (**Figure 3A**). Moreover, Met siRNA reduced the OSCC cell viability and reversed the cetuximab resistance induced by HGF treatment, while ErbB3 siRNA did not show these effects (**Figure 3B**). At the protein level, p-Akt and p-ERK1/2 expression was blocked by cetuximab treatment and was restored partially by HGF; however, the expression of these pro-

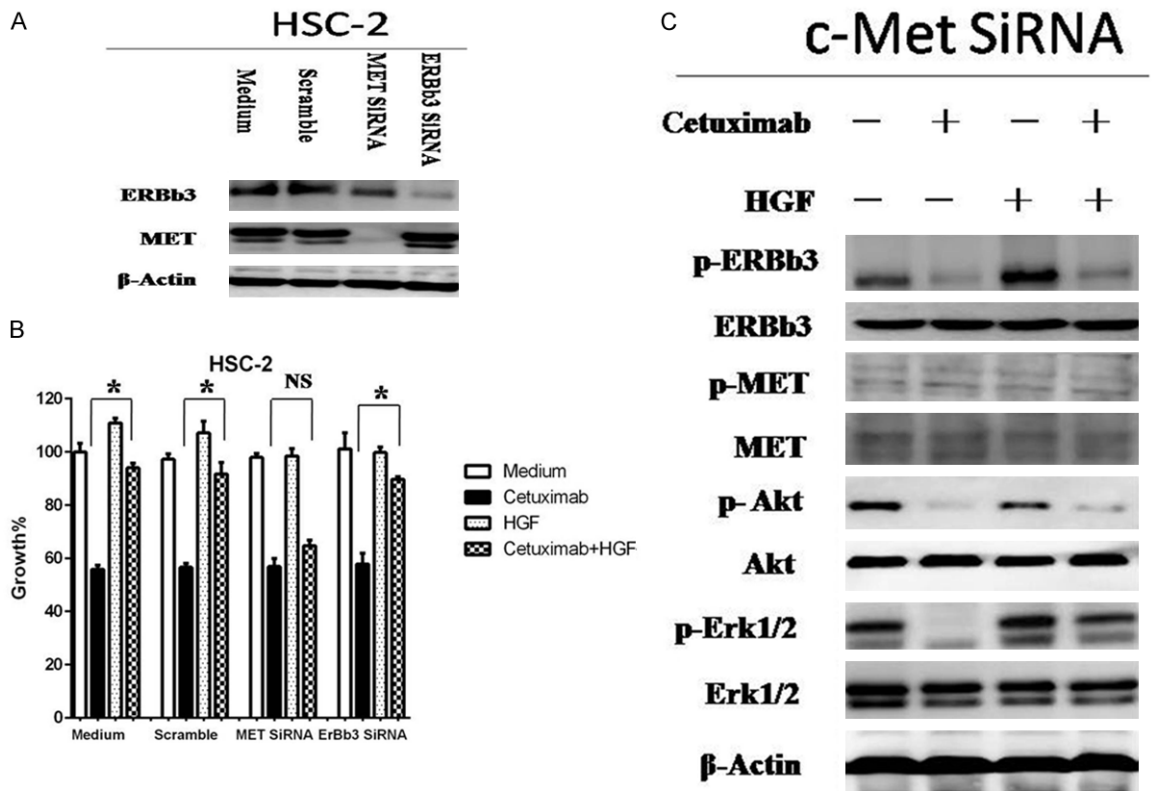

Figure 3. Met siRNA restoration of OSCC cell sensitivity to cetuximab in HGF-induced cetuximab-resistant OSCC cells. A. Western blot. HSC-2 cells were grown and transiently transfected with Met, negative control, or ErbB3 siRNA for 48 h and then subjected to western blotting. B. The MTT assay. HSC-2 cells were grown and transiently transfected with Met, negative control, or ErbB3 siRNA for 48 h and then subjected to the MTT assay. C. Western blot. HSC-2 cells were grown and transiently transfected with Met siRNA for 48 h and then treated with cetuximab (1 μg/mL) for 72 h before being subjected to western blotting.

teins was almost completely blocked by Met siRNA transfection (**Figure 3C**).

Association of met and akt expression with poor OSCC outcomes

To relate our current data clinically, we assessed and retrieved data on the differentially expressed genes in OSCC tissue samples from the NCBI-GEO database and then identified the expression of HGF, Met, ERK1 (also called MAPK3 in **Figure 4D**), ERK2 (also called MAPK1 in **Figure 4E**), and Akt in OSCC tissues. The Kaplan-Meier curves were plotted against the expression of these genes and then statistically analyzed by using the log rank test. We found that the expression of Met and Akt was associated with a poor survival of OSCC patients (P = 0.021 and 0.011, respectively; **Figure 4A** and **4F**), whereas HGF expression was associated with a better patient survival (P = 0.026; **Figure 4B**). Moreover, interactive analysis of HGF and Met expression did not yield any statistical significance for patient survival (**Figure 4C**). In addition, the expression of ERK1 and ERK2 also did not show any significant association with patient survival (P = 0.20 and 0.51; **Figure 4D** and **4E**).

Discussion

Cetuximab is a monoclonal antibody used to treat various human cancers by blockage of EGFR activation to inhibit the downstream pathways in tumor cell growth and invasion [36]. However, primary and acquired cetuximab resistance occurs in most cancer cases, even if the tumor cells express a high level of EGFR [37]. The underlying molecular mechanisms of drug resistance are complicated. For example, in colon cancer, resistance to anti-EGFR antibodies is mainly through the irregular activation of the EGFR downstream proteins by other proteins, like RAS [38], STAT [39], PTEN [40], or

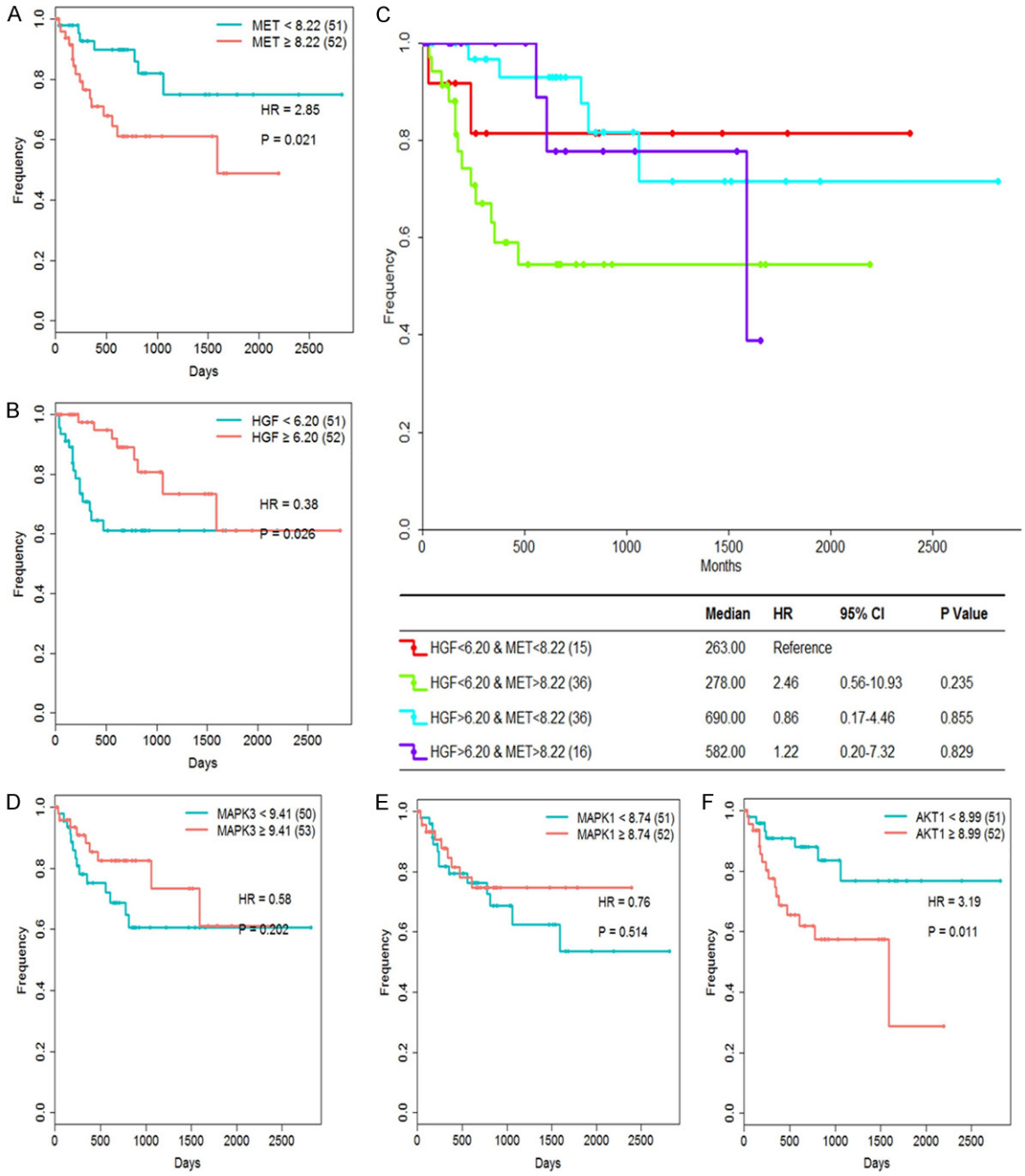

Figure 4. Association of gene expression with the survival of OSCC patients. The online database data were retrieved and statistically analyzed. A. Association of Met expression with the overall survival (OS) of OSCC patients. High Met expression predicted a poor OS (P = 0.021). B. Association of HGF expression with the OS of OSCC patients. A high HGF level was associated with a better OS (P = 0.026). C. Association of Met/HGF expression with the OS of OSCC patients. The combined expression of Met and HGF failed to predict the OS of OSCC patients. D and E. Association of ERK1 and ERK2 expression with the OS of OSCC patients. ERK1 and ERK2 expression failed to predict the OS of OSCC patients. F. Association of Akt expression with the OS of OSCC patients. A high Akt expression predicted a poor OS of OSCC patients (P = 0.011).

Met [41]. A single nucleotide polymorphism of the EGFR gene has been reported to occur in more than 40% of HNSCC patients, a change that could affect the affinity of cetuximab to bind to EGFR protein [42]. In another previous study, Smad4 expression was found to be criti-

cal in mediating cetuximab sensitivity and resistance [43]. In addition, in cetuximab-sensitive HNSCC A431 cells, cetuximab downregulated p-ERK and p-Akt expression, whereas Akt phosphorylation was unchanged in cetuximab-resistant cells [44]. In our current study, we found that HGF-induced cetuximab resistance occurred in both HSC-2 and HSC-3 cells with high levels of p-Akt and p-ERK1/2 when they were cotreated with cetuximab, indicating that Akt and ERK1/2 were responsible for mediating cetuximab resistance in OSCC.

Met protein is frequently highly expressed in various human cancers. It is activated by its ligand HGF to trigger activation of the downstream signaling pathways and multiple cellular events [45], like cancer development and metastasis. However, it has been shown in lung cancer and hepatocellular carcinoma that aberrant HGF-independent Met activation occurs and then in turn triggers the ErbB3/PI3K/Akt signaling pathway, leading to cancer progression [46, 47]. In HNSCC, a previous study has revealed that ErbB3 is required to activate EGFR and drug resistance [48]. However, in our current study, we did not provide any evidence showing the importance of ErbB3 in HGF-induced cetuximab resistance in OSCC cells. Furthermore, in non-small cell lung cancer, it has been demonstrated that there is a synergistic effect of HGF and EGF on cancer cell growth and that the cross-talk between Met and HGF, at the molecular level, is crucial for tumor cell growth [49]. In colorectal cancer, HGF induction of Met activation resulted in cetuximab resistance [50]. Furthermore, Met siRNA was able to reduce HNSCC cell viability and migration *in vitro* and tumor xenograft growth *in vivo* [51]. In addition, in human OSCC tissues, Met expression was significantly upregulated compared with that of the adjacent normal tissues [35]. Met inhibitors exhibited a strong effect on inhibition of OSCC cell growth and induction of OSCC cell apoptosis by suppression of Akt and ERK1/2 activities [52]. In our current study, HGF induced Akt and ERK1/2 phosphorylation, leading to cetuximab-resistant OSCC cells. With the combination of the Met inhibitor and Met siRNA, OSCC cell viability was significantly reduced and Akt and ERK1/2 phosphorylation was dramatically decreased. Taken together, we speculated that Met and EGFR cross-talk could synergistically promote cancer progression and that anti-Met therapy could be effective even in OSCC cells resistant to anti-EGFR therapy. In this context, the combination of cetuximab and Met inhibitor could be a better option for the treatment of OSCC patients with primary or acquired cetuximab resistance.

Indeed, different Met inhibitors are under clinical trials, and crizotinib and cabozantinib were approved by the US FDA to treat NSCLC and medullary thyroid cancer, respectively [53]. Another Met inhibitor, PHA-665752, is a compound used to inhibit Met phosphorylation and the downstream signaling cascades [54]. In ovarian cancer, PHA-665752 and Met siRNA have been shown to inhibit tumor cell growth and overcome cisplatin resistance [55]. While in colorectal cancer, the combination of PHA-665752 and cetuximab significantly decreased tumor cell proliferation compared with that of either agent alone [56]. In our current study, PHA-665752 was shown to resensitize OSCC cells to cetuximab resistance, and its combination with cetuximab had a better antitumor efficacy. At the molecular level, our current data showed that both HGF and cetuximab were able to regulate Akt and ERK1/2 activities. Indeed, Yu *et al.* have demonstrated that the PI3K/Akt/mTOR pathway is an ideal target for controlling OSCC because the combination of PI3K/Akt inhibitor with radiation improved the radiation efficacy for the treatment of OSCC [57]. Targeting of the PI3K/Akt/mTOR pathway also improved the effects of doxorubicin on its antitumor activity in OSCC [58]. Furthermore, sulfasalazine (SSZ), an anti-inflammatory drug, has been demonstrated to have a potential therapeutic ability in the treatment of OSCC by promoting autophagy-induced tumor cell death and inhibiting the PI3K/Akt and MAPK pathways [59]. Another previous study has revealed that activation of the ERK, Akt, and p38 pathways is involved in HGF- and EGF-induced OSCC cell migration [60]. Taken together, the PI3K/Akt and MAPK pathways could be involved in OSCC development and progression, and these signaling pathways are activated by different upstream factors, e.g. EGFR and Met. When OSCC cells became sensitive to cetuximab treatment, the Akt and ERK activities were significantly suppressed in OSCC cells, leading to a decrease in OSCC progression; whereas in cetuximab-resistant cells, the cetuximab-EGFR-

Akt/ERK axis was aberrant, but there was an increase in HGF-induced Met activity to induce Akt/ERK phosphorylation, leading to OSCC cell proliferation. However, blockage of these two signaling pathways using the combination of cetuximab and Met inhibitor or Met siRNA inhibited Akt and ERK phosphorylation, resulting in cell growth inhibition. Thus, our current data provide insightful information regarding the effects of the combined treatment of cetuximab and Met inhibitor as a novel therapeutic strategy in the control of OSCC.

In addition, our *ex vivo* data analysis showed that the expression of the cell growth signaling proteins Akt and Met was associated with a poor survival of OSCC patients. However, due to the small sample size, the expression of Met with HGF did not show statistically significant data that predict OSCC patient survival. Future studies with a large sample size should be performed to confirm our current findings. However, our current study does have some limitations; for instance, all the experiments were performed *in vitro*, and *in vivo* experiments are needed to assess and validate the effects of the combined treatment of cetuximab and Met inhibitor on the control of OSCC in the future. Furthermore, our *ex vivo* data lacked a sufficient sample size as well as cetuximab or Met inhibitor treatment data for OSCC patients.

Conclusions

Our current study revealed that HGF activated Met, subsequently increasing the phosphorylation of the downstream PI3K/Akt and MAPK pathways, which were responsible for the cetuximab resistance of OSCC, and that inhibition of Met expression or activity restored OSCC sensitivity to cetuximab treatment. Future studies are needed to confirm the effects of the combination treatment of cetuximab with Met inhibitor in the control of OSCC clinically.

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Disclosure of conflict of interest

None.

Abbreviations

Akt, Protein kinase B; c-Met, hepatocyte growth factor receptor; EGFR, epidermal growth factor receptor; ERK1/2, Extracellular signal-regulated kinase; HGF, Hepatocyte growth factor; OSCC, Oral squamous cell carcinoma.

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Identification of hub ubiquitin ligase genes affecting Alzheimer's disease by analyzing transcriptome data from multiple brain regions

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Abstract

The ubiquitin-proteasome system (UPS) plays crucial roles in numerous cellular functions. Dysfunction of the UPS shows certain correlations with the pathological changes in Alzheimer's disease (AD). This study aimed to explore the different impairments of the UPS in multiple brain regions and identify hub ubiquitin ligase (E3) genes in AD. The brain transcriptome, blood transcriptome and proteome data of AD were downloaded from a public database. The UPS genes were collected from the Ubiquitin and Ubiquitin-like Conjugation Database. The hub E3 genes were defined as the differentially expressed E3 genes shared by more than three brain regions. E3Miner and UbiBrowser were used to predict the substrate of hub E3. This study shows varied impairment of the UPS in different brain regions in AD. Furthermore, we identify seven hub E3 genes (CUL1, CUL3, EIF3I, NSMCE1, PAFAH1B1, RNF175, and UCHL1) that are downregulated in more than three brain regions. Three of these genes (CUL1, EIF3I, and NSMCE1) showed consistent low expression in blood. Most of these genes have been reported to promote AD, whereas the impact of RNF175 on AD is not yet reported. Further analysis revealed a potential regulatory mechanism by which hub E3 and its substrate genes may affect transcription functions and then exacerbate AD. This study identified seven hub E3 genes and their substrate genes affect transcription functions and then exacerbate AD. These findings may be helpful for the development of diagnostic biomarkers and therapeutic targets for AD.

Keywords

Alzheimer's disease, brain region, gene expression, ubiquitin-proteasome system, ubiquitin ligases

Introduction

The ubiquitin-proteasome system (UPS) includes the following enzymes: ubiquitin activating enzymes (E1), ubiquitin conjugating enzymes (E2), ubiquitin ligase enzymes (E3), deubiquitinating enzymes (DUB), and the 26S proteasome.[1,2] The UPS is a multicomponent system for protein degradation that is necessary for eukaryotic organisms to degrade approximately 80% of intracellular proteins.[3] Ubiquitin is an evolutionarily conserved peptide containing 76 amino acids; ubiquitin-related enzymes bind ubiquitin molecules to substrates through a multistep catalytic process, and the substrate protein is then degraded by the proteasome.[4] The UPS also plays crucial roles in posttranslational modification, regulation of cellular signal transduction, cell cycle processes, and repair of DNA damage.[4,5] Numerous metabolic pathways and cellular regulatory networks require precise spatial and temporal control of effector protein levels by the UPS.[6]

The UPS is involved in protein quality control and the removal of misfolded or aggregated proteins, and UPS dysfunction is correlated with neurodegenerative diseases. Damage to the UPS will affect the degradation of amyloid β (Aβ) and cause abnormal aggregation of Aβ in Alzheimer's disease (AD). Aggregated Aβ inhibits proteasome activity and subsequently causes damage to the multivesicular body sorting pathway.[7] The impaired UPS also causes increased β-secretase and γ-secretase, which accelerates the hydrolysis of amyloid precursor protein (APP) to produce Aβ and promotes Aβ aggregation.[7] Ubiquitinated APP cannot be degraded by the proteasome when the activity of the proteasome is inhibited, which results in the accumulation of ubiquitinated proteins in the cell.[8] Dysfunction of the UPS

may also promote the aggregation of tau protein in AD, and the higher-order oligomers and aggregates of tau cannot enter the narrow proteasome open channels.[9] Recent studies have suggested that UPS-dependent protein degradation is associated with synaptic plasticity and learning or memory functions, that the UPS plays a vital role in hippocampal long-term memory consolidation and that substrate-specific E3 may be the key factor in memory regulation.[10]

The above studies suggested that the impaired UPS correlated to the pathogenesis of AD, and the decreased UPS activity in AD may lead to further deterioration of the disease. It is worth noting that increasing the UPS activity will enhance the organism's ability to resist oxidative stress and prolong life in various animal models and human cell lines.[11] Our previous work revealed the differences in the expression of learning- and memory-related genes in different brain regions in AD.[12] In this study, we aimed to explore the different impairments of the UPS in multiple brain regions of AD through transcriptome data analysis and to identify hub E3 genes and their potential substrate genes.

Methods

Data collection

The microarray data of Alzheimer's disease were downloaded from the Gene Expression Omnibus (GEO) database (http://www.ncbi.nlm.nih.gov/geo). Data screening was based on the following criteria: (1) the data are genome-wide; (2) the data should include AD and control samples; (3) the data should include multiple brain regions; (4) no age difference exists between AD and controls in each brain region; and (5) raw microarray data are available. With these criteria, we finally chose the dataset of GSE5281[13] for our reanalysis. The dataset contains 87 AD samples and 74 control samples. The brain regions in this dataset included the entorhinal cortex (EC), hippocampus (HIP), medial temporal gyrus (MTG), posterior cingulate (PC), superior frontal gyrus (SFG), and primary visual cortex (PVC). There was no age difference between AD and controls in each brain region (Table 1). The datasets of GSE36980[14] and GSE48350[15] also contain AD and control transcriptome data for multiple brain regions and were used as validation sets. In order to explore the expression differences in the blood transcriptome and proteome in patients with AD, we chose the GSE63060 dataset[16] to analyze the gene expression profiles in blood, which contains 145 patients with AD and 104 age- and sex-matched controls. The GSE29676 dataset[17] contains serum proteome data from 50 patients with AD and 40 controls and was used to explore the changes in protein levels. For details on data preprocessing, see our previous reports.[12,18]

Differential expression gene analysis

Bioinformatics analysis of the microarray data was carried out by R statistical software v3.6.1 (https://www.r-project.org/) and Bioconductor Library (https://www.bioconductor.org/). Differential gene expression analysis was performed

Table 1. Age information between patients with Alzheimer's disease and controls.

Brain regions	Alzheimer's disease		Control		p^b
	N^a	Age	N^a	Age	
Entorhinal cortex	10	85.6 ± 6.3	13	80.3 ± 9.2	0.118
Hippocampus	10	77.8 ± 5.7	13	79.6 ± 9.4	0.574
Medial temporal gyrus	16	79.1 ± 6.4	12	80.1 ± 9.8	0.771
Posterior cingulate	9	77.6 ± 6.5	13	79.8 ± 9.4	0.522
Superior frontal gyrus	23	79.2 ± 7.5	11	79.3 ± 10.2	0.978
Primary visual cortex	19	80.2 ± 6.7	12	77.9 ± 6.9	0.385

[a]Number of samples.
[b]The p value was calculated by student's t-test.

using the empirical Bayesian algorithm in the limma package in R.[19] Up- and downregulated genes were defined as a log2 transformed fold-change (logFC) ≥ 1 or ≤ -1 for AD samples compared with controls. A false discovery rate (FDR)-corrected p value ≤ 0.05 was considered significant. Differentially expressed genes shared by more than three brain regions are considered to be hub genes. The heatmap package in R was used to show the gene expression profiles, and the clustering method was chosen as "ward.D2."

GO and KEGG enrichment analysis

Information on human genes and related GO terms (including biological process, cellular component and molecular function) was downloaded from the QuickGO database (http://www.ebi.ac.uk/QuickGO-Beta/). The reference human genes and pathways were downloaded from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (http://www.kegg.jp/). GO terms and KEGG pathways with less than 10 genes were removed. GO and KEGG enrichment analysis was performed using the hypergeometric test and the formula shown in a previous report.[20] The reference gene set is all detected genes in the GSE5281 dataset. An FDR-corrected p value ≤ 0.05 was considered significantly enriched.

UPS gene collection

The UPS genes were downloaded from the Ubiquitin and Ubiquitin-like Conjugation Database (http://uucd.biocuckoo.org).[21] This database collected 129,416 potential proteins from 148 different species and protein types, including E1, E2, E3, DUB, ubiquitin-binding domain-containing protein (UBD) and ubiquitin-like domain-containing protein (ULD). A total of 878 human UPS genes (including 10 E1 genes, 42 E2 genes, 700 E3 genes, and 126 DUB genes) were downloaded in this study. Proteasome-related genes were extracted from the proteasome pathway in the KEGG database. This study mainly focused on the E3

genes, and the hub E3 genes were defined as the differentially expressed E3 genes shared by more than three brain regions.

Construction of AD prediction models

The prediction models were built using the blood transcriptome and proteome data of patients with AD and controls. The biomarkers were chosen as the hub E3 genes and proteins. The receiver operating characteristic (ROC) curve and the area under the curve (AUC) of the single marker were calculated using the pROC package in R. A stepwise modeling strategy was used to screen the optimal multigene and multiprotein models. The model with the largest AUC was defined as the optimal model. For example, in the multigene prediction model, the gene with the largest AUC was first selected. Then, we used a multivariate logistic regression model to generate the combined effect of the selected gene and each of the remaining genes. Next, we selected the best two-gene model with the highest AUC and repeated the previous steps. Finally, we selected the optimal model with the highest AUC in each multigene combination model.

E3 substrate prediction and correlation analysis

Two web server tools, including E3Miner (http://e3miner.biopathway.org)[22] and UbiBrowser (http://ubibrowser.ncpsb.org),[23] were used to predict the substrate of hub E3. The hub E3 genes and the top 10 predicted substrate genes based on confidence were used to construct the E3-substrate interactive network. Network visualization was performed using Cytoscape v3.4.0 (https://cytoscape.org/). Pearson correlation analysis was used to analyze the correlation between the E3 genes and the predicted substrate genes. A p value ≤ 0.05 was considered significant. The mouse dataset GSE113436 was used to validate the expression of NSMCE1 and the predicted target genes. The data is the transcriptome profile of Nsmce1 overexpression in mouse hippocampal neuronal cells.[24]

Results

Commonly differentially expressed genes correlate to AD pathology

Differential expression analysis showed that there were a large number of downregulated genes in EC, HIP, MTG, and PC, and relatively few genes were affected in SFG and PVC (Figure 1(a)). Furthermore, there were also a considerable number of upregulated genes in the MTG and SFG. Most of these genes were differentially expressed in multiple brain regions, and few genes were differentially expressed in specific brain regions (Figure 1(b)). There were two differentially expressed genes shared by all six brain regions (APOO and PCYOX1L). A Venn diagram showed that a total of 462 genes were differentially expressed in more than three brain regions (Figure 1(b)). The KEGG pathway enrichment results suggested that these genes were mainly involved in the pathways of neurodegenerative diseases (such as

Alzheimer's disease), nervous system (such as synaptic vesicle cycle and GABAergic synapse), metabolism (such as oxidative phosphorylation, glycolysis and the TCA cycle), signal transduction (such as the Hippo signaling pathway and cAMP signaling pathway), protein degradation (such as the proteasome) and transcription (such as basal transcription factors) (Figure 1(c)). Numerous studies have reported that these pathways are correlated with the Aβ and tau protein processes, synaptic dysfunction and neurodegeneration in AD.[7,25–28] This evidence suggests that the damaged genes and pathways shared by multiple brain regions can accurately reflect the pathological changes of AD.

UPS-related functions were damaged in multiple brain regions in AD

GO and KEGG enrichment analyses were performed using the differentially expressed genes in each brain region. The results showed that there were 193, 357, 272, 283, 666, and 265 enriched GO items (Figure 2(a)) and 23, 40, 29, 38, 3, and 22 enriched KEGG pathways (Figure 2(b)) in the EC, HIP, MTG, PC, SFG, and PVC, respectively. There were 195 enriched GO items and 27 enriched KEGG pathways shared by more than three brain regions. These enriched GO items and KEGG pathways included multiple UPS-related functions. By analyzing the enriched UPS-related functions in different brain regions, we found that HIP and PC showed the highest significance, EC and MTG showed moderate significance, and SFG and PVC showed relatively low significance (Figure 2(c)).

E3 genes were downregulated in multiple brain regions

There were multiple differentially expressed UPS genes in the EC, HIP, MTG, and PC, whereas only a few genes were affected in the SFG and PVC (Figure 3(a)). Because E3 has substrate specificity, this study mainly focused on the E3 genes. These differential E3 genes were almost all downregulated in the EC, HIP, and PC, and only MTG showed more than a dozen upregulated genes. The clustering results of the 73 combined differentially expressed E3 genes in six brain regions confirmed this trend (Figure 3(b)). The Venn diagram analysis showed that there were seven differentially expressed E3 genes shared by more than three brain regions (Figure 3(c)), and these genes were identified as hub E3 genes. Among these genes, UCHL1 was downregulated in the EC, HIP, MTG, and PC; CUL3, EIF3I, and NSMCE1 were downregulated in the EC, HIP, and PC; and CUL1, PAFAH1B1, and RNF175 were downregulated in the HIP, MTG, and PC (Figure 3(d)). The clustering results suggested that the HIP and PC showed the most serious downregulation of the hub E3 genes, followed by the MTG and EC, whereas the SFG and PVC showed relatively slight damage. The clustering results of the proteasome genes also showed a consistent trend (Figure 3(e)). Furthermore, most of the combined differentially expressed E3 genes showed a low expression trend in different brain regions, and the hub E3 genes also showed a consistent downregulation trend in almost all brain regions in the two validation datasets

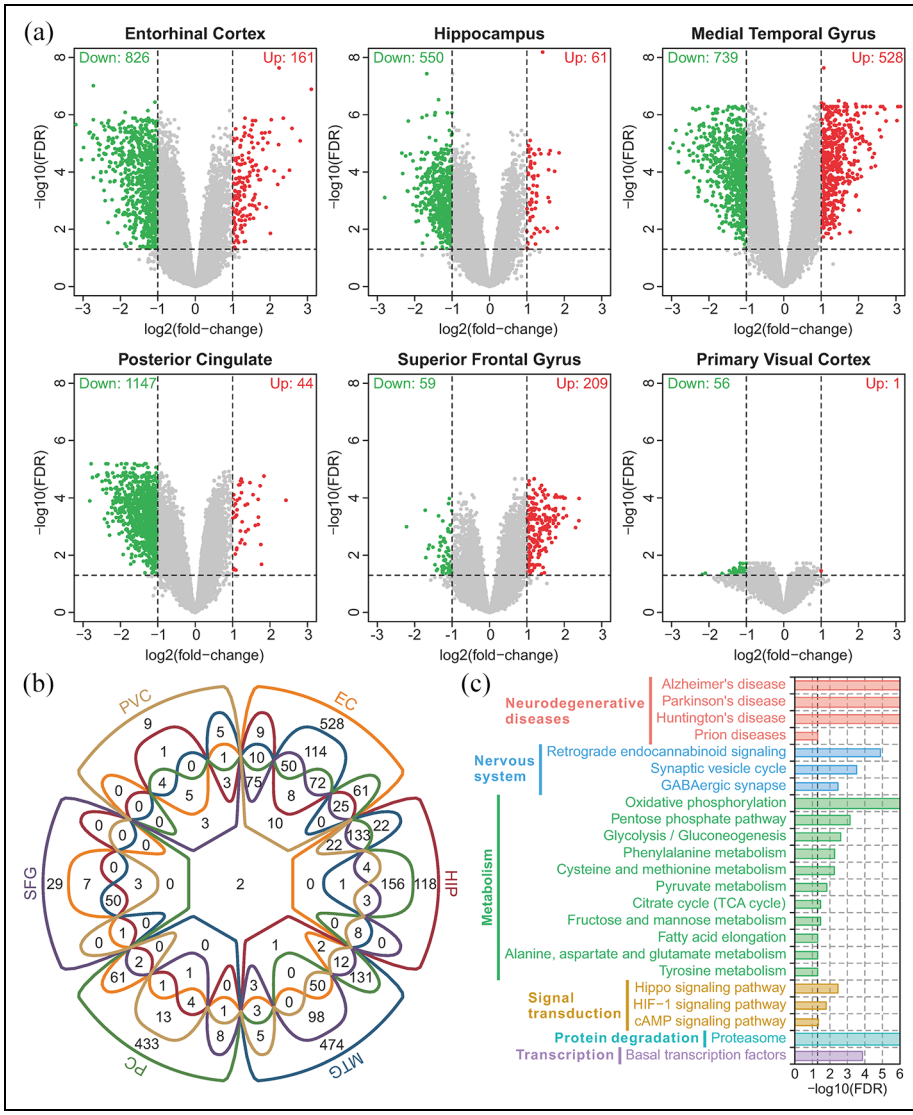

Figure 1. Overview of the differentially expressed genes and their functions: (a) volcano plot of all genes in six brain regions. Red points indicate upregulated genes, and green points indicate down regulated genes, (b) Venn diagram of differentially expressed genes in six brain regions, and (c) enriched KEGG pathways of differentially expressed genes shared by more than three brain regions.

Each color represents a pathway class. The dashed line indicates statistical significance, and the bar width indicates the enrichment percentage.

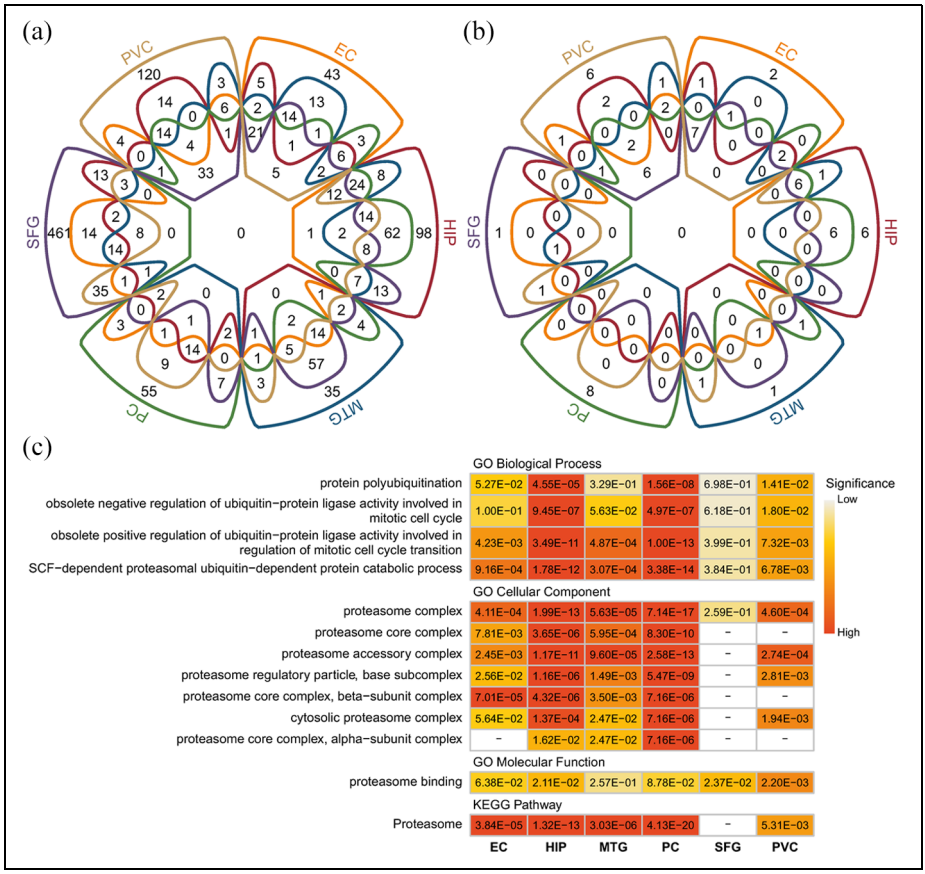

Figure 2. Enriched GO functions and KEGG pathways in six brain regions: (a) Venn diagram of enriched GO functions (including biological process, cellular component and molecular function) in six brain regions, (b) Venn diagram of enriched KEGG pathways in six brain regions, and (c) significantly enriched ubiquitin-proteasome system-related functions and pathways in more than three brain regions.

The color bar indicates the FDR-corrected *p* value. The blank indicates that there are no differentially expressed genes in the pathway corresponding to the brain region.

(Figure 3(f)–(i)). Combining the above analysis results, we infer that these hub E3 genes may play crucial roles in the pathogenesis of AD. The role of these hub E3 genes in AD pathology showed in Supplemental Table 1, and the brief signaling cascade of these hub E3 genes showed in Supplemental Figure 1. To explore the possible E2 conjugating enzyme in connection with identified E3 ligases, we used Pearson correlation analysis to explore the combination possibility between hub E3 and E2 genes. Among these hub E3 genes, CUL1, CUL3, PAFAH1B1, and RNF175 showed a strong correlation with multiple E2 genes (Supplemental Figure 2).

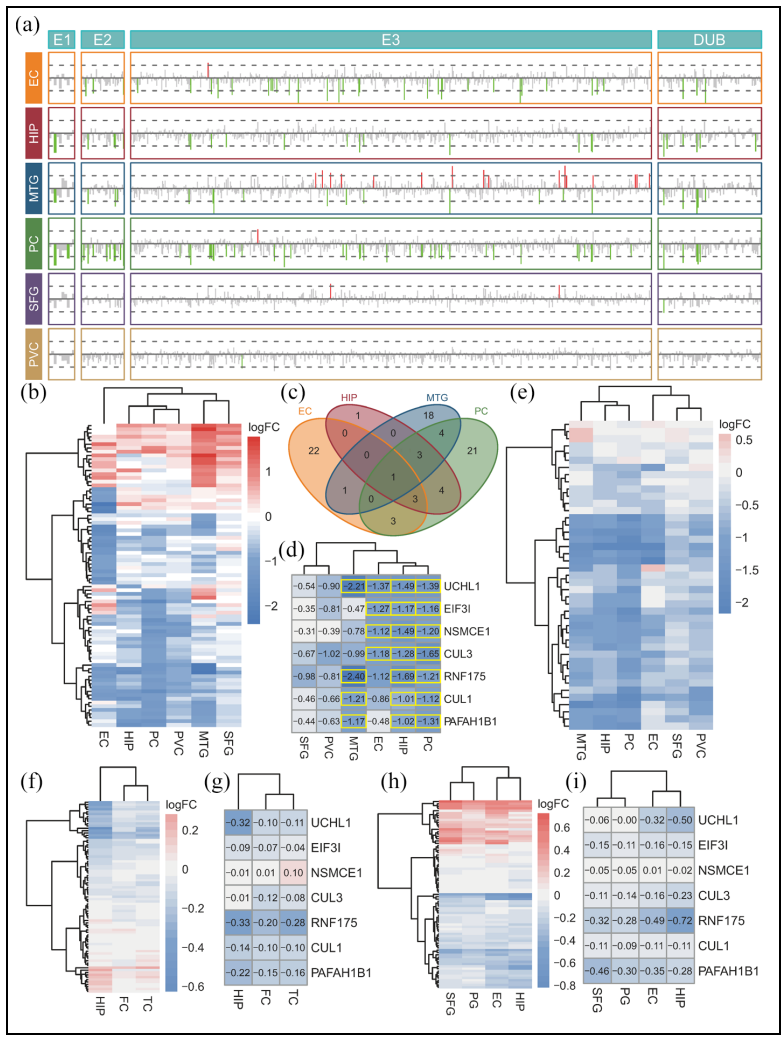

Figure 3. Expression profiles of ubiquitin-proteasome system-related genes and hub E3 genes: (a) gene expression of ubiquitin-related genes in different brain regions. The line length indicates the $\log_2$ (fold-change). The red, green and gray colors indicate that the gene is upregulated, downregulated and not statistically significant, (b) heatmap of the combined differentially expressed E3 genes in six brain regions, (c) Venn diagram of the differentially expressed E3 genes in the EC, HIP, MTG and PC, (d) heatmap of the hub E3 genes in six brain regions, (e) heatmap of the proteasome-related genes in six brain regions, (f) heatmap of the combined differentially expressed E3 genes in the validation set of GSE36980, (g) heatmap of the hub E3 genes in the validation set of GSE36980, (h) heatmap of the combined differentially expressed E3 genes in the validation set of GSE48350, and (i) heatmap of the hub E3 genes in the validation set of GSE48350.

The color bar and the number in the box of the heatmap indicate the $\log_2$ (fold-change). The yellow box indicates statistical significance.

Expression of hub E3 genes and proteins in the blood of AD

To explore the expression difference in E3 genes in blood, we mapped the AD brain transcriptome data with blood transcriptome and proteome data. There were 73 combined differentially expressed E3 genes in the GSE63060 dataset and 28 combined differentially expressed E3 proteins in the GSE29676 dataset. The clustering results showed that the total expression profiles of these genes or proteins may not distinguish patients with AD and controls (Figure 4(a) and (b)). Three hub E3 genes (CUL1, EIF3I, and NSMCE1) were significantly expressed at low levels in patients with AD compared to controls in blood, which is consistent with the expression trend in the brain (Figure 4(c)). However, the protein expression of CUL1 and RNF175 showed the opposite trend between AD and controls in blood (Figure 4(d)). We constructed prediction models to explore the diagnostic accuracy of these hub E3 genes and proteins in AD. The optimal multigene model is the combination of five hub E3 genes, including NSMCE1, PAFAH1B1, CUL1, RNF175, and EIF3I (Figure 4(e)), which had the highest AUC of 0.726 (Figure 4(f)). The optimal multiprotein model is the combination of five hub E3 proteins, including RNF175, PAFAH1B1, CUL3, CUL1, and UCHL1 (Figure 4(g)), which had the highest AUC of 0.789 (Figure 4(h)). Unfortunately, neither the multigene nor the multiprotein model achieved high prediction accuracy.

Hub E3 and its substrates aggravate AD by affecting transcription functions

The interactive network of the hub E3 genes and the predicted substrate genes showed that there were eight substrates that were regulated by multiple E3 (Figure 5(a)). The heatmap showed that there were four differentially expressed substrate genes: EDF1 was downregulated in the EC, HIP, MTG, and PC; TAF7 was downregulated in the HIP, PC, and PVC; TAF3 was upregulated in the HIP, MTG, and SFG; and PIK3C2B was upregulated in the MTG (Figure 5(b)). The KEGG enrichment results showed that these substrate genes are involved in transcription functions (Figure 5(c)). Pearson correlation analysis of the hub E3 genes and the differentially expressed substrate genes showed that EIF3I and UCHL1 positively correlated with EDF1 in all brain regions; EIF3I, NSMCE1, and UCHL1 positively correlated with TAF7 in most brain regions; NSMCE1 negatively correlated with TAF3 in the HIP and MTG; and PAFAH1B1 negatively correlated with PIK3C2B in the MTG and PC (Figure 5(d)). Based on the above results, we concluded that the downregulated E3 genes may affect the substrate genes, damage the transcription functions and aggravate AD (Figure 5(e)). In the validation dataset, there was no difference in the expression of Taf3 or Taf7 after Nsmce1 was overexpressed (Supplemental Figure 3).

Discussion

This study showed varied impairment of the UPS in different brain regions in AD. The UPS was seriously damaged in the HIP and PC, moderately damaged in the

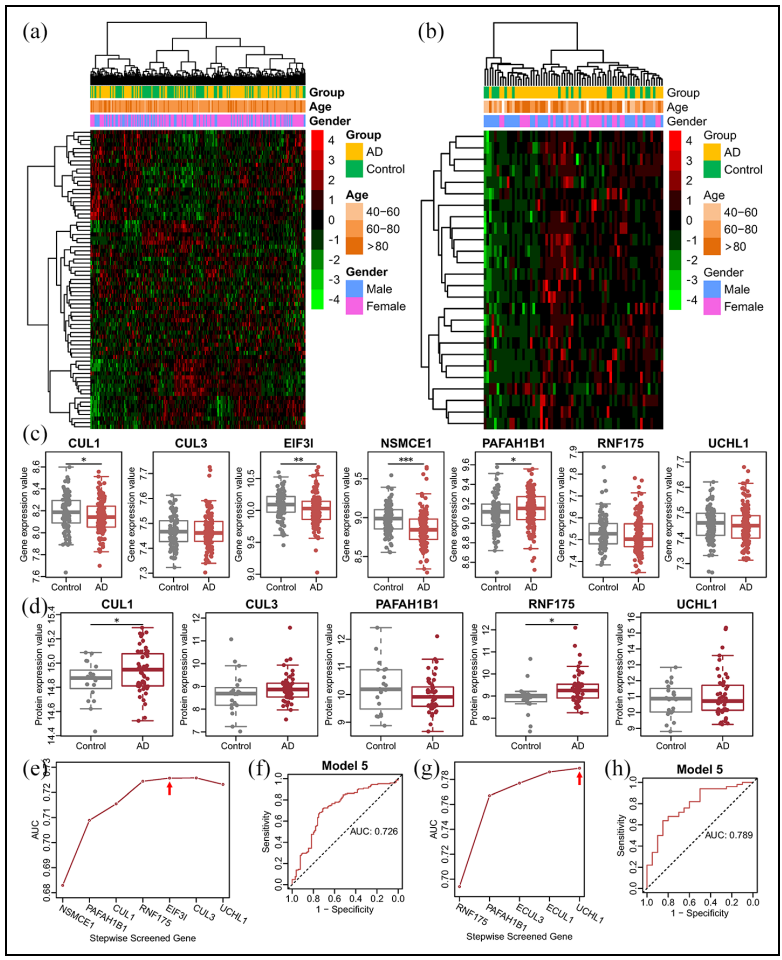

Figure 4. Expression of hub E3 genes and proteins in the blood of AD and diagnostic model construction: (a) heatmap of the combined differentially expressed E3 genes in the GSE63060 dataset, (b) heatmap of the combined differentially expressed E3 proteins in the GSE29676 dataset. The gene and protein expression values were *z*-score converted, and the color bar indicates the z-score, (c) gene expression of the hub E3 genes in the blood between patients with AD and controls, (d) protein expression of the hub E3 genes in the blood between patients with AD and controls (no protein expression data of EIF3I or NSMCE1). Statistical significance: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, (e) stepwise screened prediction models using hub E3 gene expression values in blood transcriptome data. From left to right on the x-axis (stepwise screened genes), each additional gene corresponds to a model (for example, the gene of NSMCE1 represents model 1, which contains one gene of NSMCE1, PAFAH1B1 represents model 2, which contains two genes including NSMCE1 and PAFAH1B1). The red arrow shows the optimal model, (f) ROC curve of the screened optimal multigene model, (g) stepwise screened prediction models using hub E3 protein expression values in blood proteome data, and (h) ROC curve of the screened optimal multiprotein model.

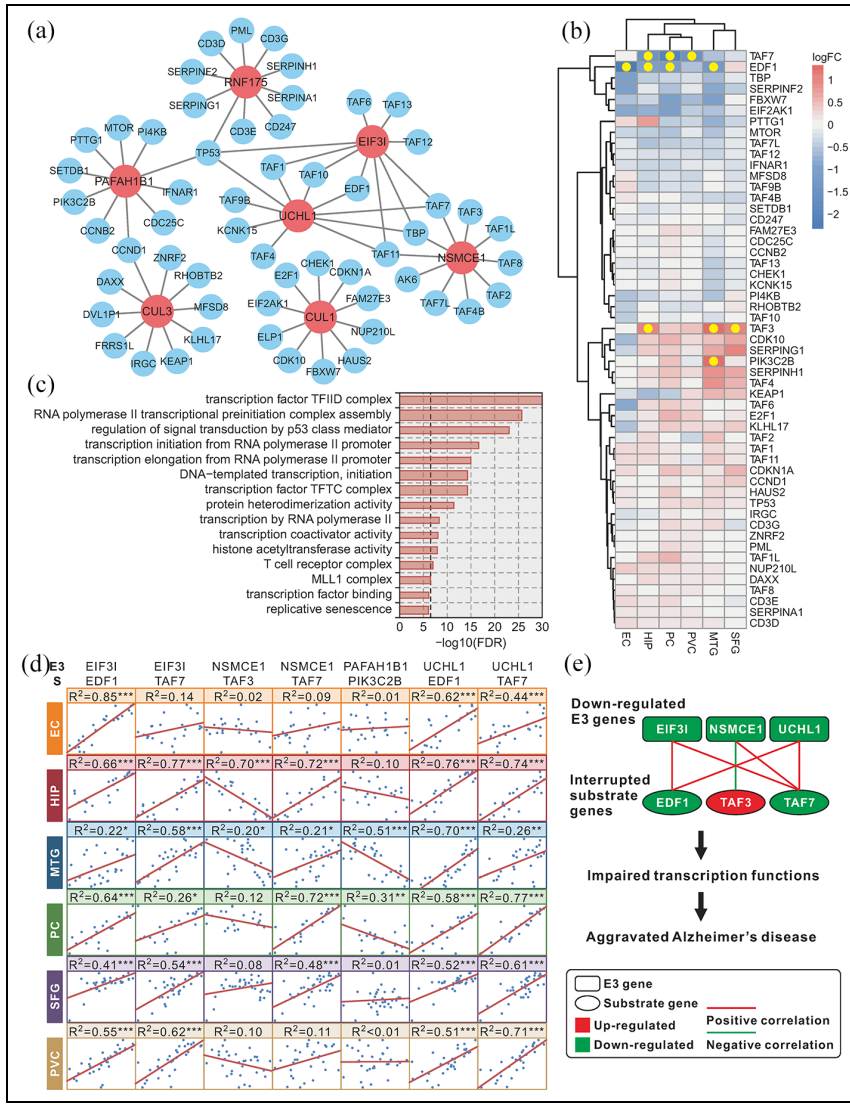

Figure 5. Network and function of hub E3 and their substrate genes: (a) hub E3 and their substrate gene interactive network. The red color indicates E3 genes, and the blue color indicates substrate genes, (b) heatmap of the hub E3 substrate genes in six brain regions. The color bar indicates the $\log_2$ (fold-change), and the yellow point indicates statistical significance, (c) top 15 enriched GO biological processes of the hub E3 substrate genes. The dashed line indicates statistical significance, and the bar width indicates the enrichment percentage, (d) Pearson correlation of hub E3 genes and differentially expressed substrate genes (represented by *S*). The R^2 indicates the square of the Pearson correlation coefficient. Statistical significance: $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, and (e) the potential mechanism of down regulated hub E3 genes aggravating AD.

EC and MTG, and slightly damaged in the SFG and PCV. The seven identified hub E3 genes were downregulated in most brain regions, and three of these genes were also expressed at low levels in blood. These hub E3 genes and their substrate genes may affect transcription functions and then exacerbate AD.

Different brain regions have specific functions, and there are strong connections among one another. The EC is the main channel between the HIP and neocortex and is involved in the formation of long-term memory.[29] EC receives information from the HIP and transmits it to the neocortex through neurotransmitters such as glutamate and is one of the brain regions affected in the early stages of AD.[30] The HIP is a part of the temporal lobe, which is essential for the formation of new memories. Usually, HIP function is impaired earlier than other cortices in patients with AD, suggesting that the first process in AD is memory impairment.[31] The MTG involves many cognitive processes, such as semantic memory and language processing, and integrates information from different senses. Studies have shown that there is active neuronal loss in the MTG in AD.[32,33] The PC is a part of the cingulate cortex, which is a highly connected and metabolically active brain region and is functionally involved in learning and spatial memory. It has been identified that there was amyloid deposition and decreased metabolism in PC during AD progression, and the volume of the PC in patients with AD is smaller than that of healthy controls.[34,35] The SFG is located in the upper part of the prefrontal cortex, accounting for approximately one-third of the frontal lobe. Impaired SFG will lead to disorders of working memory and self-awareness.[36,37] Several studies have shown that reduced metabolic functions in FC are associated with AD.[38] The PVC occupies the entire surface of the occipital lobe and acts as a receiver of visual data. Some evidence suggests that the changes in PVC are associated with normal aging, and Aβ plaques and neurofibrillary tangles usually occur in the late stages of AD.[39,40] Our previous study found that multiple learning and memory-related genes were affected in the HIP and temporal lobe in AD.[12] Combined with the findings in this study that UPS showed the most serious damage in the HIP and PC, we speculate that there may be a causal relationship between UPS impairment and reduced learning or memory functions.

The number of E1, E2, and E3 has a pyramid structure, and a previous report showed that there were approximately 2 E1 genes, 30-40 E2 genes and more than 600 E3 genes in humans.[11] Recent studies have revealed more genes in the human ubiquitination pathway (including E1, E2, E3, and DUB).[21] E3 has substrate specificity and is the most critical enzyme in the UPS. This study identified seven hub E3 genes, including CUL1, CUL3, EIF3I, NSMCE1, PAFAH1B1, RNF175, and UCHL1. The proteins encoded by CUL1 and CUL3 belong to the cullin protein family. Multisubunit Cullin-RING structure ligase is the most diverse E3 and plays important roles in AD pathology. The CUL1-dependent ubiquitination process is activated by the Neddylation process, which regulates the degradation process of APP and causes APP to be degraded by endocytosis. Studies have shown that downregulated CUL1 leads to APP aggregation and promotes Aβ production.[41] Under normal conditions, the KEAP1-CUL3 complex polyubiquitylates NRF2

and is subsequently degraded by the proteasome. However, the interaction between the KEAP1-CUL3 complex and NRF2 is unstable under oxidative stress conditions,[42] which may cause NRF2 aggregation and inhibit its downstream gene expression in AD.[43] EIF3I encodes eukaryotic translation initiation factor 3 subunit I and is involved in the transcription process of Aβ23.[44] NSMCE1 encodes a subunit of the SMC5-SMC6 complex and plays an important role in DNA repair, and overexpression of NSMCE1 inhibits APP and other AD marker genes in mice.[24] A previous study showed that heterozygous mutations in PAFAH1B1 cause learning and behavior disorders in mice.[45] Compared with cerebral malaria-resistant mice, PAFAH1B1 was reduced in cerebral malaria-susceptible mice and was associated with AD.[46] Several studies reported that UCHL1 mRNA and protein levels were reduced in AD.[47–50] A previous study showed identified UCHL1 was deregulated in four out of six brain regions, suggested that there were region-specific changes in UPS components.[50] Overexpression of UCHL1 in APP23 transgenic mice delayed AD progression.[51] However, a correlation between RNF175 and AD has not yet been reported. Furthermore, studies have reported that the substrate genes of hub E3 have certain correlations with AD, such as EDF1,[52] TAF7,[53] and TAF3.[54] These substrate genes are correlated to transcription functions.

There were several limitations to this study. Firstly, the potential substrate genes of hub E3 genes were obtained through database mining and literature search, the interactions between hub E3 genes and the substrate genes still need experimental verification. Secondly, the combination of multiple E1-E2-E3 axes is still unknown, and more in-depth research is needed. Lastly, this study performed only transcriptome level analysis. Whether the protein expression of these E3 or substrates are consistent with gene expression remains to be confirmed.

Conclusion

In summary, the damaged UPS extensively and persistently exacerbates the AD process. This study identified seven hub E3 genes (CUL1, CUL3, EIF3I, NSMCE1, PAFAH1B1, RNF175, and UCHL1) and their substrate genes (EDF1, TAF3, TAF7, etc.) affect transcription functions and then exacerbate AD. Among these E3 genes, CUL1, EIF3I, and NSMCE1 were both downregulated in blood and brain and may be used as potential markers for AD diagnosis. However, the underlying mechanism of the interactions between E3 and substrates affecting AD remains unclear. Therefore, revealing the E3-substrate regulatory mechanisms in future works will help develop diagnostic biomarkers and therapeutic targets for AD.

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Supplemental material

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Justin Liu received his Bachelor's degree in microbiology at University of California, San Diego (UCSD) which is one of the world's leading public research universities. Shortly after he graduated from UCSD, he worked as a research coordinator at Edwards Lifesciences, a public company in California. Currently, he entered a graduate program as Master student in statistics at University of California, Riverside (UCR). He had published several research papers in the field of life sciences.

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Original Article

Integrated analysis of 10 lymphoma datasets identifies E2F8 as a key regulator in Burkitt's lymphoma and mantle cell lymphoma

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Abstract: Burkitt's lymphoma (BURK), diffuse large B-cell lymphoma (DLBCL) and mantle cell lymphoma (MCL) are three main types of B-cell lymphomas. This study aimed to compare the differences of affected biological functions and pathways, as well as to explore the possible regulatory mechanisms and the potential therapeutic targets in BURK, DLBCL and MCL. We performed an integrated analysis of 10 lymphoma datasets including 352 BURK patients, 880 DLBCL patients, 216 MCL patients, and 33 controls. Our results showed that signaling pathways, amino acid metabolism and several lipid metabolism pathways varies considerably among these three types of lymphoma. Furthermore, we identified several key transcription factors (TFs) and their target genes that may promote these diseases by influencing multiple carcinogenic pathways. Among these TFs, we reported first that E2F8 displayed the most significant effects in BURK and MCL. Our results demonstrate that over-expression of E2F8 activates target genes that may promote cell cycle, mitosis, immune and other cancer related functions in BURK and MCL. Therefore, we suggest that E2F8 could be used as a biomarker and potential therapeutic target for BURK and MCL. These findings would be helpful in the study of pathogenesis, and drug discovery and also in the prognosis of B cell lymphomas.

Keywords: Burkitt's lymphoma, diffuse large B-cell lymphoma, mantle cell lymphoma, transcriptional regulation network, E2F8

Introduction

Non-Hodgkin's lymphomas (NHLs) are broadly classified into two major groups: B-cell lymphomas and T-cell/NK-cell lymphomas. B-cell lymphomas develop from abnormal B lymphocytes and account for 85-90% of all NHLs [1, 2]. Burkitt's lymphoma (BURK), diffuse large B-cell lymphoma (DLBCL) and mantle cell lymphoma (MCL) are three main types of B-cell lymphomas. BURK is often associated with Epstein-Barr Virus (EBV) infection, being endemic in some parts of Africa but encountered only sporadically in the rest of the world. BURK is largely diagnosed in children. In Cameroon, the country with the highest incidence rate worldwide, the average incidence in the northwest province is 2.58 cases per 100,000 per year in children aged 15 or younger [3]. DLBCL accounts for approximately 33% of all newly diagnosed cases of NHL worldwide. It is most commonly diagnosed during the sixth decade, and is slightly more common in males [4]. Epidemiology on MCL is primarily available for the U.S. and Europe, where the average incidence is approximately 0.5 cases per 100,000 person-years. The male-to-female ratio is around 2.5:1, and the average age of onset is 65-70 years [5, 6].

Immunosuppression is the main risk factor for NHLs [1, 7]. Other putative environmental and lifestyle-related risk factors have also been proposed, including current cigarette smoking [8] and heavy alcohol intake [9].

In addition to the traditional risk factors, recent studies have identified several NHL risk genes. A previous study found that the co-expression of MYC/BCL6 may affect the survival in patients with DLBCL [10]. Another study suggested that subset-specific B-cell-associated gene signatures were better suited to identifications than the traditional classifications (biologic, morphologic, and clinical heterogeneity) in clinical diagnosis and prognosis of DLBCL [11]. In addition, the top discriminatory genes expressed in formalin-fixed, paraffin-embedded (FFPE) tissues of DLBCL patients also showed high statistical difference and contained novel biology with potential therapeutic insights [12]. Furthermore, gene expression signatures were performed well in distinguishing BURK and DLBCL [13, 14]. Still another recent study identified that the deregulation of CUL4A, ING1 and MCPH1 affects the proliferation; and DNA damage response pathways in MCL by integrating gene expression and copy number changes [15]. Palomero *et al.* showed that over-expression of SOX11 activated PDGFA and promoted tumor microvascular angiogenesis in MCL cell lines, suggesting that the SOX11-PDGFA axis may serve as a potential therapeutic target of MCL [16]. Saba *et al.* demonstrated that B-cell receptor activation and canonical NF-kB signaling may promote tumor proliferation and affect overall survival in MCL patients [17].

Reconstruction of transcriptional regulation networks (TRNs) has been a useful approach that provides a global view of gene expression profiles and their regulations, and it also provides new insights in tumorigenesis, pathways and regulations [18, 19], as well as differentiation of embryonic stem cells [20]. Our previous study has identified several tissue-specific transcription factors (TFs) that correlated with breast cancer by analyzing tissue-specific TRNs [21]. However, to our knowledge, no other studies have compared multiple types of NHLs from the perspective of TRNs. Therefore, in the present study, we integrated 10 lymphoma microarray datasets including BURK, DLBCL and MCL in order to explore common differential changes in global gene expression, pathway and TRN analyses, as well as to identify key regulators and their targets among these three types of lymphoma.

Materials and methods

Lymphoma datasets collection

Transcriptome datasets of BURK, DLBCL and MCL were downloaded from NCBI-GEO (https://www.ncbi.nlm.nih.gov/geo/) in Jun, 2017. The data selection criteria were as follows: (1) all datasets were genome-wide; (2) the samples in each dataset must be ≥ 10; (3) all samples were non-cell-line samples; and (4) complete microarray raw or normalized data were available. Based on the above criteria, we chose 10 datasets for our integrated analysis. The details of these datasets are illustrated in **Table 1**. In total, we collected samples from 352 BURK patients, 880 DLBCL patients, 216 MCL patients, and 33 controls.

Data preprocessing and differential expression analysis

R statistical software v3.3.3 (https://www.r-project.org/) was used to perform data preprocessing and differential gene expression gene analysies. Because these datasets contain different microarray platforms and different types of lymphoma, they were first grouped into 14 batches according to study, category (different types of lymphoma and controls), and platform. Each batch contained only one study, one type of lymphoma or control, and one platform (**Table 2**). We used the Robust Multichip Average (RMA) algorithm in oligo package [22] to normalize the raw expression data and generate normalized gene expression intensity. Gene annotation, integration and renormalization of the 14 batches was carried out using a custom designed Python code. We removed probes with no gene annotation or those that matched multiple gene symbols. Next, we calculated the average expression value of multiple probe IDs matching an official gene symbol and took this value to represent the expression intensity of the corresponding gene symbol. The renormalization method and scripts are detailed in our previous publications [23, 24]. The distributions of RMA-processed and global-renormalized gene expression values across all studies are shown in Figures S1 and S2. After global

Table 1. Details of lymphoma datasets

GEO ID	Contributor	Title	Samples
Burkitt's lymphoma			
GSE4475	Hummel M, 2006	A Biologic Definition of Burkitt's Lymphoma from Transcriptional and Genomic Profiling	221 cases
GSE69053	Sha C, 2015	Transferring genomics to the clinic: distinguishing Burkitt and Diffuse large B-cell lymphoma	131 cases
Diffuse large B-cell lymphoma			
GSE31312	Li Y, 2011	Development and application of a new immunophenotypic algorithm for molecular subtype classification of Diffuse Large B-Cell Lymphoma (DLBCL): Report from an International DLBCL Rituximab-CHOP Consortium Program Study	498 cases
GSE56315	Bødker JS, 2014	Diffuse Large B-Cell Lymphoma Classification System That Associates Normal B-Cell Subset Phenotypes With Prognosis	89 cases 33 controls
GSE64555	Linton K, 2014	Microarray gene expression analysis of FFPE tissue permits molecular classification in diffuse large B-cell lymphoma	40 cases
GSE69053	Sha C, 2015	Transferring genomics to the clinic: distinguishing Burkitt and Diffuse large B-cell lymphoma	212 cases
GSE86613	Bødker JS, 2016	Expression data from diagnostic samples of diffuse large B-cell lymphomas (DLBCL), follicular lymphoma (FL) and primary and relapsed transformed FL	41 cases
Mantle cell lymphoma			
GSE21452	Staudt LM, 2010	Integrated genomic profiling in mantle cell lymphoma	64 cases
GSE36000	Jares P, 2012	Mantle Cell Lymphoma	38 cases
GSE70910	Liu D, 2015	Direct in vivo evidence for B-cell receptor and NF-KB activation in mantle cell lymphoma: role of the lymph node microenvironment and activating mutations	55 cases
GSE93291	Staudt LM, 2017	A new molecular assay for the proliferation signature in mantle cell lymphoma applicable to formalin-fixed paraffin-embedded biopsies	59 cases

Table 2. Grouping the lymphoma datasets for data integration and global renormalization

GEO ID	Category	Platform	Batch
GSE4475	Burkitt's lymphoma	GPL96	1
GSE21452	Mantle cell lymphoma	GPL570	2
GSE31312	Diffuse large B-cell lymphoma	GPL570	3
GSE36000	Mantle cell lymphoma	GPL570	4
GSE56315	Diffuse large B-cell lymphoma	GPL570	5
GSE64555	Diffuse large B-cell lymphoma	GPL570	6
GSE69053	Burkitt's lymphoma	GPL8432	7
GSE69053	Diffuse large B-cell lymphoma	GPL8432	8
GSE69053	Burkitt's lymphoma	GPL14951	9
GSE69053	Diffuse large B-cell lymphoma	GPL14951	10
GSE70910	Mantle cell lymphoma	GPL570	11
GSE86613	Diffuse large B-cell lymphoma	GPL570	12
GSE93291	Mantle cell lymphoma	GPL570	13
GSE56315	Control	GPL570	14

expression was renormalized, the distribution of gene expression values across all studies had a consistent range. The empirical Bayesian algorithm in the limma package [25] was used to detect differentially expressed genes between lymphoma patients and controls. Up- and down-regulated genes were defined as a log2 transformed fold-change (logFC) ≥ log2(1.5) or ≤ -log2(1.5), respectively. A *P* value ≤ 0.05 was considered as significant.

Gene set enrichment analysis

We used javaGSEA Desktop Application v2.2.4 to perform gene set enrichment analysis (GSEA) of the three types of lymphoma [26]. KEGG gene sets v6.0 (including 186 gene sets) was chosen to perform KEGG pathway enrichment analysis in BURK, DLBCL and MCL. The analysis parameters were set as follows: gene sets containing fewer than 15 genes or more than 500 genes were excluded, a phenotype label was set as case vs. control, and the t-statistic mean of the genes was computed in each KEGG pathway using a permutation test with 1000 replications. The up-regulated pathways were defined by a normalized enrichment score (NES) > 0 and the down-regulated pathways were defined by an NES < 0. Pathways with a *P* value ≤ 0.05 were chosen as significantly enriched.

Transcriptional regulation network analysis

We downloaded 1,691 human transcription factors (TFs) from the AnimalTFDB 2.0 web server (http://bioinfo.life.hust.edu.cn/AnimalTFDB/) [27] and mapped 915 TFs to our lym-

Table 3. Differentially expressed genes in three types of lymphoma

Lymphoma	Cases/Comtrols	Up-regulated	Down-regulated
Burkitt's lymphoma	352/33	216	205
Diffuse large B-cell lymphoma	880/33	198	89
Mantle cell lymphoma	216/33	230	67

phoma datasets. GENIE3 (Gene Network Inference with Ensemble of trees) [28] was used to reconstruct transcriptional regulatory networks for BURK, DLBCL and MCL. We used the gene expression matrix of the three types of lymphoma and the 915 mapped transcription factors as the input data and ran GENIE3 with its default parameters. Next, we extracted the top 10,000 interactions from the original output and used Cytoscape v3.4.0 to visualize the output results. The NetworkAnalyzer tool in Cytoscape was used to analyze the three resultant networks.

Tissue-specific network analysis

GIANT web server (http://giant.princeton.edu/) [29] was implemented to construct genome-scale integrated analysis of gene networks in BURK, DLBCL and MCL. We filtered the TFs with nodes containing greater than 100 in transcriptional regulatory networks of the three types of lymphoma, and then used these TFs in each type of lymphoma as input parameters to perform gene network analysis. The tissue option chosen for the GIANT web server was "lymph node". The server generated a network of query genes and target genes, as well as performing GO Biological Process enrichment analysis of the genes in the network. GO Biological Processes with a P value ≤ 0.05 was considered as significantly enriched.

Identify key regulators and target genes

We filtered key transcription factors based on the relationship confidence in the GIANT network in BURK, DLBCL and MCL, respectively. A Venn diagram was used to screen the differentially expressed target genes in the GIANT networks in the three types of lymphoma. We used univariate linear model (the "lm" function in R) to calculate the correlation between the filtered key transcription factors and differentially expressed target genes in each type of lymphoma. The absolute value of correlation coefficient ≥ 0.5 and a P value ≤ 0.05 was considered as significant. We used the "plot" function in R to show the log2 transformed fold changes and the correlation coefficients of each paired transcription factors and target genes in the three types of lymphoma.

Results

Differentially expressed genes in three types of lymphoma

In total, we found 11,511 genes in the integrated lymphoma datasets. Differentially expressed genes in the three types of lymphoma were illustrated in **Table 3**. We obtained 216 up-regulated and 205 down-regulated genes, 198 up-regulated and 89 down-regulated genes, as well as 230 up-regulated and 67 down-regulated genes in BURK, DLBCL, and MCL, respectively. The top 20 differentially-expressed genes in three types of lymphoma were shown in Table S1. In addition, we noted that 178 common genes were deregulated in the three types of lymphoma (Figure S3A).

Commonly and differentially deregulated KEGG pathways

We screened the pathways that were significantly enriched in at least one type of lymphoma. The results were shown in **Figure 1**. There were 13 commonly enriched pathways (10 up-regulated and 3 down-regulated) in BURK, DLBCL, and MCL. These pathways include cell cycle, amino sugar and nucleotide sugar metabolism, PPAR signaling pathways, pyrimidine metabolism, Fc gamma R-mediated phagocytosis, N-glycan biosynthesis and other cancer-related pathways. Furthermore, we observed tyrosine metabolism, GnRH signaling pathway, vascular smooth muscle contraction, neuroactive ligand-receptor interaction, fatty acid and other lipid metabolism were common down-regulated in BURK and DLBCL, whereas these pathways showed no change in MCL (**Figures 1** and S4). In addition, the JAK-STAT signaling pathway, adherents junction and chemokine signaling pathways were commonly deregulated in DLBCL and MCL, whereas these pathways did not reach significant difference in BURK (**Figures 1** and S4). Furthermore, there were

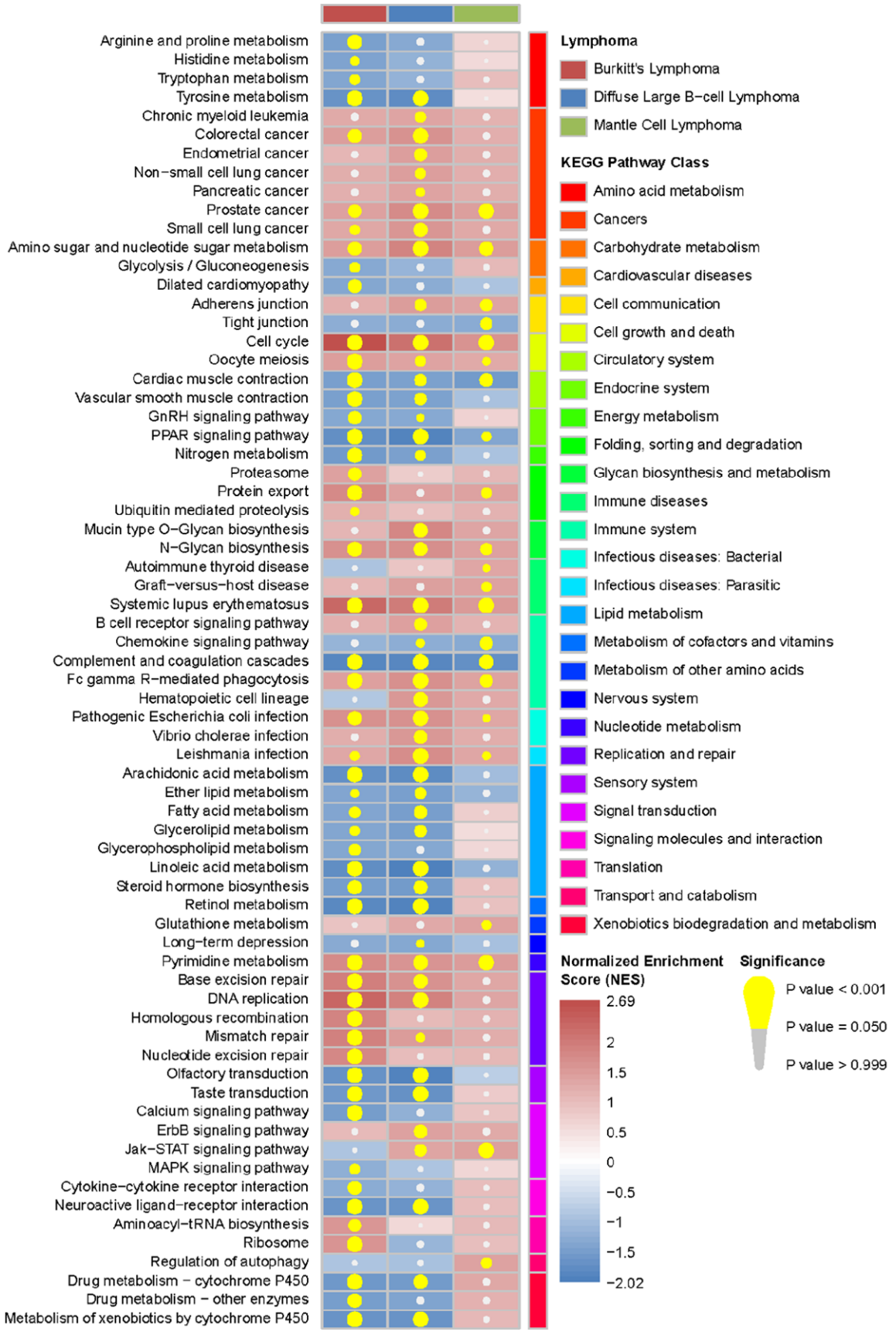

Arginine and proline metabolism
Histidine metabolism
Tryptophan metabolism
Tyrosine metabolism
Chronic myeloid leukemia
Colorectal cancer
Endometrial cancer
Non−small cell lung cancer
Pancreatic cancer
Prostate cancer
Small cell lung cancer
Amino sugar and nucleotide sugar metabolism
Glycolysis / Gluconeogenesis
Dilated cardiomyopathy
Adherens junction
Tight junction
Cell cycle
Oocyte meiosis
Cardiac muscle contraction
Vascular smooth muscle contraction
GnRH signaling pathway
PPAR signaling pathway
Nitrogen metabolism
Proteasome
Protein export
Ubiquitin mediated proteolysis
Mucin type O−Glycan biosynthesis
N−Glycan biosynthesis
Autoimmune thyroid disease
Graft−versus−host disease
Systemic lupus erythematosus
B cell receptor signaling pathway
Chemokine signaling pathway
Complement and coagulation cascades
Fc gamma R−mediated phagocytosis
Hematopoietic cell lineage
Pathogenic Escherichia coli infection
Vibrio cholerae infection
Leishmania infection
Arachidonic acid metabolism
Ether lipid metabolism
Fatty acid metabolism
Glycerolipid metabolism
Glycerophospholipid metabolism
Linoleic acid metabolism
Steroid hormone biosynthesis
Retinol metabolism
Glutathione metabolism
Long−term depression
Pyrimidine metabolism
Base excision repair
DNA replication
Homologous recombination
Mismatch repair
Nucleotide excision repair
Olfactory transduction
Taste transduction
Calcium signaling pathway
ErbB signaling pathway
Jak−STAT signaling pathway
MAPK signaling pathway
Cytokine−cytokine receptor interaction
Neuroactive ligand−receptor interaction
Aminoacyl−tRNA biosynthesis
Ribosome
Regulation of autophagy
Drug metabolism − cytochrome P450
Drug metabolism − other enzymes
Metabolism of xenobiotics by cytochrome P450
Lymphoma
Burkitt's Lymphoma
Diffuse Large B−cell Lymphoma
Mantle Cell Lymphoma
KEGG Pathway Class
Amino acid metabolism
Cancers
Carbohydrate metabolism
Cardiovascular diseases
Cell communication
Cell growth and death
Circulatory system
Endocrine system
Energy metabolism
Folding, sorting and degradation
Glycan biosynthesis and metabolism
Immune diseases
Immune system
Infectious diseases: Bacterial
Infectious diseases: Parasitic
Lipid metabolism
Metabolism of cofactors and vitamins
Metabolism of other amino acids
Nervous system
Nucleotide metabolism
Replication and repair
Sensory system
Signal transduction
Signaling molecules and interaction
Translation
Transport and catabolism
Xenobiotics biodegradation and metabolism
Normalized Enrichment Score (NES)
2.69
2
1.5
1
0.5
0
−0.5
−1
−1.5
−2.02
Significance
P value < 0.001
P value = 0.050
P value > 0.999

Figure 1. GSEA results of enriched KEGG pathways in Burkitt's lymphoma, diffuse large B-cell lymphoma and mantle cell lymphoma.

16, 10 and 5 pathways that were only significantly enriched in BURK, DLBCL, and MCL, respectively. These results suggest a relatively large difference in these three types of lymphoma.

Expression profiles of transcription factors and transcriptional regulatory networks

There were 29, 14 and 16 differentially expressed transcription factors in BURK, DLBCL, and MCL, respectively (**Figure 2A**). Among them, 11 deregulated transcription factors were shared by all the three types of lymphoma (Figure S3B). Transcriptional regulatory networks (TRNs) of three types of lymphoma were shown in Figures S5, S6, S7. Network properties of the three TRNs were listed in Table S2. We found that MCL TRN had the highest clustering coefficient of 0.120; the clustering coefficient in BURK TRN was 0.049 and in DLBCL it was 0.079. Furthermore, MCL TRN showed the highest connected component of 101, followed by DLBCL TRN (76) and BURK TRN (56). In addition, MCL TRN had the most multi-edge node pairs at 130, followed by DLBCL TRN (89) and BURK TRN (68). We sorted transcription factors for each of the TRNs according to the number of nodes, and then extracted the transcription factors with more than 100 nodes. In total, we got 8, 11 and 24 top transcription factors for BURK, DLBCL, and MCL, respectively (**Figures 2B-D**, S5, S6, S7). We revealed that no top TF was shared by all three types of lymphoma, only E2F8 was shared by BURK and MCL, and CDX2 was shared by BURK and DLBCL (**Figure 2E**). We combined these differentially expressed TFs and top-node TFs in each type of lymphoma and found the intersected TFs in BURK was E2F8, and in MCL were PRDM1, E2F8, VDR and IRF4 (Figure S8). No deregulated top-node TF was found in DLBCL.

GIANT networks and enriched biological functions

Figure 3 shows the GIANT networks of top-node TFs and their biological functions in each of the three types of lymphoma. We listed the top 20 enriched GO Biological Processes in each type of lymphoma (only 5 significantly enriched items in DLBCL). In BURK, only E2F8 showed the highest relationship confidence with target genes. In DLBCL, there were 6 genes (YY1, STAT1, HMGB1, PMS1, RNF138 and YBX1) which showed a relatively high relationship confidence with target genes. Furthermore, WHSC1, E2F8, HMGB3, E2F1 and FOXM1 had the highest relationship confidence with target genes in MCL. The enriched biological functions in BURK were mainly cell division, mitosis, microtubule cytoskeleton organization, nuclear division, chromosome segregation and other cell cycle related functions (**Figure 3A**). In DLBCL, there were 5 enriched DNA replication and transcription related biological processes. In addition, several immune-related biological functions (interleukin-13 production, interleukin-4 production, vitamin D receptor signaling pathway and other functions) and cell cycle related functions were significantly enriched in MCL. Based on the GIANT networks, we therefore speculate that these top-node TFs may play key roles in BURK and MCL.

Key transcription factors and their target genes in three types of lymphoma

We combined target genes in the GIANT network and deregulated genes in BURK, DLBCL and MCL (**Figure 4A-C**). There were 15, 3 and 8 target genes that were deregulated in BURK, DLBCL and MCL, respectively. Based on the above results, we calculated the correlations between the key TFs and deregulated target genes in each type of lymphoma. Interestingly, these target genes were all up-regulated. In BURK, there were significant correlations between E2F8 and 15 downstream target genes (**Figure 4D**). However, the key TFs in DLBCL showed no expression changes and relatively lower correlations with target genes (**Figure 4E**). Furthermore, the key TFs and target genes in MCL presented the highest correlations of all (**Figure 4F**). Among these key TFs, E2F8 was significantly up-regulated, while E2F1 and HMGB3 showed boundary significance.

Possible regulatory mechanisms in three types of lymphoma

Through integration the results of differentially expressed genes, affected pathways and biological functions, transcriptional regulation net-

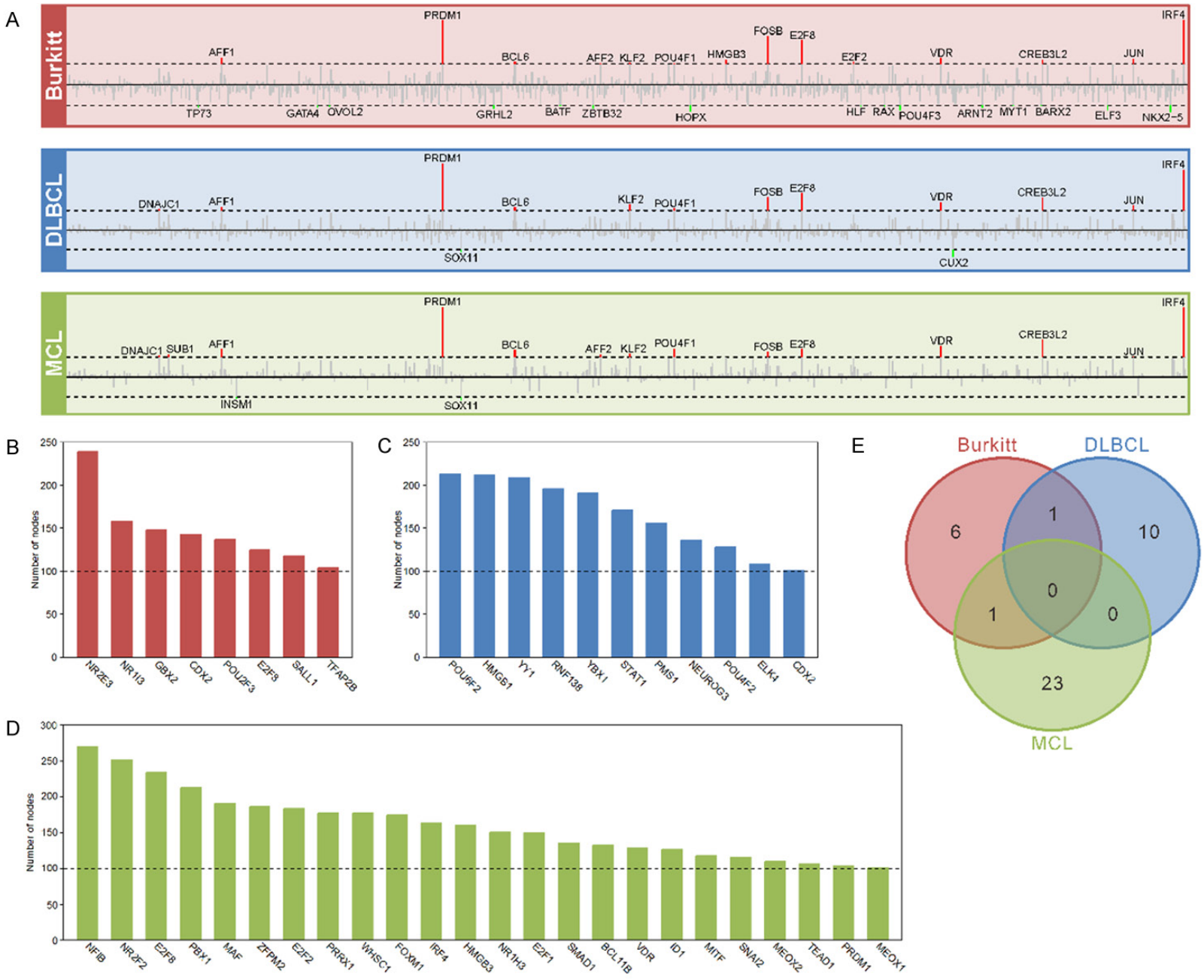

A
Burkitt
PRDM1
IRF4
FOSB
E2F8
AFF1
BCL6
AFF2
KLF2
POU4F1
HMGB3
E2F2
VDR
CREB3L2
JUN
TP73
GATA4
OVOL2
GRHL2
BATF
ZBTB32
HOPX
HLF
RAX
POU4F3
ARNT2
MYT1
BARX2
ELF3
NKX2-5
DLBCL
DNAJC1
SOX11
CUX2
MCL
SUB1
INSM1
B
C
D
Number of nodes
NR2E3
NR1I3
GBX2
CDX2
POU2F3
E2F3
SALL1
TFAP2B
POU5F2
HMGB1
YY1
RNF138
YBX1
STAT1
PMS1
NEUROG3
POU4F2
ELK1
CDX2
NFIB
NR2F2
E2F8
PBX1
MAF
ZFPM2
E2F2
PRRX1
WHSC1
FOXM1
IRF4
HMGB3
NR1H3
E2F1
SMAD1
BCL11B
VDR
ID1
MITF
SNAI2
MEOX2
TEAD1
PRDM1
MEOX1
E
Burkitt
DLBCL
MCL
6
1
10
0
1
0
23

Figure 2. Differentially expressed and multiple connected transcription factors in three types of lymphoma. A. log2 transformed fold change of all transcription factors in three different types of lymphoma. B-D. transcription factors with nodes ≥ 100 in Burkitt's lymphoma, diffuse large B-cell lymphoma and mantle cell lymphoma, respectively. E. Venn diagram of transcription factors with nodes ≥ 100 in three types of lymphoma.

works and TF-target interactions, we propose the possible regulatory mechanisms in each type of lymphoma (**Figure 5**). In BURK, the up-regulated E2F8 activated 15 downstream target genes, and these genes further affected multiple cell cycle related pathways, eventually resulting in cancer. However, in DLBCL, the effect of these transcription factors may not be carcinogenic. Furthermore, in MCL, the highly-expressed E2F8, E2F1 and HMGB3 activated 8 downstream target genes, and influenced cell cycle and immune functions, also resulting in cancer.

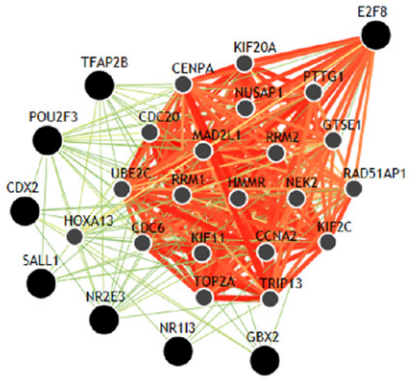
Discussion

In the present study, we observed there were 13 commonly enriched pathways and 31 specific enriched pathways in BURK, DLBCL and MCL. Furthermore, through reconstruction of TRNs and GIANT networks analysis in the three types of lymphoma, we identified several key TFs and their target genes in BURK, DLBCL and MCL, respectively. In particular, E2F8 was up-regulated in all the three types of lymphoma and was also the top TF (nodes ≥ 100) in BURK and MCL. We further found that several E2F8 target genes were over-expressed in BURK and MCL, and that these genes play crucial roles in cell cycle and immune-related functions. However, the top TFs may not cause a carcinogenic effect in DLBCL.

Previous studies have demonstrated some important pathways result in a carcinogenic effect. Multiple tumor-related metabolic pathways (energy metabolism, amino acid metabolism, fatty acid metabolism and choline phospholipid metabolism) were identified in BURK mouse models [30]. Fatty acid oxidation and up-regulated carnitine palmitoyltransferase I (CPTI) have been found to promote cancer growth, regulate gene expression and apoptosis in most of cancers, including BURK and DLBCL [31]. In addition, up-regulated fatty acid synthase (FASN) affected lipogenesis and multiple signaling pathways in several cancer cell lines and was suggested as a potential therapeutic target [32]. Our study showed several lipid metabolism pathways (fatty acid metabolism, glycerolipid metabolism, steroid hormone biosynthesis, etc.) were significantly down-regulated in BURK and DLBCL, whereas these pathways showed no change or were slightly elevated in MCL (**Figure 1**). The JAK-STAT signaling pathway is another tumor-related signaling pathway. Deregulation of JAK-STAT has been found in several lymphoma cell lines [33, 34], primary mediastinal B-cell lymphoma [35], DLBCL [36], MCL and follicular lymphoma [37]. Furthermore, the JAK-STAT signaling pathway was also found to be influenced in other hematological malignancies (leukemia, multiple myeloma, myeloproliferative neoplasms, etc.) and suggested that several JAK inhibitors have broad clinical application prospects [38]. In our study, the JAK-STAT signaling pathway was significantly up-regulated in DLBCL and MCL, whereas it exhibited no change in BURK. Furthermore, our study also showed several significantly down-regulated amino acid metabolism pathways (tyrosine metabolism etc.), sensory systems (olfactory transduction and taste transduction) and xenobiotic biodegradation and metabolism (drug metabolism - cytochrome P450 and metabolism of xenobiotics by cytochrome P450) in BURK and DLBCL. However, all these pathways showed no change or even modest increases in MCL. Based on these findings, we propose that these impairment pathways have a certain carcinogenic effect and vary widely in different types of lymphoma.

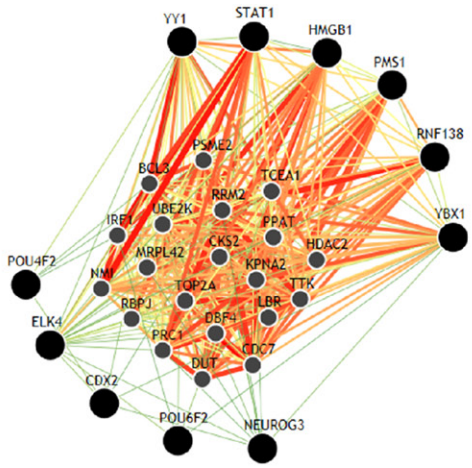
This study identified 11 up-regulated TFs in all the three types of lymphoma. Among them, PRDM1 and IRF4 were most over-expressed and also found in the top 20 list of differentially expressed genes (**Figure 2** and Table S1). Most of these TFs play dominant roles in TRNs and their target genes are related to cell cycle, mitosis, immune and other biological functions (**Figures 3** and S5, S6, S7). PRDM1 belongs to the PRDM gene family of transcription repressors containing Kruppel-type zinc fingers. The SET-related PR (PRDI-BF1-RIZ) domain is a master transcriptional regulator for terminal differentiation of B cells into plasma cells [39, 40]. Previous studies have demonstrated that

A



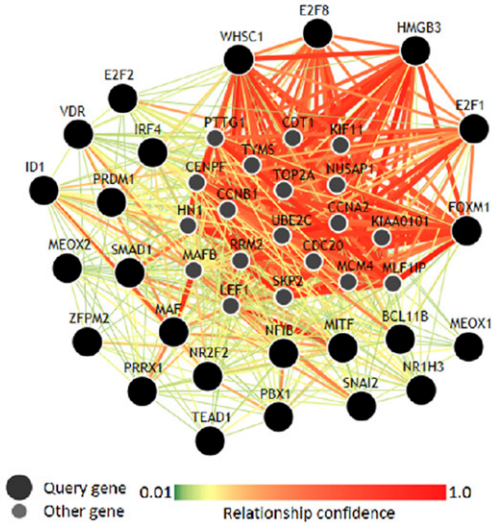
GO Biology Processes	FDR
GO0000226: microtubule cytoskeleton organization	3.54E-04
GO0051783: regulation of nuclear division	3.54E-04
GO0007067: mitosis	3.67E-04
GO0007088: regulation of mitosis	3.99E-04
GO0000280: nuclear division	4.28E-04
GO0051301: cell division	4.54E-04
GO0007017: microtubule-based process	4.55E-04
GO0048285: organelle fission	5.56E-04
GO0007059: chromosome segregation	5.58E-04
GO0031145: anaphase-promoting complex-dependent proteasomal ubiquitin-dependent protein catabolic process	6.74E-04
GO0051983: regulation of chromosome segregation	7.06E-04
GO0090068: positive regulation of cell cycle process	1.51E-03
GO0051302: regulation of cell division	1.58E-03
GO0000278: mitotic cell cycle	1.69E-03
GO0040001: establishment of mitotic spindle localization	4.17E-03
GO0051293: establishment of spindle localization	8.69E-03
GO0051653: spindle localization	9.20E-03
GO0007346: regulation of mitotic cell cycle	1.35E-02
GO0030261: chromosome condensation	1.41E-02
GO0007091: metaphase/anaphase transition of mitotic cell cycle	1.56E-02

B



GO Biology Processes	FDR
GO0000122: negative regulation of transcription from RNA polymerase II promoter	3.22E-03
GO0035458: cellular response to interferon-beta	5.97E-03
GO0006266: DNA ligation	9.91E-03
GO0010225: response to UV-C	1.04E-02
GO0035456: response to interferon-beta	1.11E-02

C



GO Biology Processes	FDR
GO0010564: regulation of cell cycle process	4.85E-03
GO0007346: regulation of mitotic cell cycle	1.69E-02
GO0032616: interleukin-13 production	1.71E-02
GO0070561: vitamin D receptor signaling pathway	1.73E-02
GO0032656: regulation of interleukin-13 production	1.85E-02
GO0010839: negative regulation of keratinocyte proliferation	1.91E-02
GO0071295: cellular response to vitamin	1.91E-02
GO0032673: regulation of interleukin-4 production	2.00E-02
GO0010837: regulation of keratinocyte proliferation	2.00E-02
GO0000122: negative regulation of transcription from RNA polymerase II promoter	2.01E-02
GO0000278: mitotic cell cycle	2.15E-02
GO0032633: interleukin-4 production	2.18E-02
GO0071305: cellular response to vitamin D	2.30E-02
GO0007059: chromosome segregation	2.42E-02
GO0031145: anaphase-promoting complex-dependent proteasomal ubiquitin-dependent protein catabolic process	2.55E-02
GO0051656: establishment of organelle localization	2.58E-02
GO0007067: mitosis	2.60E-02
GO0033280: response to vitamin D	2.69E-02
GO0050680: negative regulation of epithelial cell proliferation	2.71E-02
GO0000280: nuclear division	2.72E-02

Figure 3. Genome-scale integrated analysis of gene networks in Burkitt's lymphoma (A), diffuse large B-cell lymphoma (B) and mantle cell lymphoma (C). The top 20 enriched biological functions of the genes in the networks are shown in the right panel.

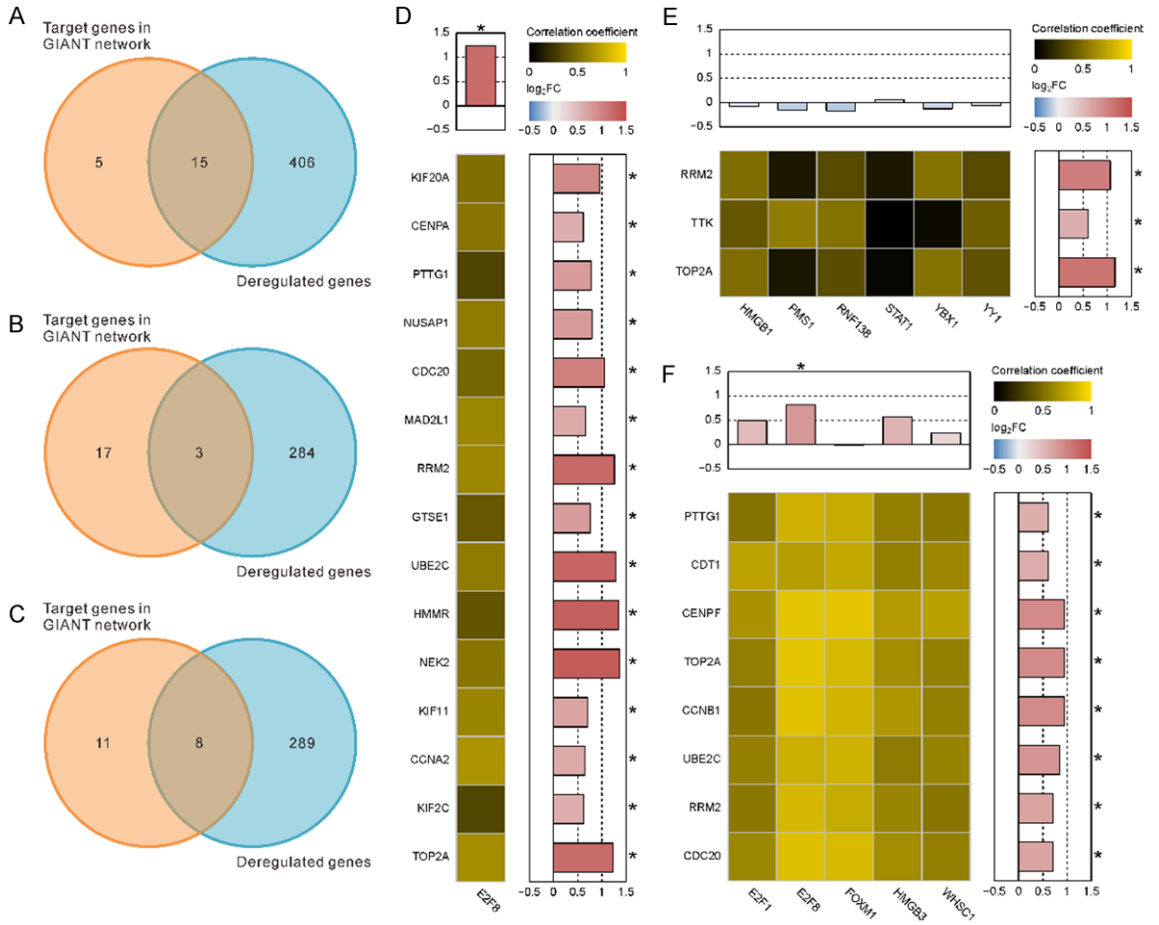


Figure 4. Correlation of key transcription factors and deregulated target genes in three types of lymphoma. A-C. Venn diagram of target genes in GIANT network and deregulated genes in Burkitt's lymphoma, diffuse large B-cell lymphoma and mantle cell lymphoma, respectively. D-F. correlation of key transcription factors and deregulated target genes in Burkitt's lymphoma, diffuse large B-cell lymphoma and mantle cell lymphoma, respectively. The black/yellow color bar indicates the correlation coefficients of key transcription factors and deregulated target genes. The blue/red color bar indicates the log2 transformed fold change of these genes. Significantly differentially expressed genes are indicated by an asterisk.

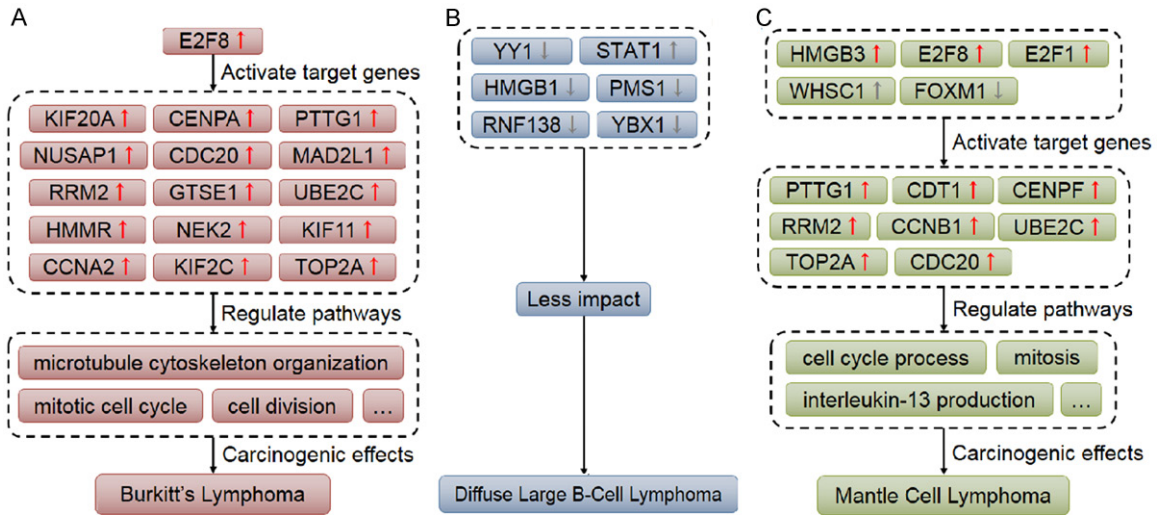


Figure 5. Possible regulatory mechanisms of key transcription factors, target genes, and pathways in three types of lymphoma. A. Up-regulated E2F8 activated several target genes and then affected microtubule cytoskeleton organization, mitotic cell cycle, cell division, and other cancer-related pathways, finally caused Burkitt's lymphoma. B. The expression of these transcription factors may have less effect on diffuse large B-cell lymphoma. C. Up-regulated HMGB3, E2F8 and E2F1 activated several target genes and then affected cell cycle process, mitosis, interleukin-13 production, and other cancer-related pathways, finally resulting in mantle cell lymphoma.

PRDM1 is over-expressed in Hodgkin lymphomas [39], T-cell lymphoma [41] and DLBCL [40, 42]. Furthermore, other studies have demonstrated that over-expression of miR-9 or let-7a reduced PRDM1 levels in multiple cell lines and thus these factors are suggested as potential therapeutic targets for lymphomas [39, 40]. IRF4 (also called MUM1) is a lymphocyte-specific member of the interferon regulatory factor (IRF) family of transcription factors and plays crucial roles in the activation of innate and adaptive immune systems [43]. IRF4 is involved in the t(6;14)(p25;q32) translocation, which causes the juxtaposition of the IRF4 gene (mapping at 6p25) to the IgH locus on 14q32 [43]. Over-expression of the IRF4 is detected in multiple myelomas, HIV-related lymphomas, lymphoplasmacytic lymphomas, peripheral T-cell lymphomas, DLBCL and MCL [43-46] and may represent a possible therapeutic target [46]. Furthermore, we also identified several top-node TFs in the TRNs of the three types of lymphoma, including E2F1, E2F8, HMGB3 etc. A previous study found E2F1 is highly expressed in BURK cell lines and sporadic BURK specimens, suggesting E2F1 as a promising therapeutic target for sporadic BURK [47]. However, recent research showed over-expression of E2F1 has no relationship with prognosis in T lymphoblastic lymphoma or leukemia [48]. The expression of E2F1 in this study was also up-regulated in BURK, DLBCL and MCL, despite all did not reach a significant difference.

E2F8 is a recently identified member of the E2F family. It controls cell cycle genes through homodimers or heterodimers from DNA-binding domains [49]. Furthermore, E2F8 plays a variety of regulatory roles including regulation of placenta architecture and embryonic development, mediation of the DNA damage-correlated cell-cycle, promotion of angiogenesis and lymphangiogenesis, and also influences polyploidization [49]. The traditional view is that E2F8 functions as a transcriptional repressor. Its mediated transcriptional repression is a critical mechanism during postnatal liver development [50] and it also represses a large set of E2F target genes in skin cancer [51]. However, new perspectives noted that E2F8 acts as a transcription activator which confers cisplatin resistance by promoting MASTL mediated mitotic progression in ER+ breast cancer cells [52]. Another breast cancer study demonstrated that E2F8 transcriptionally upregulates CCNE1 and CCNE2 *via* directly interacting with their respective gene promoters, which accelerates the transition from G1 to S phase of breast cancer cells [53]. Furthermore, a recent study suggested that E2F8 promotes papillary thyroid cancer progression by activating CCND1 and other cell cycle related genes [54]. In our study, E2F8 was up-regulated in all the three types of lymphoma, and was the most important TFs in BURK and MCL TRNs (**Figures 2**, S5 and S7). In addition, over-expressed E2F8 target genes influenced cell cycle and disrupted immune functions in BURK and MCL (**Figure 4**). However, to our knowledge, no other study reports the role of up-regulated E2F8 in lymphoma. Our study also discloses that HMGB3 is a key regulator in MCL, although the expression did not achieve significance (**Figure 4**). High mobility group-box 3 (HMGB3) is a member of the HMG superfamily and is classified with HMGB1 and HMGB2 into the HMGB subfamily. It plays important roles in DNA recombination, repair, replication, and transcription [55]. Recent studies showed that high-expression of HMGB3 promotes growth and migration in colorectal cancer [56] and reduces patient survival in esophageal squamous cell carcinoma [57]. Knockdown of HMGB3 also showed a potential therapeutic effect in gastric cancer cells [55]. However, no studies have reported the association between HMGB3 and lymphoma so far.

In summary, this study identifies a series of commonly and specifically deregulated pathways in BURK, DLBCL and MCL, respectively. Furthermore, several key TFs may activate their target genes, influencing cell cycle, mitosis, immune functions and other oncogenic pathways in BURK and MCL. However, these top TFs

may have less impact on DLBCL. It is noteworthy that we are the first to report that E2F8 plays a dominant role in BURK and MCL. Therefore, we suggest that E2F8 can be used as a biomarker and a potential therapeutic target for BURK and MCL. Future studies are required to uncover its potential mechanisms.

Acknowledgements

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Disclosure of conflict of interest

None.

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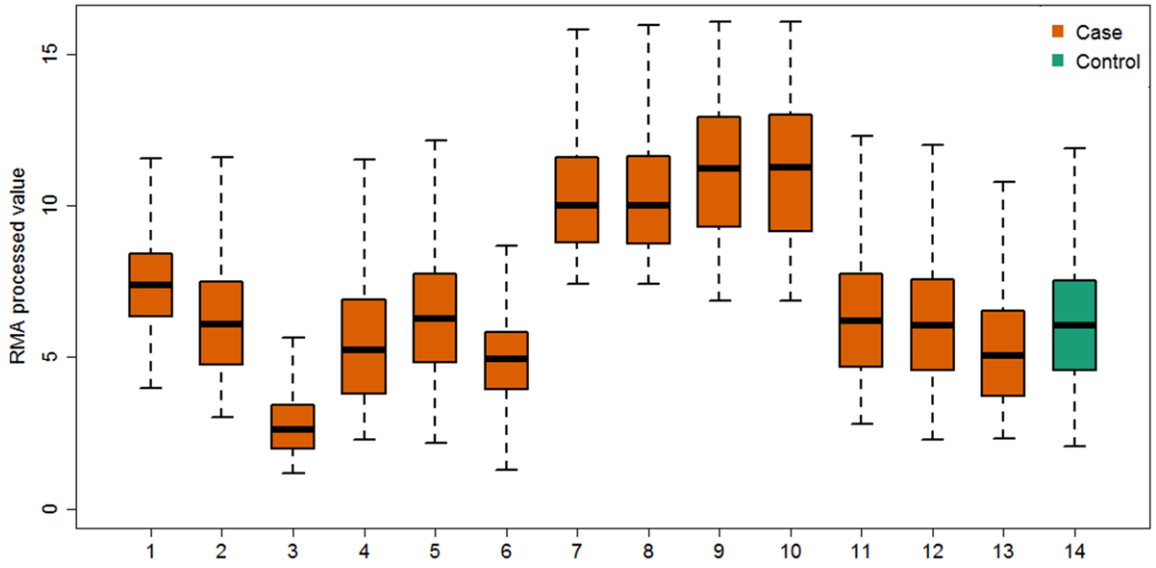

Figure S1. The distribution of RMA processed gene expression values of lymphoma datasets. The orange box shows the distribution of gene expression values in lymphoma patients and the green box shows the distribution of gene expression values in controls. There was a relatively large deviation in the distribution of gene expression values across these studies.

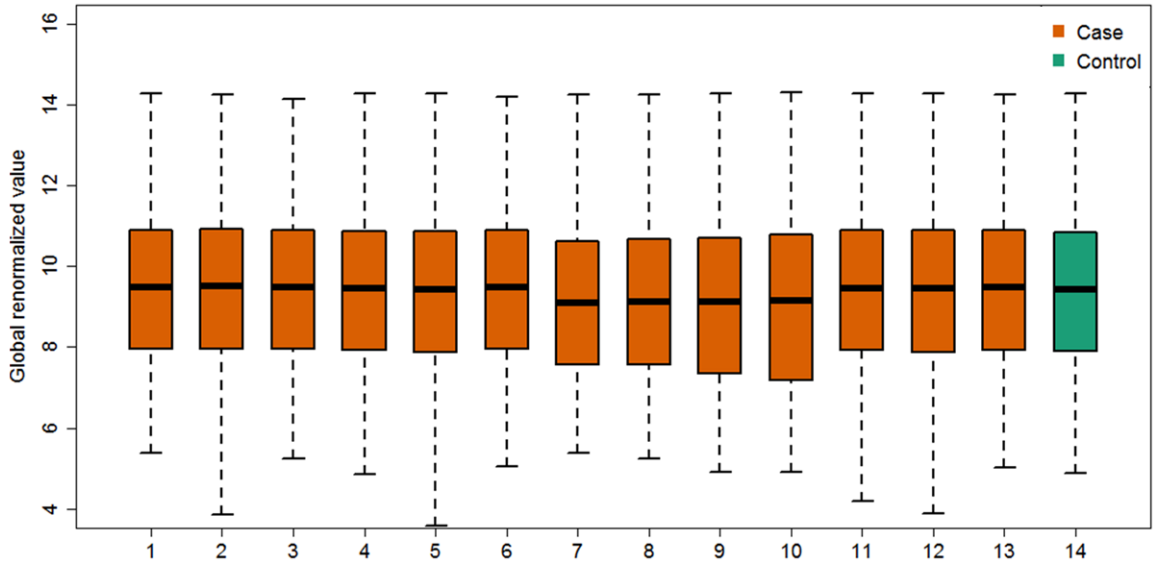

Figure S2. The distribution of global renormalized (after RMA processed) gene expression values of lymphoma datasets. The orange box shows the distribution of gene expression values in lymphoma patients and the green box shows the distribution of gene expression values in controls. The distribution of gene expression values across these studies had a consistent range.

Table S1. Top 20 differentially expressed genes in three types of lymphoma

Burkitt's lymphoma			Diffuse large B-cell lymphoma			Mantle cell lymphoma		
Symbol	logFC	*P* value	Symbol	logFC	*P* value	Symbol	logFC	*P* value
CYorf15B	2.38	4.61E-23	KRT6A	-2.88	1.27E-13	KRT13	-3.13	1.61E-14
HIST2H2BE	2.06	9.57E-25	SPRR1B	-2.36	7.19E-13	KRT6A	-2.78	9.96E-14
RPS4Y1	2.03	3.19E-07	SPRR3	-2.27	4.91E-11	KRT14	-2.40	4.06E-11
CENPE	1.92	7.45E-18	HIST2H2BE	2.14	1.68E-40	SPRR3	-2.37	3.97E-13
KRT14	-1.90	2.30E-09	PRDM1	1.98	1.57E-29	STATH	-2.10	1.71E-07
MMP1	-1.87	9.04E-11	SPRR1A	-1.97	1.96E-10	PRDM1	2.06	6.92E-22
C7orf10	1.85	3.36E-13	STATH	-1.89	1.87E-10	HIST2H2BE	2.05	8.03E-18
KIF18B	1.84	1.34E-17	RPS4Y1	1.84	8.86E-07	IRF4	2.04	8.32E-25
ORM1	-1.83	1.35E-09	CENPE	1.84	2.02E-29	SPRR1B	-1.99	6.25E-12
IRF4	1.81	1.58E-17	IRF4	1.79	9.99E-21	SPRR1A	-1.97	1.31E-11
PRDM1	1.80	2.31E-21	MMP1	-1.78	4.14E-07	C7orf10	1.95	7.34E-18
EIF1AY	1.79	6.70E-11	OLFM4	-1.77	4.07E-12	HTN3	-1.84	5.57E-07
KRT6A	-1.69	1.72E-07	KRT13	-1.75	1.93E-07	GPR15	1.79	2.92E-37
SPC25	1.68	7.87E-18	HTN3	-1.70	3.00E-10	FCGR2A	1.77	5.21E-19
FCGR2A	1.65	4.99E-20	S100A7	-1.66	5.65E-08	S100A2	-1.72	1.51E-09
KRT13	-1.64	5.16E-09	KRT14	-1.65	1.77E-05	CENPE	1.69	7.88E-11
CDKN3	1.62	9.13E-17	CYorf15B	1.62	3.78E-08	CYorf15B	1.64	1.71E-05
HIST1H2BG	1.58	2.20E-20	FCGR2A	1.62	1.92E-23	HTN1	-1.59	6.62E-06
PLAC8	1.55	5.54E-06	SPC25	1.62	3.54E-27	RPS4Y1	1.58	3.02E-03
TPX2	1.52	3.58E-15	S100A2	-1.61	5.68E-09	ZBP1	1.58	6.16E-16

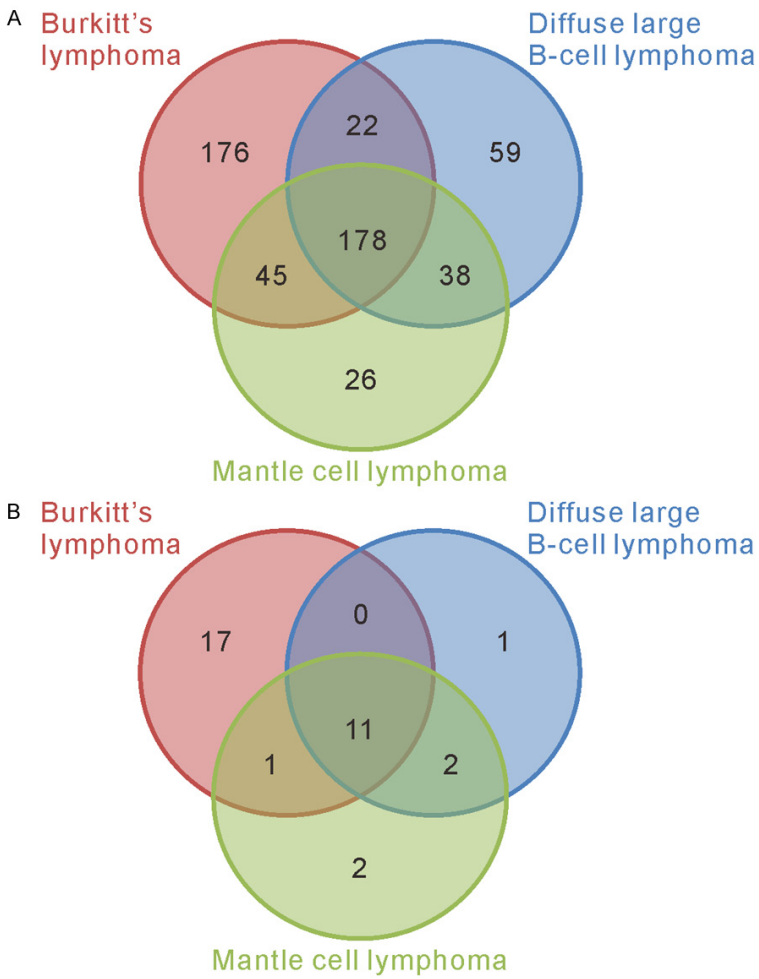

Figure S3. Venn diagram of all differentially expressed genes (A) and differentially expressed transcription factors (B) in three types of lymphoma.

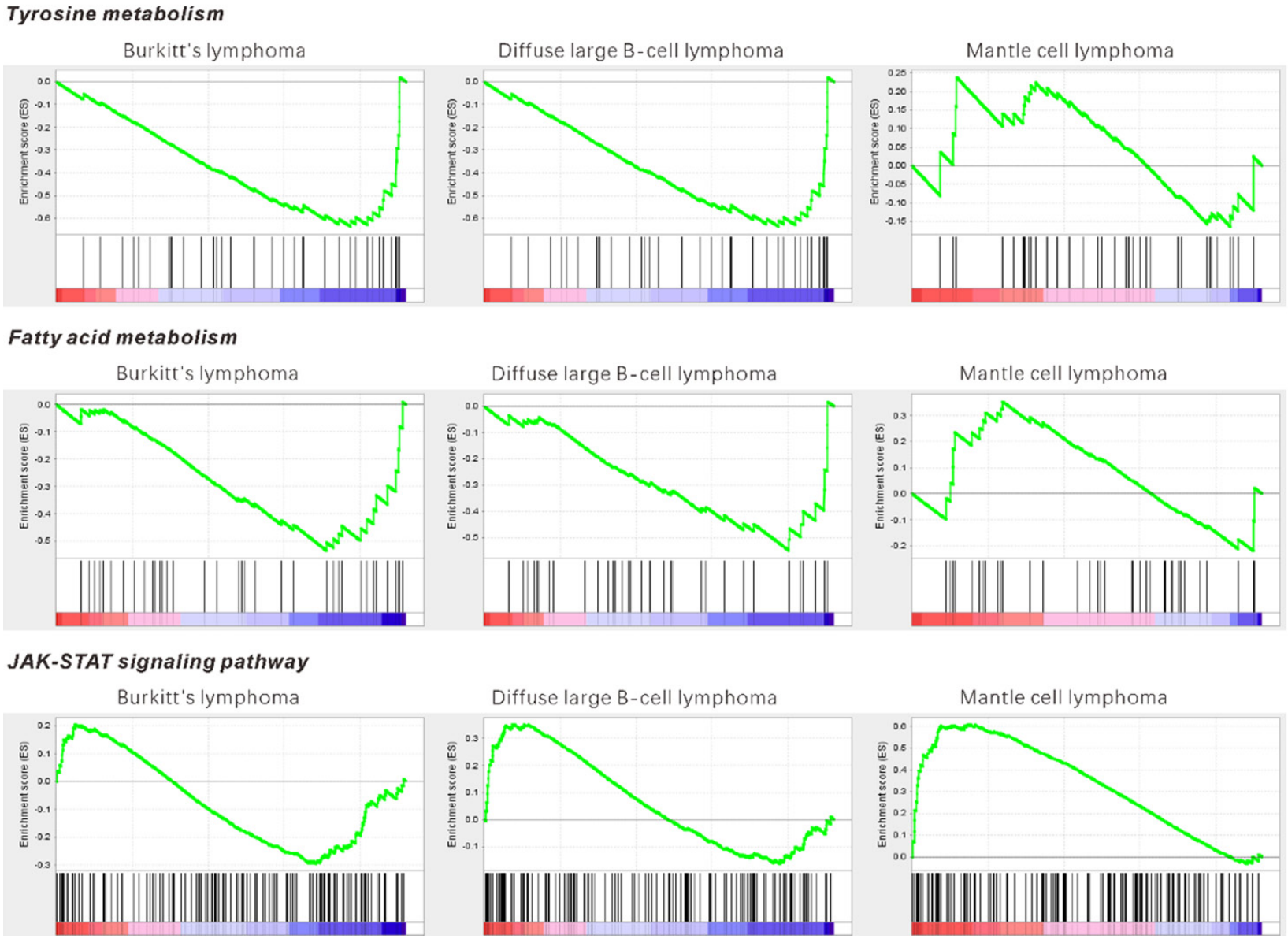

Figure S4. GSEA enrichment results of tyrosine metabolism, fatty acid metabolism and JAK-STAT signaling pathway in three types of lymphoma. A green curve below zero indicates that the pathway is down-regulated, while a green curve above zero indicates that this pathway is up-regulated.

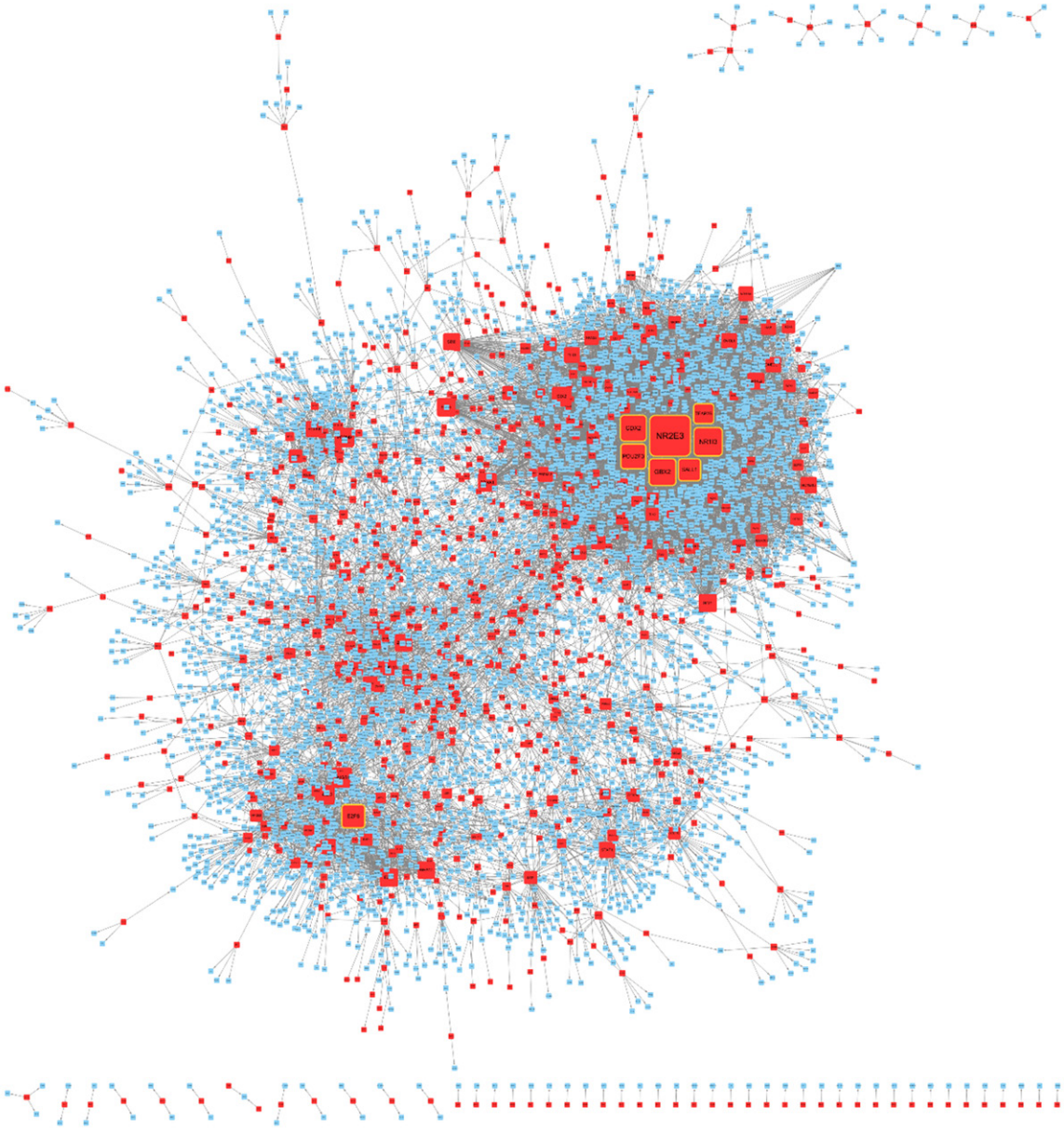

Figure S5. Transcriptional regulatory networks of Burkitt's lymphoma. Red boxes represent transcription factors and blue boxes represent target genes. The box size represents the number of nodes of a gene. Arrows indicate the direction of regulation. Transcription factors with greater than 100 nodes are marked with a yellow box.

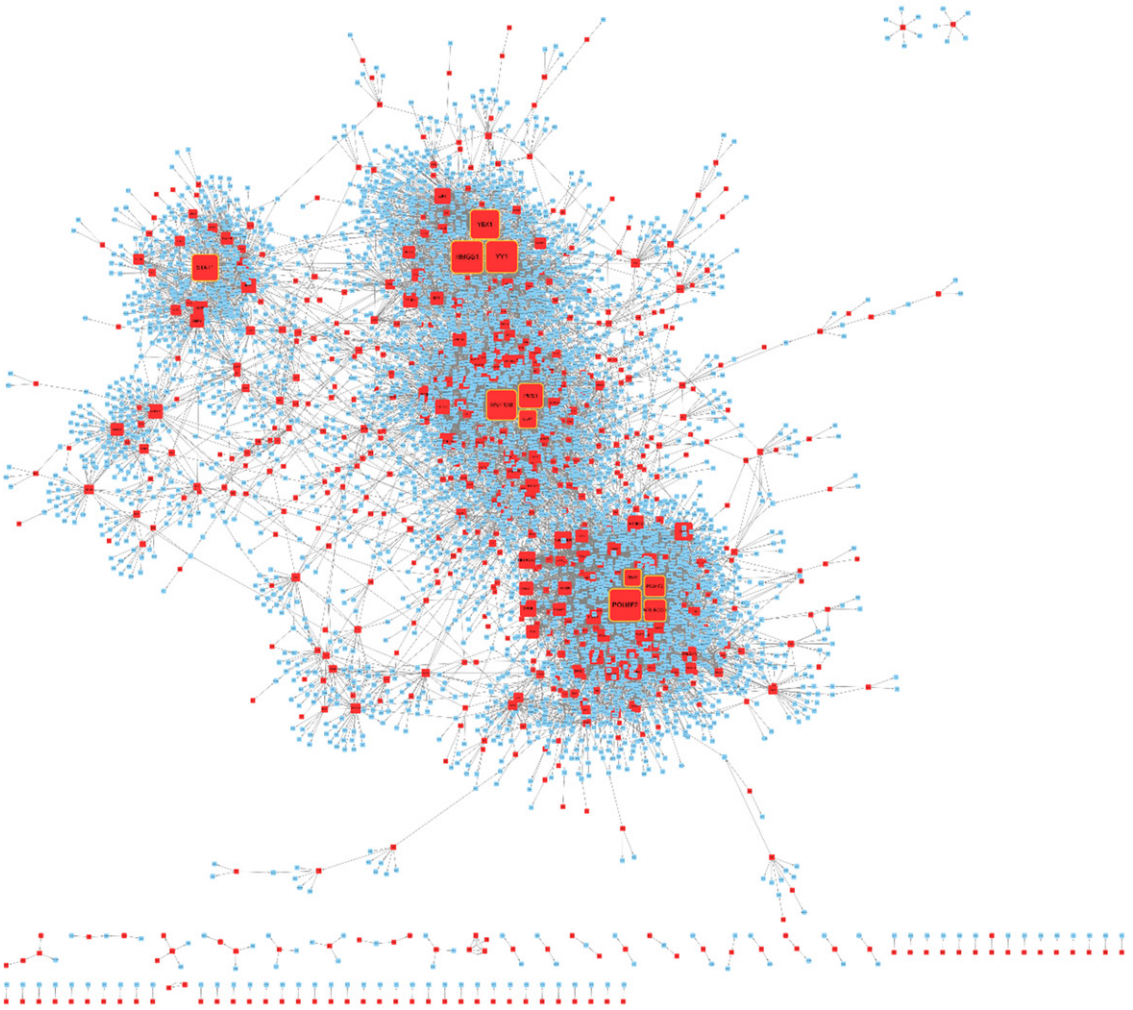

Figure S6. Transcriptional regulatory networks of diffuse large B-cell lymphoma. Red boxes represent transcription factors and blue boxes represent target genes. The box size represents the number of nodes of a gene. Arrows indicate the direction of regulation. Transcription factors with greater than 100 nodes are marked with a yellow box.

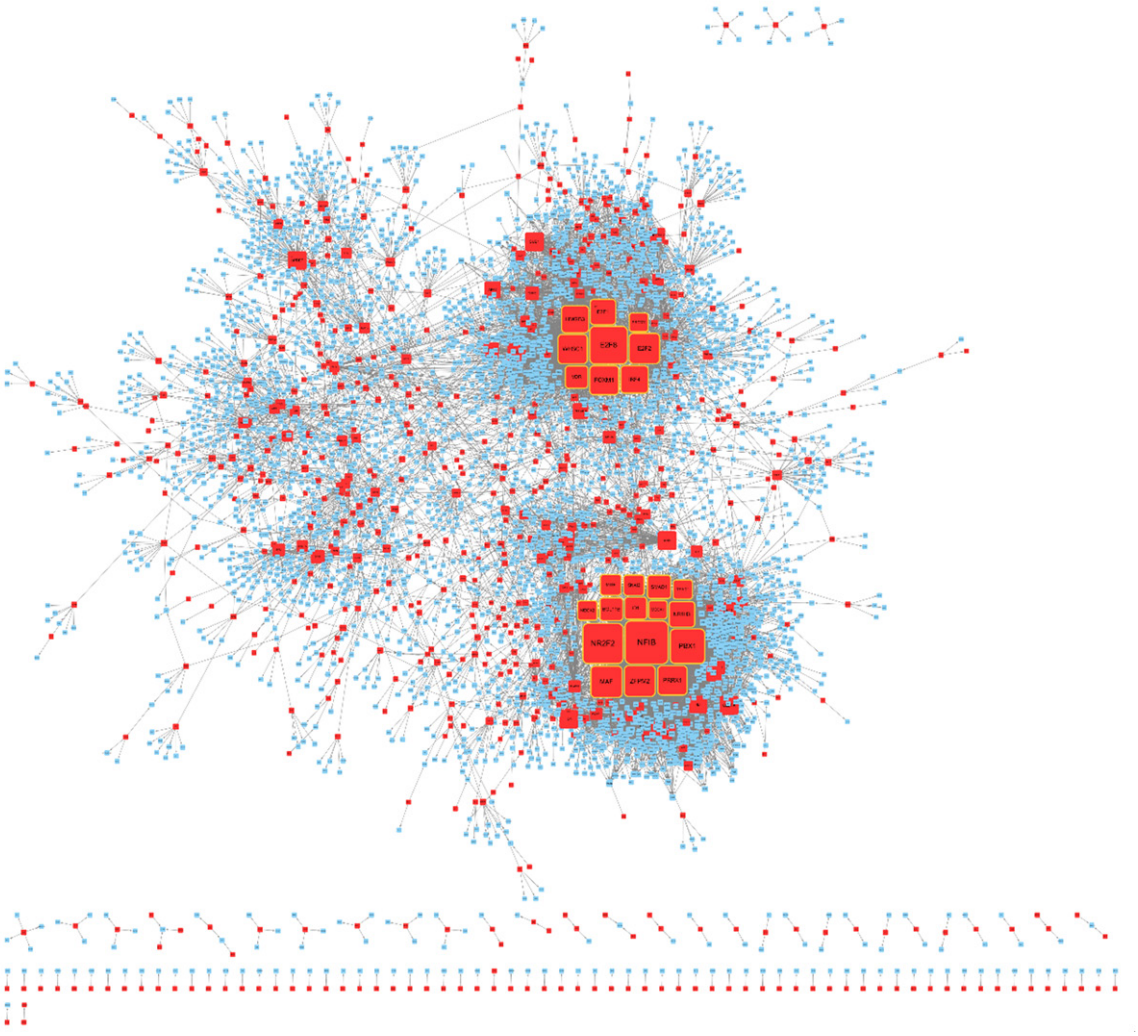

Figure S7. Transcriptional regulatory networks of mantle cell lymphoma. Red boxes represent transcription factors and blue boxes represent target genes. The box size represents the number of nodes of a gene. Arrows indicate the direction of regulation. Transcription factors with greater than 100 nodes are marked with a yellow box.

Table S2. Transcriptional regulatory network properties in three types of lymphoma

Parameters	Burkitt's lymphoma	Diffuse large B-cell lymphoma	Mantle cell lymphoma
Clustering coefficient	0.049	0.079	0.120
Connected component	56	76	101
Characteristic path length	7.027	5.624	8.316
Ave. number of neighbors	3.775	4.079	4.368
Number of nodes	5262	4859	4519
Multi-edge node pairs	68	89	130

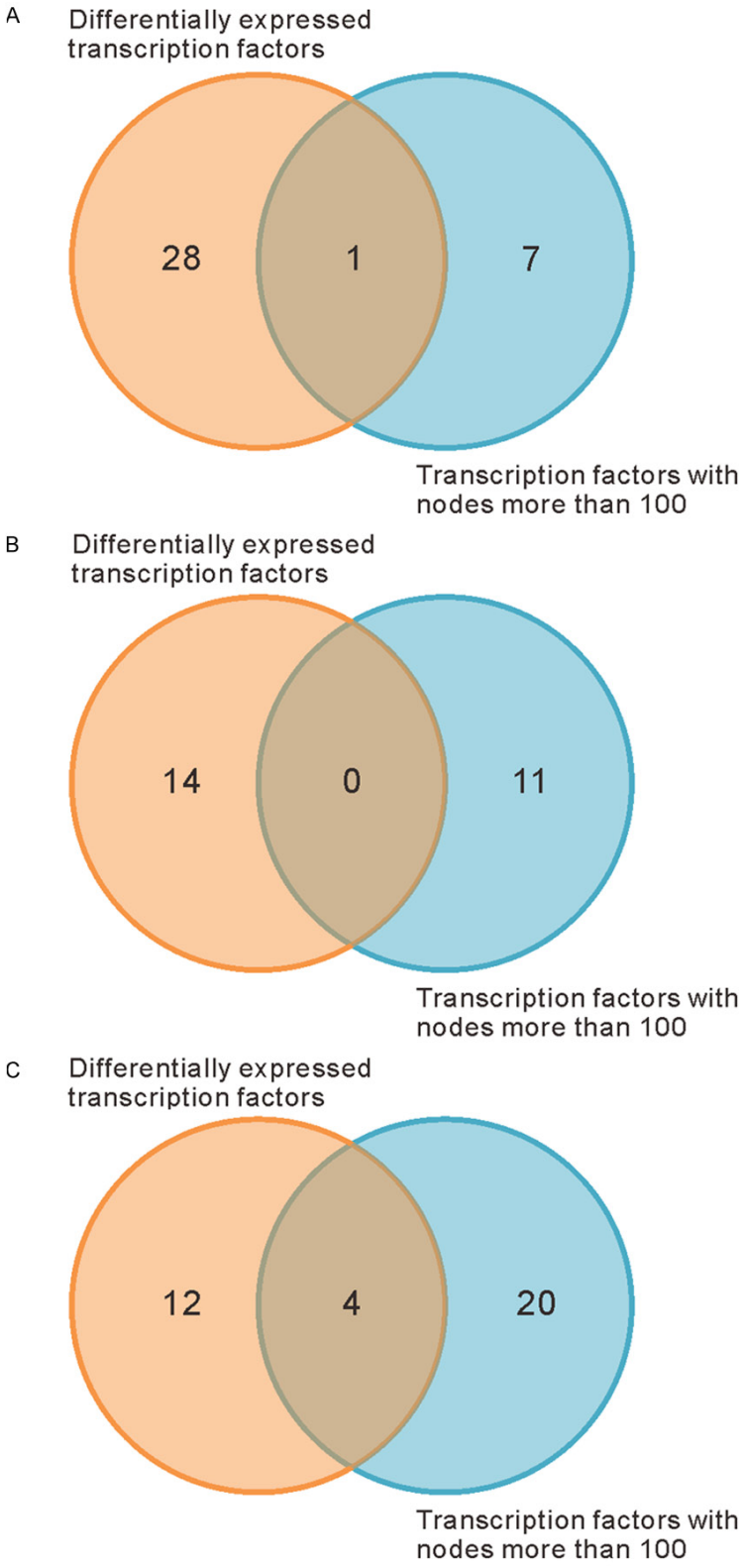

Figure S8. Venn diagram of differentially expressed transcription factors and transcription factors with greater than 100 nodes in Burkitt's lymphoma (A), diffuse large B-cell lymphoma (B) and mantle cell lymphoma (C). The common transcription factor in Burkitt's lymphoma is E2F8. The common transcription factors in mantle cell lymphoma are PRDM1, E2F8, VDR and IRF4.